VERY EASY & CUTE SECTION WITH MANY MINI ANSWERS

VSAQ

SECTION-B

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VSAQ CHAPTERS

28 MARKS

★★★ 1.	SOLUTIONS	(2+2 Marks)	56 - 57
★★★ 2.	ELECTRO CHEMISTRY	(2+2 Marks)	58 - 59
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★★★ 7.	ALCOHOLS, PHENOLS &ETHERS	(2 Marks)	68 - 69
★★ 8.	ALDEHYDES, KETONES	(2 Marks)	70 - 71
★★ 9.	AMINES	(2 Marks)	72 - 73
★★★ 10.	BIOMOLECULES	(2+2 Marks)	74 - 75

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Super Scoring Pages

★★★SSP★★★

Page 56,57,58,59,60,62,63,64,66,68,70,72,74,75

Total SS Pages : 14

1 SOLUTIONS

2+2Marks

75. Define molarity.

A: **1) Molarity (M):** It is the 'number of moles of the solute' dissolved 'per litre' of the solution.

2) Formula: $\text{Molarity(M)} = \frac{\text{No.of moles of solute}}{\text{Volume of solution in Litre}}$

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76. Define molality. [AP 15]

A: **1) Molality(m):** It is the 'number of moles of the solute' dissolved in 'one kilogram (Kg) of the solvent'

2) Formula: $\text{Molality(m)} = \frac{\text{No.of moles of solute}}{\text{Mass of solvent in Kg}}$

---

77. Define mole fraction. [AP 16,18,19]

A: **1) Mole Fraction(χ):** It is the 'ratio of the number of moles of one component' to the total number of moles of all the components present in the solution.

2) Formula: Mole fraction $\chi = \frac{\text{No.of moles of one component}}{\text{Total no.of moles of all the components}}$

---

78. State Henry's law. **[BMP]**[AP 18,20]

A: **1) Henry's law:** The partial pressure (p) of a gas in vapour phase is directly proportional to the mole fraction (χ) of the gas in the solution.

2) Formula: Partial pressure of a gas $p_{gas} = K_H \chi_{gas}$.

☺ Henry said $p = K_H \cdot \chi$

---

79. Define osmotic pressure. **[BMP]**[AP 15,16,18]

A: **1) Osmotic pressure (Π)** is the pressure required just to stop the Osmosis. (or)
It is the minimum pressure required to stop the flow of solvent particles through a semipermeable membrane from a pure solvent into a solution.

2) Formula: $\Pi = CRT$

---

80. What are isotonic solutions? [AP 17,22,24]

A: **1) Isotonic solutions** are the solutions which have the 'same osmotic pressure' at a given temperature .

2) Ex: Blood is isotonic with 0.9% (w/v) NaCl.

81. What is Ebullioscopic constant?

A: **1) Ebullioscopic constant(K_b) :** It is the **elevation in boiling point** of a solution when one mole of non-volatile solute is added to 1kg of solvent.

2) Formula: $\Delta T_b = K_b m$ **Units:** $k.kgmol^{-1}$

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82. What is Cryoscopic constant? [AP 23]

A: **1) Cryoscopic constant (K_f):** It is the **depression in the freezing point** of a solution when one mole of non-volatile solute is dissolved in 1kg of solvent.

2) Formula: $\Delta T_f = K_f \times m$ **Units:** $k.kgmol^{-1}$

---

83. Calculate the amount of benzoic acid (C_6H_5COOH) required for preparing 250 mL of 0.15 M solution of benzoic acid in methanol.

A: Molarity(M)= 0.15M, Volume (V)=250mL,

GMM (C_6H_5COOH) =122, Weight of C_6H_5COOH (w) =?

GMM of C_6H_5COOH
$(12\times6)+(5\times1)+(12\times1)+(16\times2)+(1\times1) = 122$

Formula: Molarity $M = \frac{w}{GMM} \times \frac{1000}{V(mL)}$

$$\Rightarrow w = \frac{M \times GMM \times V(mL)}{1000} = \frac{0.15 \times 122 \times 250}{1000} = 4.575 \text{ g}$$

---

84. Calculate the molarity of a solution containing 5 g of NaOH in 450 ml solution. [AP 15]

Sol: Given weight of solute (NaOH) is w=5 g; GMM of NaOH =40

Moles of NaOH, $n = \frac{weight}{GMM} = \frac{5}{40}$

Volume of the solution in mL = 450 mL, Molarity=?

☺ EASY PROBLEM ☺

Molarity $M = n \times \frac{1000}{V(mL)} = \frac{5}{40} \times \frac{1000}{450} = 0.278 \text{ mol dm}^{-3}$

---

85. Calculate molality of 2.5 g of ethanoic acid (CH_3COOH) in 75g of benzene.

Sol: Given weight of ethanoic acid = 2.5g

GMM of ethanoic acid $C_2H_4O_2 = (12\times2)+(1\times4)+(16\times2) = 60$

Moles of $C_2H_4O_2$, $n = \frac{weight}{GMM} = \frac{2.5}{60}$

weight of solvent (benzene)= 75 g Molality (m) =?

☺ IMP. PROBLEM ☺

$$\text{Molality(m)} = n \times \frac{1000}{\text{Mass of solvent (in g)}} = \frac{2.5}{60} \times \frac{1000}{75} = 0.556 \text{ mol.kg}^{-1}$$

2 ELECTRO CHEMISTRY

2+2Marks

86. State Faraday's first law of electrolysis. [BMP][AP 16,18,20,25]

A: Faraday's first law of electrolysis:

"The mass (m) of the substance deposited or liberated at the electrode is directly proportional to the quantity of electric charge (Q) passing through the electrolyte".

Thus, $m \propto Q$.

☺ FSL ☺ FIRST BEST 'Q'

---

87. State Faraday's second law of electrolysis. [AP 20]

A: Faraday's second law of electrolysis:

"When same quantity of electric charge is passed through different electrolytes, connected in series the amount of different substances deposited at the electrodes is directly proportional to their equivalent masses".

Mathematically, $\frac{m_1}{m_2} = \frac{E_1}{E_2}$

☺ FSL ☺ SECOND BEST 'Q'

---

88. What is a fuel cell? How is it different from a conventional galvanic cell?

A: 1) **A fuel cell** is a device that converts combustion energy of fuel directly into electrical energy.

2) Conventional galvanic cell converts chemical energy of the spontaneous redox reaction into electrical energy.

3) Fuel cells relatively generate less pollution when compared to conventional galvanic cells.

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89. What is metallic corrosion? Give one example. [BMP] [AP 15]

A: 1) Metallic Corrosion: It is the process of slow eating up of metals by gases and water vapours present in atmosphere due to the formation of certain compounds like oxides, sulphides, carbonates,etc

2) Ex: Rusting of Iron (Fe converts into its oxide)

☺ Even Corrosion gets U 2 Marks! ☺

---

90. Give the electro-chemical reaction that represents the corrosion or rusting of iron.

A: 1) Corrosion reaction:

Cathode: $O_2 + 4H^+ + 4e^- \rightarrow 2H_2O$ **Anode:** $2Fe \rightarrow 2Fe^{+2} + 4e^-$

2) Overall reaction: $2Fe + O_2 + 4H^+ \rightarrow 2Fe^{+2} + 2H_2O$

91. Define molar conductivity Λ_m and how is it related to conductivity (k)?

A: 1) **Molar conductivity** is defined as the conductance of the solution containing one gram-mole of the electrolyte, when the electrodes are unit distance apart.

2) Molar conductivity $\Lambda_m = \frac{k \times 1000}{M}$. Here k is conductivity, 'M' is molarity.

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92. What is cell constant of a conductivity cell?

A: 1) **Cell constant (*l*/a)** is the ratio of distance between electrodes and area of cross section.

2) Its unit is Cm^{-1} or m^{-1}.

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93. How is a galvanic cell represented on paper as per IUPAC convention? Give one example.

A: 1) As per IUPAC convention, a galvanic cell is represented as follows.
For the Daniel Cell : $Zn/Zn^{+2}(C_1)||Cu^{+2}(C_2)/Cu$

2) The electrode on which oxidation takes place is written on the left hand side and the other electrode on which reduction takes place is written on the right hand side.

3) Oxidation electrode and reduction electrode are separated by double line which represents salt bridge. C_1 and C_2 represent concentrations of Zn^{+2} and Cu^{+2} ions respectively.

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94. What is standard hydrogen electrode?

A: 1) Standard hydrogen electrode is a reference electrode

2) H_2 gas adsorbed on platinum at one atmosphere placed in contact with 1MHCl concentration in solution is treated as standard hydrogen electrode (SHE).

3) Potential of this electrode is taken as zero.

4) By using the standard hydrogen electrode (SHE) we can measure the potential of single electrode.

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95. Write the Nernst equation for the EMF of the cell Ni(s) / Ni^{2+}(aq) // Ag^{+}(aq) / Ag(s)

A: 1) $E_{cell} = E^0_{cell} - \frac{RT}{2F} ln \frac{[Ni^{2+}]}{[Ag^{+}]^2}$ (or) $E_{cell} = E^0_{cell} - \frac{0.059}{2} log \frac{[Ni^{2+}]}{[Ag^{+}]^2}$

2) **Anode:** $Ni(s) \longrightarrow Ni^{2+}(aq) + 2e^{-}$.....(1)

3) **Cathode:** $2Ag^{+}(aq) + 2e^{-} \longrightarrow 2Ag(s)$ .....(2);

4) **Net Equation:** $Ni(s) + 2Ag^{+}(aq) \longrightarrow Ni^{2+}(aq) + 2Ag(s)$

3 CHEMICAL KINETICS

2 Marks

96. What is a pseudo first order reaction? Give one example. **[BMP]** [AP 16,23]

A: 1) **Pseudo first order reaction** is a reaction in which one of the reactants, is present in large amount and its concentration does not get altered during the course of the reaction and it behaves as first order reaction.

2) **Ex: Hydrolysis of Ethyl acetate:** $CH_3COOC_2H_5 + H_2O \xrightarrow{H^+} CH_3COOH + C_2H_5OH$

Rate of reaction $= k[CH_3COOC_2H_5]$

Here the H_2O is taken in excess. So, the rate doesn't depend on $[H_2O]$

---

97. Give two examples for gaseous first order reactions.

A: 1) $N_2O_{5(g)} \rightarrow 2NO_{2(g)} + \frac{1}{2}O_{2(g)}$, order =1 [Decomposition of N_2O_5]

2) $C_2H_{4(g)} + H_{2(g)} \rightarrow C_2H_{6(g)}$, order =1 [Hydrogenation of ethene]

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98. Give two examples for zero Order reactions. [AP 19]

A: 1) $H_2 + Cl_2 \xrightarrow{h\gamma} HCl$, rate= $k[H_2]^0[Cl_2]^0$, order =0

2) $2HI \xrightarrow{gold} H_2 + I_2$, rate = $k[HI]^0$, order =0

---

99. Define molecularity of a reaction with an example.

A: 1) **Molecularity of a reaction:** It is the number of reacting molecules taking part in a single elementary step.

2) **Examples:**

$C_2H_5Cl \rightarrow C_2H_4 + HCl$ (Unimolecular reaction)

$H_2 + I_2 \rightarrow 2HI$ (Bimolecular reaction)

$2NO_{(g)} + O_{2(g)} \rightarrow 2NO_{2(g)}$ (Trimolecular reaction)

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100. Write the equation for the rate of the reaction $5Br^-_{(aq)} + BrO^-_{3(aq)} + 6H^+_{(aq)} \rightarrow 3Br_{2(aq)} + 3H_2O(l)$

A: Rate $= -\frac{1}{5}\frac{\Delta[Br^-]}{\Delta t} = -\frac{\Delta[BrO_3^-]}{\Delta t} = -\frac{1}{6}\frac{\Delta[H^+]}{\Delta t} = +\frac{1}{3}\frac{\Delta[Br_2]}{\Delta t} = +\frac{1}{3}\frac{\Delta[H_2O]}{\Delta t}$

---

101. What is rate law? Illustrate with an example.

A: 1) **Rate Law:** It is an experimentally determined expression which relates the rate of reaction with concentration of reactants.

2) **Consider a hypothetical reaction:** $aA + bB \rightarrow$ products.

The rate law expression for this reaction is, Rate $= k[A]^m[B]^n$.

The order of the above reaction is $(m+n)$.

102. What are elementary reactions?

A: 1) The reactions taking place in one step are called elementary reactions.

2) **Ex:** $2HI \rightarrow H_2 + I_2$

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103. What are complex reactions? Name one complex reaction.

A: 1) **Complex reactions:** These are the reactions which involve sequence of elementary reactions for the conversion of reactants into products.

2) Ex 1: Oxidation of ethane to CO_2 and H_2O passes through a series of intermediate steps. Here alcohol, aldehyde and acid are formed.

Ex 2: $2H_2O_2 \rightarrow 2H_2O + O_2$

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104. What is half-life of a reaction? Illustrate your answer with an example.

A: 1) **Half-life of a reaction ($t_{1/2}$):** It is the time required to reduce half of the initial concentration of reactant.

2) For a zero order reaction, rate constant is given by $k = \frac{[R_0]-[R]}{t}$

At $t = t_{1/2}$ we have $[R] = \frac{[R_0]}{2}$

$\therefore$ The rate constant at $t_{1/2}$ is $k = \frac{[R_0]-\left[\frac{R_0}{2}\right]}{t_{1/2}} \Rightarrow k = \frac{[R_0]}{2[t_{1/2}]} \Rightarrow t_{1/2} = \frac{[R_0]}{2k}$

3) For a first order reaction, $t_{1/2} = \frac{0.693}{k}$

---

105. Calculate the overall order of a reaction which has the rate expression

(a) Rate = $k[A]^{1/2}[B]^{3/2}$ **(b) Rate = $k[A]^{3/2}[B]^{-1}$**

Sol: (a) Rate = $k[A]^x[B]^y$; Order = x + y $\therefore$ Order = $\frac{1}{2}+\frac{3}{2}=\frac{4}{2}=2$ i.e., second order

(b) Order = $\frac{3}{2}+(-1)=\frac{3}{2}-1=\frac{1}{2}$ i.e., half order.

---

106. A first order reaction is found to have a rate constant $k = 5.5 \times 10^{-14}\ s^{-1}$. Find the half-life of the reaction. [AP 15]

Sol: 1) Half-life for a first order reaction is $t_{1/2} = \frac{0.693}{k}$

2) $\therefore t_{1/2} = \frac{0.693}{5.5\times10^{-14}s^{-1}} = 1.26\times10^{13}s$

4 The d&f-Block Elements

2+2 Marks

107. What are transition elements? Give examples.

A: 1) **Transition elements:** These are the elements with partially filled d-orbital either in their atoms or ions.

2) **Ex:** Chromium (Cr 24) : [Ar] $4s^1 3d^5$; for Cr^{+3} : [Ar]$4s^0 3d^3$.

Manganese (Mn 25) : [Ar]$4s^2 3d^5$. for Mn^{+2} : [Ar]$4s^0 3d^5$.

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108. Why Zn^{2+} is diamagnetic whereas Mn^{2+} is paramagnetic? [BMP] |AP 23|

A: 1) Electronic configuration of $Zn^{+2}$is[Ar] $4s^0 3d^{10}$

It has no unpaired electrons. So it is diamagnetic.

Atomic numbers
Zn = 30;Mn = 25;Ar = 18

2) E.C of Mn^{+2} is [Ar]$4s^0 3d^5$

It has five unpaired electrons. So it is paramagnetic.

☹ Zn-Paired-Dia
☺ Mn-Unpaired-Para

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109. Calculate the 'spin only' magnetic moment of $Fe^{2+}_{(aq)}$ ion. [BMP] |AP 17,22,24|

A: 1) Electronic configuration of Fe^{+2} ion = [Ar] $3d^6 4s^0$

⇅	1	1	1	1

The number of unpaired electrons is n=4.

2) ∴ 'Spin only' magnetic moment $(\mu) = \sqrt{n(n+2)} = \sqrt{4(4+2)} = \sqrt{24} = 4.9$ BM

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110. Aqueous Cu^{2+} ions are blue in colour, where as Aqueous Zn^{2+} ions are colourless. Why? |AP 16,25|

A: 1) E.C of Cu^{+2} is[Ar] $4s^0 3d^9$. It contains one unpaired electron.

Hence it exhibits blue colour in aqueous solution.

Atomic numbers
Cu =29
Zn = 30

2) E.C of Zn^{+2} is [Ar] $4s^0 3d^{10}$.

It contains no unpaired electron. Hence it is colourless in aqueous solution.

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111. What is an alloy? Give example. |AP 18|

A: 1) **Alloy:** It is a homogeneous mixture of a metal with other metals or metalloids or non-metals, having 'similar physical properties' to that of the original metal.

Alloys add more desirable properties to the original metal.

2) **Ex 1:** Gold is an alloy in Jewellery; Steel is an alloy of Iron and Carbon

Ex 2: Brass: 60-80% Cu; 20-40% Zn; Bronze: 75-90% Cu; 10-25% Sn

112. What is mischmetal? Give its composition and uses. [AP 16,19]

A: **1) Mischmetal** is a well known alloy of Lanthanoid metal.

2) Composition: 95% Lanthanoid metals (Ce, La, Nd, Pt), 5% iron and traces of S, C, Ca, Al.

3) Uses: Ignitium sources - Cigarette lighter, fire starter, bullets.

Mischmetal is a Bullet Metal

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113. What is actinoid contraction?

A: **1) Actinoid contraction:** It is the gradual decrease in the size of atoms or ions from left right in actinoid series.

2) It is due to poor screening effect of 5f orbitals.

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114. Why do the transition metals form a large number of complex compounds?

A: Transition metals form a large number of complex compounds due to their **small size, higher effective nuclear charge** and presence of **partially filled d-orbitals.**

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115. How do transition metals exhibit catalytic activity?

A: 1) Transition elements exhibit catalytic activity due to

(i) Variable oxidation states

(ii) Free valencies on their surfaces (iii) presence of **partially filled d-orbitals.**

2) **Ex:** Fe is the catalyst used in Haber's process in the manufacture of ammonia.

V_2O_5 is the catalyst used in Contact process in the manufacture of H_2SO_4.

Pt is the catalyst used in Ostwald's process in the manufacture of nitric acid.

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116. Why is it difficult to obtain M^{3+} oxidation state in Ni,Cu and Zn

A: 1) The element Ni, Cu and Zn are stable in +2 oxidation state and hence difficult to obtain +3 oxidation state.

2) Reason for +2 oxidation stability:

Ni and Cu in +2 oxidation state (Ni^{+2}) are stable due to highest negative enthalpy of hydration.

Zn in +2 oxidation state (Zn^{+2}) is stable due to completely filled d-subshell.

5 COORDINATION COMPOUNDS

2 Marks

117. What are coordination compounds? Give two examples.

A: **1) Coordination compounds** are the metal atoms that are bond to few anions or neutral molecules by sharing electrons through coordinate covalent bonds.

2) Ex: $[Co(NH_3)_6]Cl_3$, $K_4[Fe(CN)_6]$

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118. What is a double salt? Give example.

A: **1) Double salts** contain two or more compounds which are stable in solid state. But they dissociate into its constituent ions when dissolved in water . In this process, the individual properties of constituent ions are not lost.

2) Ex: Mohr salt - $FeSO_4$, $(NH_4)_2SO_4$. $6H_2O$

Potash alum - $K_2SO_4.Al_2(SO_4)_3.24H_2O$

Carnallite - $KCl.MgCl_2.6H_2O$

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119. What is the difference between a double salt and a complex compound?

A: 1) A double salt differs from complex compound in its dissociation.

2) A double salt dissociates completely into its constituent ions when dissolved in water whereas a complex compound do not dissociate into its constituent ions when dissolved in water.

3) **Ex: Double salt:** Carnalite $KCl.MgCl_2.6H_2O \rightarrow K^+ + Mg^{+2} + 2Cl^-$

Complex salt: Potassium ferrocyanide $K_4[Fe(CN)_6] \rightarrow 4K^+ + [Fe(CN)_6]^{4-}$

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120. What is a chelate ligand ? Give example.

A: **1) Chelate ligand:** It is a ring-like structure obtained when a bidentate or a polydentate ligand uses its two or more donor atoms to bind a single metal ion.

2) Ex: Oxalate $(C_2O_4^{-2})$, en-(ethylene diammine) $\begin{pmatrix} CH_2-NH_2 \\ | \\ CH_2-NH_2 \end{pmatrix}$

---

121. What is an ambidentate ligand? Give example. **[BMP]** [AP 16]

A: **1) Ambidentate ligand:** It is a ligand which contains two donor atoms but only one of them forms a coordinate bond at a time with central metal atom / ion.

2) Ex:

$M \leftarrow SCN$	$M \leftarrow NCS$
thiocyanato	isothiocyanato
(S donor atom)	(N donor atom)

122. $CuSO_4.5H_2O$ is blue in colour where as anhydrous $CuSO_4$ is colourless. Why?

A: 1) In $CuSO_4.5H_2O$, water acts as ligand. As a result it causes crystal field splitting. Hence, d-d transition is possible in $CuSO_4.5H_2O$ and it shows colour.

2) In the anhydrous $CuSO_4$ due to the absence of water (ligand), crystal field splitting is not possible. Hence it is colourless.

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123. What is the oxidation state of cobalt in (i) $K[Co(CO)_4]$ and (ii) $[Co(NH_3)_6]^{3+}$

A: i) $K[Co(CO)_4]$: Oxidation state of cobalt is –1

ii) $[Co(NH_3)_6]^{3+}$: Oxidation state of cobalt is +3

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124. $[Cr(NH_3)_6]^{3+}$ is paramagnetic while $[Ni(CN)_4]^{2-}$ is diamagnetic. Why?

A: 1) $[Cr(NH_3)_6]^{3+}$ is **paramagnetic** due to the **presence** of three **unpaired electrons.**

2) $[Ni(CN)_4]^{2-}$ is **diamagnetic** due to the **absence** of **unpaired electrons.**

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125. $[Fe(CN)_4]^{2-}$ and $[Fe(H_2O)_6]^{2+}$ are of different colours in dilute solutions. Why?

A: 1) In both the complexes, Fe has +2 oxidation state . It has four unpaired electrons.

In the presence of **weak H_2O ligand**, the unpaired electrons **do not pair up.**

In the presence of **strong CN^- ligand**, the unpaired electrons **pair up .**

2) They have **different colours** due to the **difference in the number of unpaired electrons.**

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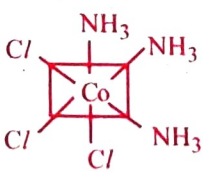
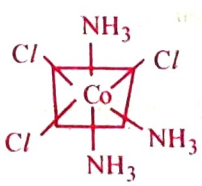
126. How many geometrical isomers are possible in the following coordination entities? (i) $[Cr(C_2O_4)_3]^{3-}$ (ii) $[CO(NH_3)_3Cl_3]$

A: i) No geometrical isomers are possible for $[Cr(C_2O_4)_3]^{-3}$.

ii) Two geometrical isomers are possible for $[CO(NH_3)_3Cl_3]$



Facial (fac)



Meridonial (mer)

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6 HALOALKANES & HALOARENES

2 Marks

127. What are Enantiomers? **[BMP][AP 16,23]**

A: 1) **Enantiomers** are a pair of stereo isomers which are non -super impossible mirror images.

2) **Ex:** d-Lactic acid & *l*-Lactic acid

☺ బంబం-ప్రతిబంబం ☺

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128. What type of isomerism is exhibited by o-, m- and p-dichlorobenzenes?

A: Isomerism exhibited by o-, m- and p-dichlorobenzenes is **Position isomerism.**

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129. What are ambident nucleophiles? **[AP 16,22,23]**

A: 1) **Ambident nucleophiles: These** are the nucleophiles with two donor atoms.

2) **Ex:** Cyanide ion, Nitrite ion.

Ambi means
☺ **Both** ☺

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130. What is the stereochemical outcome of S_N1 and S_N2 reactions?

A: 1) The stereochemical outcome of S_N1 reaction is a **racemic mixture** of two optical isomers (50:50).

2) The stereochemical outcome of S_N2 reaction is **Walden inversion** (inversion of configuration).

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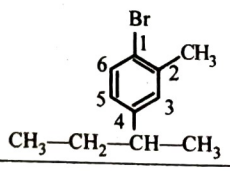
131. Write the structure of the following compounds **[AP 15]**

(i) 2-chloro-3-methylpentane **(ii)1-bromo-4-sec-butyl-2-methylbenzene**

A: (i) $CH_3-\underset{1}{CH}(Cl)-\underset{2}{CH}(CH_3)-\underset{3}{CH_2}-\underset{4}{CH_3}$ $\quad$ (ii) 

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132. Write the isomers of the compound having molecule formula C_4H_9Br?

A: 1) $CH_3-CH_2-CH_2-CH_2Br$ 1–Bromobutane

2) $CH_3-CH_2-CH(Br)-CH_3$ 2–bromobutane

3) $CH_3-CH(CH_3)-CH_2Br$ 1–bromo-2 methyl propane

4) $CH_3-C(CH_3)(Br)-CH_3$ 2–bromo-2 methyl propane

133. Which compound in each of the following pairs will react faster in S_N2 reaction with OH?
(i) CH_3Br or CH_3I (ii) $(CH_3)_3CCl$ (or) CH_3Cl

A: i) CH_3I reacts faster because C-I has **less dissociation energy** than that of C- Br bond.
ii) CH_3Cl reacts faster because it has **less steric hinderance** as compared to $(CH_3)_3CCl$.

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134. Between $C_6H_5CH_2Cl$ and $C_6H_5CHClC_6H_5$, which is more easily hydrolysed by aqueous KOH?

A: 1) $C_6H_5CHClC_6H_5$ will be more easily hydrolysed than $C_6H_5CH_2Cl$.
2) Because its carbocation(2°) is stabilised by two phenyl groups (C_6H_5-) due to resonance under S_N1 reaction.

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135. Explain why the alkyl halides though polar are immiscible with water?

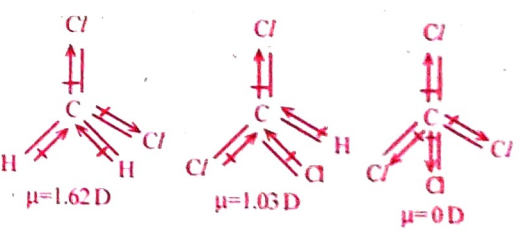
A: Alkyl halides (R-X) are polar due to the presence of polar C-X bond.
Alkyl halides are immiscible with water because
a) Alkyl halides are not able to form hydrogen bonds with water molecules.
b) Alkyl halides are not able to break hydrogen bonds in water.

---

136. Which one of the following has highest dipole moment?
(i) CH_2Cl_2 (ii) $CHCl_3$ (iii) CCl_4

A: CH_2Cl_2 has highest dipole moment

i) In CH_2Cl_2, the resultant dipolements of two C-Cl bonds is reinforced by two C-H bond moments.
Its dipole moment is $\mu = 1.62D$.



ii) In $CHCl_3$, the resultant dipole moment of two C-Cl bonds is opposed by resultant dipolemoment of C-Cl and C-H bonds. Here the net dipolemoment is $\mu = 1.03D$

iii) CCl_4 is a symmetrical molecule . Its dipole moment is $\mu = 0$.

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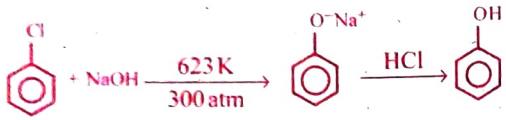
7 ALCOHOLS, PHENOLS & ETHERS

2 Marks

137. Give the reagents used for the preparation of phenol from chlorobenzene. [BMP]

A: 1) Phenol can be obtained by heating chlorobenzene in the presence of a catalyst, with 10% NaOH solution and HCl at 350°C under 200-300 atm pressure.

2) Reaction:



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138. Explain why propanol has higher boiling point than that of the hydrocarbon butane.

A: 1) The boiling point of propanol is higher because of the presence of intermolecular hydrogen bonding in the molecules. But, it is not present in butane due to the absence of polar –OH group.

2) The **boiling point (391K) of propanol** is **more** than that of **butane (309K).**

---

139. Alcohols are comparatively more soluble in water than hydrocarbons of comparable molecular masses. Explain this fact.

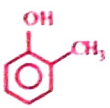
A: 1) Alcohols can form H-bonds with water and break the H-bonds already existing between water molecules. Hence, they are soluble in water.

2) But hydrocarbons cannot form H-bonds with water and hence they are insoluble in water.

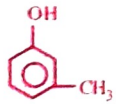
---

140. Give the structures and IUPAC names of monohydric phenols of molecular formula C_7H_8O.

A:



2-methylphenol
(o-Cresol)



3-methylphenol
(m-cresol)

4-methylphenol
(p-cresol)

---

141. Preparation of ethers by acid dehydration of secondary or tertiary alcohols is not a suitable method. Give reason.

A: Ethers cannot be prepared by the dehydration of secondary (or) tertiary alcohols, because alkenes are formed easily in these reactions. This happens due to **steric hinderance.**

---

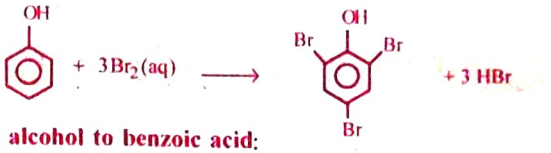
142. Write the equations for the following reactions

i) Bromination of phenol to 2,4,6-tribromophenol

ii) Benzyl alcohol to benzoic acid

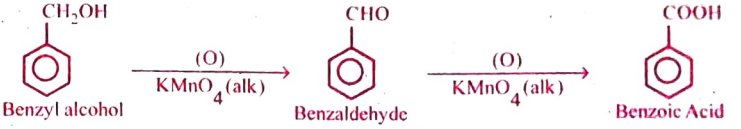
A: **i) Bromination of phenol to 2,4,6-tribromophenol:**

Phenol when treated with excess of aqueous bromine solution forms 2,4,6-tribromophenol.



ii) Benzyl alcohol to benzoic acid:

Benzyl alcohol oxidises into Benzaldehyde and then to Benzoic acid in the presence of alkaline $KMnO_4$.



---

143. Write the structures for the following compounds
1) Ethoxy Ethane 2) Ethoxybutane 3) Phenoxy ethane

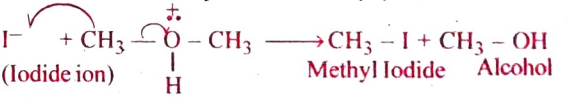
A: 1) **Ethoxy Ethane :** $C_2H_5 - O - C_2H_5$
2) **Ethoxybutane :** $C_2H_5 - O - CH_2 - CH_2 - CH_2 - CH_3$
3) **Phenoxy ethane or Ethoxy benzene or Ethyl phenyl ether :** $C_6H_5 - O - C_2H_5$

---

144. Write the mechanism of the reactions of HI with methoxymethane.

A: 1)The ether molecule is initially protonated by the halogen acid (HI) to form protonated ether.

2)The protonated ether is then attacked by the halide ion (I^-) which acts as nucleophile as follows:



---

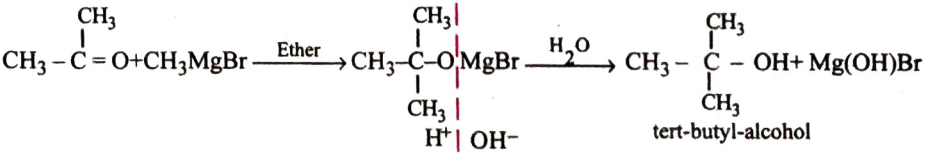
145. Name the reagents used in the following reactions:
i) Oxidation of primary alcohol to carboxylic acid
ii) Oxidation of primary alcohol to aldehyde

A: i) Acidified $KMnO_4$ (or) acidified $K_2Cr_2O_7$ is used in the oxidation of primary alcohol to carboxylic acid
ii) CrO_3 (or) pyridine chlorochromate (pcc) is used as a reagent in the oxidation of primary alcohol to aldehyde.

---

146. Identify the reactant needed to form t-butyl alcohol from acetone.

A: **CH_3MgBr** (Methyl magnesium bromide) is needed to form t-butyl alcohol from acetone.



8 ALDEHYDES, KETONES & CARBOXYLIC ACIDS

2 Marks

147. List the reagents suitable to reduce carboxylic acid to alcohol. **[BMP]**

A: **LiA/H_4 (or) copper chromate** are the needed reducing reagents .

$$CH_3COOH + 4(H) \xrightarrow{LiAlH_4} C_2H_5OH + H_2O$$

---

148. Write the reaction showing α-halogenation of carboxylic acid and give its name.

A: **HVZ Reaction:**

Carboxylic Acid having atleast one α- hydrogen is treated with Halogens (Br (or) C*l*) in the presence of red phosphorus undergo Halogenation to form α-Halo Carboxylic Acid.

$$\text{Ex: } \underset{\text{Carboxylic acid}}{R-CH_2-COOH} \xrightarrow[(ii)\ H_2O]{(i)\ X_2/\text{Red phosphorus}} \underset{\alpha\text{–halo carboxylic acid}}{R-\overset{X}{\overset{|}{C}}H-COOH} + HX$$

---

149. Arrange the following compounds in increasing order of their property indicated.

i) acetaldehyde, acetone and t-butylmethyl ketone reactivity towards HCN

ii) fluoroacetic acid, monochloroacetic acid, acetic acid and dichloroacetic acid (acid strength)

A: i) **Reactivity towards HCN:** t-butylmethyl ketone < acetone < acetaldehyde.

ii) **Acid strength:** dichloroacetic acid >fluoroaceticacid > monochloroacetic acid > acetic acid.

---

150. How do you distinguish acetophenone and benzophenone?

A: 1) **Acetophenone** undergoes iodoform reaction due to the **presence of acetyl group CH_3–CO .**

2) But **benzophenone** does not undergo iodoform reaction due to the **absence** of **CH_3–CO** group

$$\underset{\text{Acetophenone}}{C_6H_5COCH_3} + 3I_2 + 4NaOH \xrightarrow{\text{heat}} C_6H_5COONa + \underset{\text{Iodoform}}{CHI_3} + 3H_2O + 3NaI$$

---

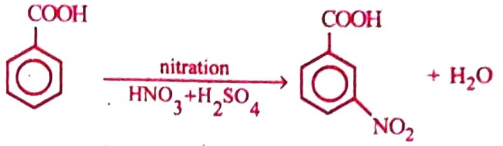
151. An organic acid with molecular formula $C_8H_8O_2$ on decarboxylation forms Toluene. Identify the organic acid.

A: The organic acid formed is phenyl acetic acid (2-Phenyl ethanoic acid)

152. Explain the position of electrophilic substitution in benzoic acid.

A: 1) COOH group on benzene, deactivates the ring at ortho and para positions. Hence substitution takes place at meta position.

2)Ex:



(m-nitrobenzoic acid)

---

153. Compare the acidic strength of acetic acid, Chloroacetic acid, benzoic acid and Phenol. [AP 22][IPE'14]

A: **1) The relative values of k_a and pk_a indicate the acidic strength:**

Higher k_a value and lower pk_a value indicates that the acid is strong.

2) Order of acidic strength: Chloroacetic acid > benzoic acid > acetic acid > phenol

The acidic strength of acetic acid, chloroacetic acid, benzoic acid and phenol:

Acid	k_a (298 k)	pk_a (298 k)
1) CH_3COOH (Acetic acid)	1.75×10^{-5}	4.76
2) $Cl–CH_2–COOH$ (Chloroacetic acid)	1.36×10^{-3}	2.87
3) C_6H_5COOH (Benzoic acid)	6.3×10^{-5}	4.0
4) C_6H_5OH (Phenol)	1.1×10^{-10}	10

---

154. Write equations showing the conversion of

i) acetic acid to acetyl chloride **ii) benzoic acid to benzamide**

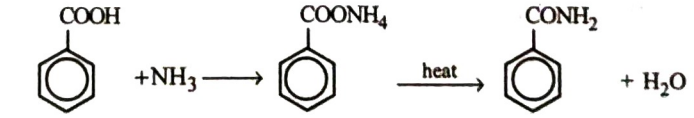
A: **i) acetic acid to acetyl chloride:**

$$3CH_3COOH + PCl_3 \longrightarrow 3CH_3COCl + H_3PO_3$$

$$CH_3COOH + PCl_5 \longrightarrow CH_3COCl + POCl_3 + HCl$$

$$CH_3COOH + SOCl_2 \longrightarrow CH_3COCl + SO_2 + HCl$$

ii) benzoic acid to benzamide



Benzoic Acid — Ammonium benzoate — benzamide

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9 AMINES

2 Marks

155. Arrange the following bases in increasing order of their basic strength. Aniline, p-nitroaniline and p-toluidine. [BMP]

A: Increasing order (low to high) of basic strength: p-nitroaniline < Aniline < p-toluidine.

---

156. Write equations for carbylamine reaction of an aliphatic amine.

A: 1) When a primary amine is heated with alcoholic caustic potash and chloroform, an offensive smelling compound called carbylamine (Ethylisocyanide) is formed. This test is used for the detection of primary amines.

2) Reaction: $\underset{\text{Ethanamine}}{C_2H_5NH_2} + CHCl_3 + 3KOH \xrightarrow[\Delta]{\text{alc.}} \underset{\text{Ethyl carbylamine}}{C_2H_5NC} + 3KCl + 3H_2O$

---

157. Gabriel phthalimide synthesis exclusively forms primary amines . Explain.

A: Gabriel phthalimide synthesis is preferred for synthesizing aliphatic primary amines because aryl halides do not undergo nucleophilic substitution with the anion formed by phthalimide.

---

158. Arrange the following bases in decreasing order of pK_b values
$C_2H_5NH_2$, $C_6H_5NHCH_3$, $(C_6H_5)_2NH$ and $C_6H_5NH_2$.

A: Decreasing order of pK_b : $C_6H_5NH_2 > C_6H_5NHCH_3 > C_2H_5NH_2 > (C_6H_5)_2NH$

pK_b values: 9.38 9.30 3.29 3.00

[Decreasing order of Basicity: $(C_6H_5)_2NH > C_2H_5NH_2 > C_6H_5NHCH_3 > C_6H_5NH_2$ **]**

---

159. Write the IUPAC names of the following compounds and classify into primary, secondary and tertiary amines.

(i) $(CH_3)_2CHNH_2$ (ii) $CH_3(CH_2)_2NH_2$ (iii) $(CH_3CH_2)_2NCH_3$.

A: (i) $CH_3-\underset{\displaystyle CH_3}{\underset{|}{CH}}-NH_2$ Propan-2-amine (Primary amine)

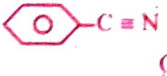
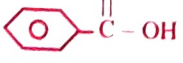
(ii) $CH_3-CH_2-CH_2-NH_2$ 1-Propanamine (Primary amine)

(iii) $CH_3-CH_2-\underset{\displaystyle CH_2-CH_3}{\underset{|}{N}}-CH_3$ N-ethyl-N-Methyl-ethanamine (Tertiary amine)

---

160. Give structures of A,B and C in the following reaction.

$C_6H_5N_2Cl \xrightarrow{CuCN} A \xrightarrow{H_2O/H^+} B \xrightarrow{NH_3} C$

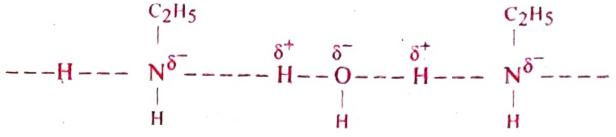
A: **A :** C_6H_5CN (or)  **B :** C_6H_5COOH(or) 

C : $C_6H_5COONH_4$ (or)

161. Explain why ethylamine is more soluble in water where as aniline is sparingly soluble.

A: 1) Ethylamine forms hydrogen bonds with water molecules and hence dissolves in water.

2) Aniline, due to its large hydrocarbon part, develops hydrophobic character. As a result it does not dissolve in water.



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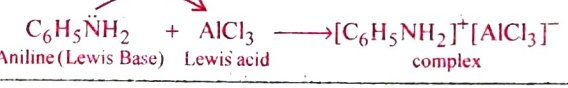
162. Why aniline does not undergo Friedel-Craft reactions?

A: **1) Aniline** doesnot undergo Friedel-Crafts reaction. Actually, aniline being a Lewis base forms a complex with $AlCl_3$ which is a Lewis acid.

2) The amino group is not in a position to activate the benzene ring towards electrophilic substitution. Hence, alkylation (or) acylation which lead to Fridel Crafts reaction is not possible.



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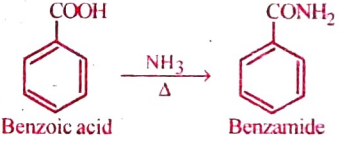
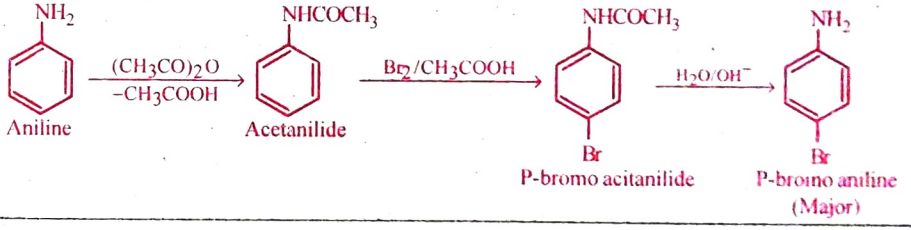
163. Accomplish the following conversions:

(i) Benzoic acid to benzamide (ii) Aniline to p-bromoaniline

A: i) **Benzoic acid to Benzamide:**



ii) **Aniline to p-bromo aniline**:



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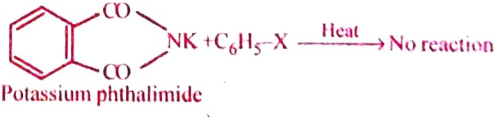
164. Why aromatic primary amines cannot be prepared by Gabriel phthalimide?

A: 1) Aryl halides does not react with potassium phthalimide. Because C-X bond in haloarene is difficult to be cleaved due to partial double bond character.

2) They do not undergo nucleophilic substitution with the anion formed by phthalimide. So, aromatic primary amines cannot be prepared by Gabriel phthalimide synthesis.



10 BIOMOLECULES

2+2Marks

165. What are essential and non-essential amino acids? Give one example for each. [BMP]

A: 1) **Essential Amino acids:** These are the amino acids which cannot be synthesised in the body and must be taken in diet. **Ex:** Valine, leucine, phenylalanine .

2) **Non-essential amino acids:** These are the amino acids which can be synthesised in the body **Ex:** Glycine alanine, glutamic acid.

---

166. Name the vitamin responsible for the coagulation of blood.

A: **Vitamin K** is responsible for the coagulation of blood.

---

167. Differentiate between globular and fibrous proteins. [BMP][AP 20]

A:

Globular Proteins	Fibrous Proteins
1) Polypeptide chains are arranged as coils.	1) Polypeptide chains are parallel to each other.
2) They have spherical shape.	2) They have thread like structure.
3) They are water soluble.	3) They are insoluble in water.
4) **Ex:** Insulin, albumins etc.,	4) **Ex:** Keratin, Myosin etc.,

---

168. What is glycogen? How is it different from starch?

A: 1) **Glycogen** is a polysaccharide which is stored in animal body (liver, muscles, brain).

Starch is a polysaccharide which is stored in plants.

2) **Glycogen** is more highly branched than Starch (amylopectin).

---

169. What are anomers?

A: 1) **Anomers** are those isomers which differ in orientation of –OH at C-1 (first carbon)'
2) **Ex:** α-D-glucose and β-D-glucose

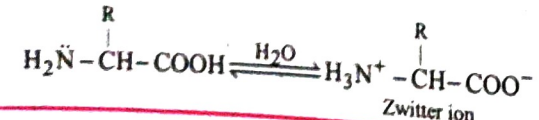
---

170. What is Zwitter ion? Give an example. [AP 15,22]

A: 1) **Zwitter ion:** It is a 'dipolar neutral ion' containing negative and positive charges.

2) **Ex:** Free amino acids in cells.

3) In Amino acids, proton transfer takes place from –COOH group to NH_2 group. This results in the formation of Zwitter ion.



☺ U C ☺ Zwitter has Twitter

171. Write the names of three types of RNA.

A: **Three types of RNA:**

a) m-RNA (messenger RNA) b)r-RNA (ribosomal RNA) c) t-RNA (transfer RNA)

---

172. Why are sugars classified as reducing and non-reducing sugars?

A: 1) Sugars are classified as reducing and non-reducing sugars to **test** whether the sugar has a free aldehyde or ketone group that can act as a reducing agent.

2) **Reducing sugars** reduce Fehling's solution or Tollen's reagent .
They contain free aldehyde or Ketonic group. **Ex:** Glucose & Maltose

3) **Non-reducing sugars** do not reduce Fehling's solution or Tollen's reagent.
They do not ~~have~~ free aldehyde or ketonic group. **Ex:** Sucrose & Cellulose

---

173. Name the different types of carbohydrates on the basis of their hydrolysis. Give one example for each.

A: **Classification of Carbohydrates (Saccharides) on the basis of their hydrolysis:**

a) **Monosaccharides** cannot be broken further into smaller units on hydrolysis
Ex: Glucose and fructose.

b) **Disaccharides** give 2 units of monosaccharides on hydrolysis.
Ex: Sucrose, lactose

c) **Oligosaccharides**: These carbohydrates give 2-10 units of monosaccharides on hydrolysis.
Ex: Raffinose(which gives glucose, fructose and galactose on hydrolysis)

d) **Polysaccharides**: These carbohydrates give large number of monosaccharides on hydrolysis.
Ex: Starch, cellulose etc.,

---

174. What is the difference between a nucleoside and a nucleotide?

A: These two are fundamental building blocks of nucleic acids DNA/ RNA

Nucleoside	Nucleotide
1) Nucleoside = Base + Sugar	1) Nucleotide = Base + Sugar + Phosphoric acid
2) Nucleosides are primarly structural components.	2) Nucleotides also act as energy units (ATP)
3) Nucleoside is a unit formed by attachment of a base to C-1 position of sugar.	3) Nucleotide is linked to phosphoric acid to C-5 position of sugar.
4) Ex: Adenine + Ribose= adenosine	4) Ex: Adenosine + Phosphoric acid = Adenylic acid

Small book, Big Marks!

AMAZING MIDDLE SECTION

SAQ

SECTION-C

Bullet Bytes, Makes You Bright!

SAQ CHAPTERS

28 PAGES - 32 MARKS

★★☆	1. SOLUTIONS	(4 Marks)	27 - 29
★★★	2. ELECTRO CHEMISTRY	(4+4 Marks)	30 - 33
★★☆	3. CHEMICAL KINETICS	(4 Marks)	34 - 36
★★☆	4. THE d&f -BLOCK ELEMENTS	(4 Marks)	37 - 38
★★	5. COORDINATION COMPOUNDS	(4 Marks)	39 - 40
★★	6. HALOALKANES & HALOARENES	(4 Marks)	41 - 43
★★★	7. ALCOHOLS, PHENOLS & ETHERS	(4+4 Marks)	44 - 46
★★	8. ALDEHYDES, KETONES & CARBOXYLIC ACIDS	(4 Marks)	47- 48
★★☆	9. AMINES	(4 Marks)	49- 51
★★★	10. BIOMOLECULES	(4 Marks)	52 - 54

Super Scoring Pages

★★★SSP★★★

Pages 28, 29, 30, 32, 34, 35, 37, 39, 41, 42, 44, 47, 49, 50, 52, 53

DEAR DUDE!

Please note that SAQs in each chapter are Jumbled, and

Easy SAQs are brought to the beginning.

1 SOLUTIONS

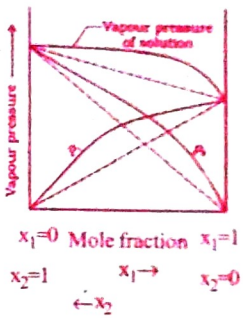
4 Marks

12. What is meant by positive deviations from Raoult's law and how is the sign of Δ_{mix} H related to positive deviation from Raoult's law ? [BMP]

A: 1) Positive deviation occurs when the vapour pressure of a solution is higher than the predicted value by Raoult's law.

2) In such cases, intermolecular interactions between solute and solvent particles (A and B) are weaker than those between solute-solute (A-A) and solvent-solvent (B-B).

3) It leads to increase in vapour pressure resulting in positive deviation.



4) For a solution showing positive deviation

(i)$P_{total} > P_A+P_B$ (ii)$\Delta H_{mix} > 0$ (+Ve) (iii)$\Delta V_{mix} > 0$ (+Ve)

5) **Ex:** (i) Ethyl alcohol and water (ii) Acetone and benzene.

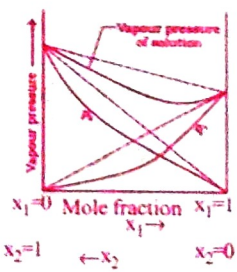
---

13. What is meant by negative deviation from Raoult's law and how is the sign of Δ_{mix} H related to negative deviation from Raoult's law ?

A: 1) Negative deviation occurs when the vapour pressure of a solution is lower than the predicted value by Raoult's law.

2) In such cases, intermolecular interactions between solute and solvent particles (A and B) are stronger than those between solute-solute (A-A) and solvent-solvent (B-B).

3) It leads to decrease in vapour pressure resulting in negative deviation.



4) For a solution showing negative deviation

(i) $P_{total} < P_A + P_B$; (ii)$\Delta H_{mix} < 0$, (–Ve) (iii)$\Delta V_{mix} < 0$, (–Ve)

5) **Ex:** (i) Nitric acid and water (ii) Hydrochloric acid and water.

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14. What is relative lowering of vapour pressure? How is it useful to determine the molar mass of a solute?

A: **1) Lowering of Vapour Pressure(LVP):** It is the difference between the vapour pressure of pure solvent (p^o) and the vapour pressure of the solution (p^s).

Thus, $\Delta p = p^o - p^s$

2) Relative Lowering of Vapour Pressure(RLVP): It is the ratio between lowering of vapour pressure ($p^o - p^s$) and vapour pressure of pure solvent (p^o).

Thus, $R.L.V.P = \frac{p^o - p^s}{p^o}$

3) According to Raoult's law, RLVP = mole fraction of the solute (χ_s.)

$$\frac{p^o - p^s}{p^o} = \chi_s \Rightarrow \frac{p^o - p^s}{p^o} = \frac{w_s}{M_s} \times \frac{M_o}{w_o}$$

4) From the above relation, molar mass of the given solute is $M_s = \frac{w_s \times M_o \times p^o}{(p^o - p^s) w_o}$

---

15. Calculate the mass of a non-volatile solute (molar mass 40g mol^{-1}) which should be dissolved in 114g Octane to reduce its vapour pressure to 80%. [AP 20]

A: Vapour pressure is reduced 80% means vapour pressure of the solution reduces to 80.

Vapour pressure of solvent P^o=100, Vapour pressure of solution P^s=80

Mass of solvent (octane) (w_o)=114g, Molar mass of Octane (C_8H_{18}) is M_o=114 g/mol

Molar mass of solute is M_s=40 g/mol, Mass of solute is w_s =?

From Raoult's law, $\frac{P^o - P^s}{p^o} = \frac{w_s}{M_s} \times \frac{M_o}{w_o}$

$$\Rightarrow \frac{100-80}{100} = \frac{w_s}{40} \times \frac{114}{114} \Rightarrow \frac{20}{100} = \frac{w_s}{40} \Rightarrow w_s = \frac{20 \times 40}{100} = 8\ g$$

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16. Define mass percentage, volume percentage and mass to volume percentage solutions.

A: **(i) Mass percentage (w/w)**: Mass of the component (in g) present in 100g of solution.

$$\text{Mass \% of a component} = \frac{\text{Mass of the component in the solution}}{\text{Total mass of the solution}} \times 100$$

(ii) Volume percentage (V/V): Volume of the component (in mL) present in 100 mL of solution.

$$\text{Volume \% (V/V) of a component} = \frac{\text{Volume of the component in the solution}}{\text{Total volume of the solution}} \times 100$$

(iii) Mass by volume % (w/V): The mass of solute in (g) present in 100 mL of solution.

$$\text{Mass by volume \% (w/V) of solute} = \frac{\text{Mass of the solute in the solution}}{\text{Total volume of the solution}} \times 100$$

---

17. A solution of sucrose in water is labeled as 20% w/w.
What would be the mole fraction of each component in the solution ?

A: 20% w/w sucrose solution means 20g of sucrose is present in 100g of solution.
Weight of sucrose solute = 20g , GMM of sucrose $(C_{12}H_{22}O_{11})$= 342 g mol^{-1}.

$$\text{Moles of sucrose, } n_A = \frac{\text{weight}}{\text{GMM}} = \frac{20}{342} = 0.0585$$

Weight of solution = 100g, Mass of solvent = 100–20 = 80g,
GMM of water (H_2O) = 18 g mol^{-1}.

$$\text{Moles of water, } n_B = \frac{\text{weight}}{\text{GMM}} = \frac{80}{18} = 4.4445$$

$$\text{Mole fraction of sucrose } (\chi_A) = \frac{n_A}{n_A + n_B} = \frac{0.0585}{0.0585 + 4.4445} = \frac{0.0585}{4.503} = 0.013$$

$$\text{Mole fraction of water } (\chi_B) = 1 - \chi_A = 1 - 0.013 = 0.987 \qquad \left[\because \chi_A + \chi_B = 1\right]$$

---

18. A solution of glucose in water is labeled as 10% w/w. What would be the molality of the solution ? [AP 20,25]

A: 10% w/w glucose solution means 10g of glucose is dissolved in 100g of solution.

Weight of Glucose solute (w_s)=10 g, GMM of glucose =180,

$$\text{Moles of glucose } n = \frac{\text{weight}}{\text{GMM}} = \frac{10}{180},$$ Weight of solvent (W)= 100–10=90g

$$\text{Molality of the solution } m = n \times \frac{1000}{W\,(g)} = \frac{10}{180} \times \frac{1000}{90} = 0.617m$$

∴ Molality of the given solution =0.617m

[☞ Remaining SAQs are in Page 87]

2 ELECTRO CHEMISTRY

4+4 Marks

19. What are Galvanic cells? Explain the working of Galvanic cell with a neat sketch taking Daniel cell as example. **[BMP]**[AP 15,18]

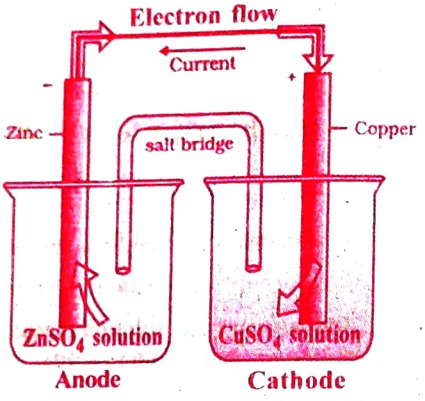
A: **1) Galvanic Cell:** It is an electro chemical cell which converts chemical energy into electrical energy by redox reactions (Spontaneous).

2) Construction: (i) Daniel Cell consists of two half cells (a)anode half cell (b)cathode half cell.

(ii) The two half cells are connected by a salt bridge containing a saturated solution of KNO_3 in agar-agar gel.

(iii) The anode half cell consists of zinc rod dipped in $ZnSO_4$ solution.

(iv) The cathode half cell consists of copper rod dipped in $CuSO_4$ solution.



DANIEL CELL

3) Representation: $Zn_{(s)} / Zn^{+2}_{(aq)} // Cu^{+2}_{(aq)}/Cu_{(s)}$

4) Working: When the two half cells are connected externally through a voltmeter, current flows out from the cell due to potential difference.

5) Reactions:

Oxidation reaction at anode : $Zn \rightleftharpoons Zn^{+2} + 2e^-$

Reduction reaction at cathode: $Cu^{+2} + 2e^- \rightleftharpoons Cu$

6) Net cell reaction: $Zn_{(s)} + Cu^{+2}_{(aq)} \rightleftharpoons Zn^{+2}_{(aq)} + Cu_{(s)}$

---

20. State and explain Kohlrausch's law of independent migration of ions.

[BMP] [AP 15,16,19,22,23,25]

A: **1) Kohlrausch law of independent migration of ions** states that " limiting molar conductivity of an electrolyte can be represented as the sum of the individual contributions of the anion and the cation of the electrolyte, at infinite dilution."

2) Thus, if $\lambda^0_{Na^+}$ and $\lambda^0_{Cl^-}$ are the limiting molar conductivities of the sodium and chloride ions respectively, then the limiting molar conductivity for sodium chloride is given by the equation

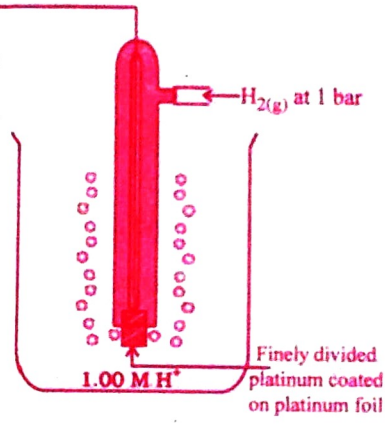
$$\Lambda^o_{(NaCl)} = \lambda^0_{Na^+} + \lambda^0_{Cl^-}$$

3) In general, if an electrolyte on dissociation gives V_+ cations and V_- anion then its limiting molar conductivity is given by $\Lambda^o_m = V_+\lambda^0_+ + V_-\lambda^0_-$

4) Here, λ^0_+ and λ^0_- are the limiting molar conductivities of the cation and anion respectively.

21. Give the construction and working of a standard hydrogen electrode with a neat diagram.

A: Standard hydrogen electrode (SHE) :



1) SHE is a reference electrode and its reduction potential is arbitrarily assigned as zero volt at all temperatures.

2) By using 'SHE' we can measure the potential of single electrode.

3) H_2 gas adsorbed on platinum at one atmosphere placed in contact with 1M H^+ concentration in solution is treated as standard hydrogen electrode (SHE).

4) Potential of this electrode is taken as zero.

Hydrogen electrode is represented as $Pt, H_{2(g)}\ 1\ atm / H^+_{(aq)} (C = 1)$

---

22. What are the products obtained at the cathode and the anode during the electrolysis of the following when platinum electrodes are used in the electrolysis

(a) Molten KCl (b) Aq. $CuSO_4$ solution (c) Aq.K_2SO_4 solution

A: a) Electrolyte: $\underset{\text{Molten}}{2KCl} \rightarrow 2K^+ + 2Cl^-$

at cathode: $2K^+ + 2e^- \rightarrow \mathbf{2K}$

at anode : $2Cl^- \rightarrow \mathbf{Cl_2} + 2e^-$

b) Electrolyte: $CuSO_{4(aq)} \rightarrow Cu^{+2}_{aq} + SO^{-2}_{4(aq)}$

at cathode: $Cu^{+2} + 2e^- \rightarrow \mathbf{Cu(s)}$

at anode: $H_2O \rightarrow 2H^+ + \frac{1}{2}\mathbf{O_2} + 2e^-$

c) Electrolyte: $K_2SO_{4(aq)} \rightarrow 2K^+_{aq} + SO^{-2}_{4(aq)}$

at cathode: $2H_2O + 2e^- \rightarrow 2OH^- + \mathbf{H_2}$

at anode: $H_2O \rightarrow 2H^+ + \frac{1}{2}\mathbf{O_2} + 2e^-$

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23. State and explain Nernst equation with the help of a metallic electrode and a non-metallic electrode.

A: Nernst Equation: The equation that gives the dependence of the electrode potential on the concentration of ions with which electrode is reversible is known as Nernst equation.

Nernst Equation: Electrode potential $E = E^{\circ} - \frac{2.303RT}{nF}\log\frac{[Products]}{[Reactants]}$

(a) For Metal Electrode: Electrode reaction: $M^{n+}_{(aq)} + ne^- \rightleftharpoons M_{(s)}$

$$E = E^{\circ} - \frac{2.303RT}{nF}\log\frac{1}{[M^{n+}]} = E^{\circ} + \frac{2.303RT}{nF}\log[M^{n+}]$$

At 25°C, $\frac{2.303RT}{F} = \frac{2.303\times 8.314\times 298}{96500} = 0.059$

$\therefore E = E^{\circ} + \frac{0.059}{n}\log C$ [Here $C = [M^{n+}]$ = Metal ion concentration]

(b) For Non -metal Electrode: Electrode reaction: $\frac{1}{2}A_{2(g)} + ne^- \rightleftharpoons A^{-n}_{(aq)}$

$$E = E^{\circ} - \frac{2.303RT}{nF}\log[A^{-n}]$$

At 25°C, $E = E^{\circ} - \frac{0.059}{n}\log C$ [Here $C = [A^{-n}]$ Anion concentration]

Notation: E°= Standard reduction potential of the cell at 25°C.

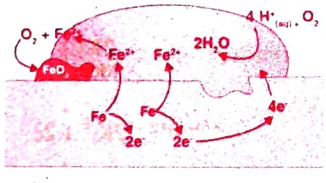
n= Number of electrons involved in cell reaction. R= Gas constant R (8.314 J K^{-1} mol^{-1});

T= Temperature =273 + 25= 298K; F = Faraday constant= 96,487 ≃ 96500 c mol^{-1};

---

24. What is metallic corrosion? Explain it with respect to iron corrosion.

A: 1) Metallic corrosion: It is the process of slow eating up of metals by gases and water vapours present in atmosphere due to the formation of certain compounds like oxides, sulphides, carbonates,etc.,



2) Corrosion of Iron is known as Rusting. Rusting is an electrochemical process that occurs when iron is exposed to oxygen and moisture.

3) Electrochemical Mechanism:

i) Formation of Anodic and Cathodic Regions

ii) On the iron surface, tiny regions act as:Anode→ oxidation occurs;Cathode → reduction occurs

4) The corrosion process is electrochemical in nature. On the iron surface, iron becomes oxidized in the presence of water.

5) On the metal surface, oxidation takes places at anode and reduction takes place at cathode.

At anode: $Fe_{(s)} \longrightarrow Fe^{+2}_{aq} + 2e^-$ (Iron atoms lose electrons and go into solution)

At cathode: $\frac{1}{2}O_{2(aq)} + H_2O_{(l)} + 2e^- \longrightarrow 2OH^-_{aq}$ (Oxygen reacts with water to form hydroxide ions)

Overall reaction: $Fe_{(s)} + \frac{1}{2}O_{2(aq)} + H_2O_{(l)} \longrightarrow Fe^{+2}_{aq} + 2OH^-_{aq}$

6) Formation of Rust: Ferrous ions react with hydroxide ions $Fe^{+2}_{aq} + 2OH^- \longrightarrow Fe(OH)_2$

On further oxidation $Fe(OH)_2 \longrightarrow Fe(OH)_3$

The Ferrous ions finally form hydrated ferric oxide $Fe_2O_3.xH_2O$ (rust).

PROBLEMS

25. A solution of $CuSO_4$ is electrolysed for 10 minutes with a current of 1.5 amperes. What is the mass of copper deposited at the cathode? [AP 15,18,20]

Sol: Given time t = 10 min. = 600s, current i= 1.5 amperes

charge = current × time =1.5 × 600 = 900C

According to the reaction: $Cu^{2+}(aq) + 2e^- = Cu(s)$

We require 2F or 2×96500 C deposit 1 mole or 63 g of Cu.

For 900C, the mass of Cu deposited = (63 ×900)/(2×96500) = 0.2938 g

---

26. Calculate Λ_m^0 for $CaCl_2$ and $MgSO_4$ from the given data Ca^{2+} = 119.0S cm² mol⁻¹. Cl^- = 76.3S cm² mol⁻¹ Mg^{2+} = 106.0 S cm² mol⁻¹ SO_4^{2-} =160.0S cm² mol⁻¹

Sol: We know from Kohlrausch law that

$$\Lambda^0_{m(CaCl_2)} = \lambda^0_{Ca^{2+}} + 2\lambda^0_{Cl^-} = 119.0 + 2(76.3) = 119.0 + 152.6 = 271.6 \text{ S cm}^2 \text{ mol}^{-1}.$$

$$\Lambda^0_{m(MgSO_4)} = \lambda^0_{Mg^{2+}} + \lambda^0_{SO_4^{2-}} = 106.0 + 160.0 = 266 \text{ S cm}^2 \text{ mol}^{-1}.$$

---

27. The standard emf of Daniel cell is 1.1V. Calculate the standard Gibbs energy for the cell reaction: [AP 15]

$Zn(s) + Cu^{2+}(aq) \rightarrow Zn^{2+}(aq) + Cu(s)$

Sol: $Zn(s) + Cu^{2+}(aq) \rightarrow Zn^{2+}(aq) + Cu(s)$

From above equation n=2, F=96500C, $E^o_{Cell} = 1.1$ V.

$$\Delta G^o_F = -nFE^o_{Cell}$$

$$\Delta G^o_F = -2 \times 96500 \times 1.1 = -21200 \text{J.mole}^{-1} = -212.3 \text{ KJ.mole}^{-1}$$

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[☞ Remaining SAQs are in Page 88]

3 CHEMICAL KINETICS

4 Marks

28. What is 'molecularity' of a reaction? How is it different from the 'order' of a reaction? Name one bimolecular and one trimolecular gaseous reactions. [BMP][AP 19,20,23]

A: **1) Molecularity** is the number of reacting molecules taking part in a single elementary step.

2) Order of a reaction is the 'sum of powers of the concentration terms' of the reactants in the rate law equation.

3) Differences between Molecularity and Order :

Molecularity	Order
i) It is determined **theoretically**.	i) It is determined **experimentally**.
ii) It can have **integral values** only.	ii) It can have **fractional values** also.
iii) It can be unimolecular, bimolecular....	iii) It can be zero order, first order....
iv) It is applicable only to **elementary reactions**.	iv) It is applicable to **elementary and complex reactions.**

4) Bimolecular reaction: Dissociation of hydrogen iodide into H_2 and I_2.

$2HI \rightarrow H_2 + I_2$

5) Trimolecular reaction: Formation of NO_2 from NO and O_2

$2NO + O_2 \rightarrow 2NO_2$

---

29. Derive the integrated rate equation for a zero order reaction.

A: 1) Consider a zero order reaction : $R \rightarrow$ product

$$\therefore \text{Rate} = \frac{-d[R]}{dt} = k[R]^0 = k \Rightarrow d[R] = -kdt$$

2) Integrating both sides, we get $[R] = -kt + I$.....(1), I is the constant of integration

At $t = 0$, we have $[R] = [R]_0$, where $[R]_0$ is initial concentration of the reactant.

3) Equation (1) can be written as $[R]_0 = -k \times 0 + I \Rightarrow [R]_0 = I$

Substituting the value of I in the equation (1), we have $[R] = -kt + [R]_0$

[This is in the form y=mx+C of a line, with slope m= –k]

4) $\Rightarrow kt = [R]_0 - [R] \Rightarrow k = \frac{[R]_0 - [R]}{t}$. This is the integrated equation for zero order reaction.

Here, k= rate constant ; $[R]_0$ = Initial concentration of the reactant

t= time, [R]= Final concentration of reactants at the time t.

30. Derive an integrated rate equation for a first order reaction. [AP 22]

A: 1) Consider a first order reaction: R→ Product

$$\text{Rate} = \frac{-d[R]}{dt} = k[R] \Rightarrow \frac{d[R]}{[R]} = -kdt$$

2) Integrating this equation, we get $\ln[R] = -kt + I$ .....(1), where I is the constant of integration

When t = 0 we have $[R] = [R]_0$, where $[R]_0$ is the initial concentration of the reactant.

3) Equation (1) can be written as $\ln[R]_0 = -k\times 0 + I \Rightarrow \ln[R]_0 = I$

Substituting the value of I in equation (1), we have

4) $\ln[R] = -kt + \ln[R]_0 \Rightarrow kt = \ln[R]_0 - \ln[R] = \ln\frac{[R]_0}{[R]} \Rightarrow k = \frac{1}{t}\ln\frac{[R]_0}{[R]} \Rightarrow k = \frac{2.303}{t}\log\frac{[R]_0}{[R]}$

This is the integrated rate equation for first order reaction.

Here, k= rate constant ; $[R]_0$ = Initial concentration of the reactant

t= time, [R]= Final concentration of the time t.

---

31. What is half-life ($t_{1/2}$) of a reaction? Derive the equations for the 'half-life' value of zero and first order reactions.

A: **Half -life of a reaction ($t_{1/2}$)** is the time in which the concentration of a reactant is reduced to one half of its initial concentration.

(i) For a zero order reaction $k = \frac{[R]_0 - [R]}{t}$

At $t = t_{1/2}$ we have $[R] = \frac{[R]_0}{2}$

$$\therefore \text{Rate constant is } k = \frac{[R]_0 - \frac{1}{2}[R]_0}{t_{1/2}} \Rightarrow k = \frac{\frac{[R]_0}{2}}{t_{1/2}} \Rightarrow t_{1/2} = \frac{[R]_0}{2k}$$

(ii) For a first order reaction $k = \frac{2.303}{t}\log\frac{[R]_0}{[R]}$

At $t_{1/2}$ we have, $[R] = \frac{[R]_0}{2}$

$$\therefore \text{Rate constant is } k = \frac{2.303}{t_{1/2}}\log\frac{[R]_0}{[R]_0/2} \Rightarrow k = \frac{2.303}{t_{1/2}}\log 2$$

$$\Rightarrow t_{1/2} = \frac{2.303}{k}\log 2 = \frac{2.303}{k}\times 0.3010 = \frac{0.693}{k} \qquad \therefore t_{1/2} = \frac{0.693}{k}$$

32. Explain the terms

a) Activation energy (E_a) **b)Collision frequency (Z)**

c) Probability factor (P) with respect to Arrhenius equation.

A: **a) Activation Energy:** It is the minimum excess energy that reactant molecules should acquire to participate in chemical reaction or to cross threshold energy barrier. $Ea_a = E_T - E_R$.
In other words, activation energy is the difference between threshold energy and the average kinetic energy of the reactant molecules.

b) Collision Frequency is the number of collisions that take place per second per unit volume of the reaction mixture.

c) Probability Factor: For a collision to be effective, the colliding molecules must have energy greater than threshold energy and also must have proper orientations.
For the collision to be effective, another factor, P is called probability factor has to be introduced.
Thus Arrhenius equation is modified to the form $k = PZ_{AB}\, e^{-Ea/RT}$
where P= Probability factor

---

33. What is Arrhenius equation? Derive an equation which describes the effect of rise of temperature (T) on the rate constant (k) of a reaction.

A: **1) Arrhenius equation** gives the temperature dependence of the rate of a chemical reaction.
Arrhenius equation: Rate constant $k = A\, e^{-Ea/RT}$.....(1)
Where A is frequency factor, E_a activation energy.

2) Taking natural logarithm on both sides of equation (1), we get $\ln k = \frac{-E_a}{RT} + \ln A$ .....(2)

3) At temperature T_1, equation (2) is $\ln k_1 = \frac{-E_a}{RT_1} + \ln A$ .....(3)

4) At temperature T_2, equation (2) is $\ln k_2 = \frac{-E_a}{RT_2} + \ln A$ .....(4)

Here K_1 and K_2 are the values of rate constants at temperature T_1 and T_2 respectively.

5) Subtracting equation (3) from (4), we get $\ln k_2 - \ln k_1 = \frac{E_a}{RT1} - \frac{E_a}{RT_2}$

$$\ln \frac{k_2}{k_1} = \frac{E_a}{R}\left[\frac{1}{T_1} - \frac{1}{T_2}\right] \Rightarrow 2.303 \log \frac{k_2}{k_1} = \frac{E_a}{R}\left[\frac{1}{T_1} - \frac{1}{T_2}\right] \Rightarrow \log \frac{k_2}{k_1} = \frac{E_a}{2.303R}\left[\frac{1}{T_1} - \frac{1}{T_2}\right]$$

4 The d&f-Block Elements

4 Marks

34. Write the characteristics properties of transition elements. [BMP] [AP 15]

A: The characteristics properties of Transition metals:

1) Transition metals are 'hard and heavy' metals.
2) They show 'variable oxidation' states.
3) They are 'good conductors of heat and electricity'.
4) They exhibit 'paramagnetic character'.
5) They form 'alloys'.
6) They form 'coloured complex compounds'.

☺ TRANSITION PROPERTIES ☺
H
O
T
P
A
C

---

35. What is lanthanoid contraction? What are the consequences of lanthanoid contraction?

A: Lanthanoid contraction: It is the steady decrease in the atomic radii even with increase in atomic number from Lanthanum to Lutetium due to **poor screening effect of 4f–electrons.**

Consequences of Lanthanoid contraction:

1) **Basic character of oxides and hydroxides:** Due to the lanthanoid contraction, the covalent nature of La-OH bond increases and thus, the basic character of oxides and hydroxides decreases from $La(OH)_3$ to $Lu(OH)_3$.
2) Due to lanthanide contraction, size of 4d series and 5d series elements remains the same. Thus, pairs of elements such as Zr/Hf, Nb/Ta and Mo/W are almost identical in size. Due to almost similar size, such pairs have very similar properties which makes their separation difficult.
3) **Separation of lanthanoids:** Due to lanthanoid contraction, there is a difference in some properties of lanthanoid like solubility, degree of hydration and complex formation. These differences enable the separation of lanthanoids by 'ion exchange method'.

---

36. What are interstitial compounds? How are they formed? Give two examples. [AP 15]

A: 1) Interstitial compounds are compounds formed when small atoms (H,C,B,N) occupy the empty spaces (interstitial sites-voids) in the metals of crystal lattice structure.

2) Formation: In metals such as Fe, Ti atoms are arranged in a closely packed structures with tiny gaps (voids). Small atoms can fit in this voids without disturbing the overall structure much.

Ex 1: Fe-C (Iron- carbon steel),

Ex 2: Ti-H (Titanium - Hydride), Fe_3H

37. Write the characteristics of interstitial compounds.

A: Characteristics of interstitial compounds:

1) They are very hard and rigid. **Ex:** Some borides approach diamond in hardness.
2) They have high melting points, higher than those of their pure metals.
3) They retain metallic conductivity
4) They are chemically inert.

---

38. Why do the transition metal ions exhibit characteristic colours in aqueous solution. Explain giving examples.

A: 1) Transition metal ions exhibit **colour property** in aqueous solution due to the **presence of unpaired d-electrons.**

2) These unpaired d-electrons from a **lower energy** are excited to a higher energy d-orbitals.
3) The energy of excitation corresponds to the frequency of light absorbed in visible region.
4) The colour observed corresponds to the complementary colour of light absorbed.
5) The frequency of light absorbed is determined by the nature of the ligand.

Ex1: Ti^{2+} [Ar] $3d^1$ Purple

V^{2+} [Ar] $3d^3$ Violet

These have **unpaired electrons** in their d orbitals and hence they are **coloured.**

Ex2: Sc^{3+} [Ar] $3d^0$ colourless

Cu^{+} [Ar] $3d^{10}$ colourless

These have **paired electrons** in their d orbitals and hence they are **colourless.**

---

39. Describe the preparation of potassium dichromate from iron chromite ore.

A: Preparation of potassium dichromate ($K_2Cr_2O_7$) from chromite ore ($FeCr_2O_4$):

1) Potassium dichromate ($K_2Cr_2O_7$) is obtained by the fusion of chromite ore ($FeCr_2O_4$) with sodium (or potassium) carbonate (Na_2CO_3) in the presence of excess of air.

$4FeCr_2O_4 + 8Na_2CO_3 + 7O_2 \rightarrow 8Na_2CrO_4 + 2Fe_2O_3 + 8CO_2$.

(yellow solution)

2) The yellow solution of sodium chromate is filtered and acidified with sulphuric acid to give a solution from which orange sodium dichromate ($Na_2Cr_2O_7.2H_2O$) can be crystallised.

$2Na_2CrO_4 + 2H^+ \rightarrow Na_2Cr_2O_7 + 2Na^+ + H_2O$

3) $Na_2Cr_2O_7$ is more soluble than potassium dichromate. So the solution of sodium dichromate is treated with potassium chloride to prepare $K_2Cr_2O_7$.

$Na_2Cr_2O_7 + 2KCl \rightarrow K_2Cr_2O_7 + 2NaCl$

Orange crystals of potassium dichromate crystallise out.

[☞ Remaining SAQs are in Page 93]

5 COORDINATION COMPOUNDS

4 Marks

40. Explain Werner's theory of coordination compounds with suitable examples.

[AP 16, 17,18,20,22,23,24,25]

A: 1) **Werner's theory** explains the structure, bonding, and isomerism of complex compounds. Here the central metal atom shows a) Primary valency b) Secondary valency.

2) **Primary Valency:**

i) It is equal to the oxidation number of the central atom.

ii) It is satisfied only by the negative ions.

☺ IPE ☺ WELCOMES WERNER EVERY YEAR

3) **Secondary Valency:**

i) It is equal to the co-ordination number of the central atom.

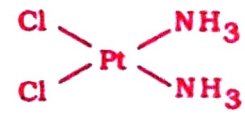
ii) It is satisfied by negative ions, neutral molecues and rarely by positive ions.

4) **Example:** $[Pt(NH_3)_2Cl_2]$

i) Here, primary valency of Pt is 2.
It is satisfied by two Cl^- ions.

ii) Secondary valency of Pt is 2.
It is satisfied by two NH_3 molecules.

iii) Shape of this complex is Square.



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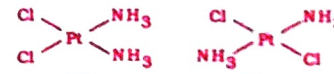
41. Explain geometrical isomerism in coordination compounds giving suitable examples.

A: **Geometrical isomerism** occur due to different geometric arrangements of the ligands.

i) Square planar complexes of the type $[MX_2L_2]$:

Here **4 ligands occupy 4 corner places** of a square around the central ion.



Square planar structure

a) cis isomer: Two identical X ligands are arranged adjacent to each other.

b) trans isomer: Two identical X ligands are arranged opposite to each other.

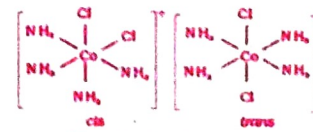
Ex: cis $[Pt(NH_3)_2Cl_2]$, trans $[Pt(NH_3)_2Cl_2]$

ii) Octahedral complexes:

Here **6 ligands occupy 6 corner places** of an octahedral.

cis & trans isomer of the type $[MX_2L_4]$:

- **cis isomer:** Two X ligands are arranged adjacent to each other.
- **trans isomer:** Two X ligands are arranged opposite to each other.



Octahedral structure

Ex: cis $[CoCl_2(NH_3)_4]^+$, trans$[CoCl_2(NH_3)_4]^+$

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42. What are homoleptic and heteroleptic complexes? Give one example for each.[BMP]

A: 1) **Homoleptic complexes** are the complexes in which a metal is bound by only one kind of ligands.

2) **Ex:** $[Co(NH_3)_6]^{+3}$

3) **Heteroleptic complexes** are the complexes in which a metal is bound by more than one kind of ligands.

4) **Ex:** $[Co(NH_3)_4Cl_2]^+$

---

43. Explain the following terms by giving an example

i) Unidentate ligand **ii)Bidentate ligand**

iii) Polydentate ligand **iv)Ambidentate ligand**

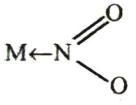
A: **i) Unidentate:** It is a ligand with **one donor** atom. **Ex:** $:NH_3$

ii) Bidentate: It is a ligand with **two donor** atoms. **Ex:** $\ddot{N}H_2 - \ddot{N}H_2$

iii) Polydentate: It is a ligand with **more than two** donor atoms. **Ex:** EDTA

iv) Ambidentate ligand: It is a ligand with two donor atoms, but only one of them forms a coordinate bond at a time with central metal atom / ion.

Ex: nitro–N (or) Nitrito-N (N donor atom)



nitro-O (or) Nitrito-O (O donor atom) $M \leftarrow O - N = O$

---

44. Using IUPAC norms write the formulas for the following :

(i) Tetrahydroxozincate (II) ion **(ii) Hexaaminecobalt (III) sulphate**

(iii) Potassium tetrachloropalladate(II) **(iv) Potassium tri(oxoalato)chromate(III)**

A: i) Tetrahydroxozincate (II) ion: $[Zn(OH)_4]^{-2}$

ii) Hexaaminecobalt (III) sulphate : $[Co(NH_3)_6]_2(SO_4)_3$

iii) Potassium tetrachloropalladate(II): $K_2[PdCl_4]$

iv) Potassium trioxalatochromate(III): $K_3[Cr(C_2O_4)_3]$

---

45. Using IUPAC norms write the systematic names of the following: |AP 15,19|

(i) $[Co(NH_3)_6]Cl_3$ **(ii)** $[Pt(NH_3)_2Cl(NH_2\ CH_3)]Cl$

(iii) $[Ti(H_2O)_6]^{3+}$ **(iv)** $[NiCl_4]^{2-}$

A: i) $[Co(NH_3)_6]Cl_3$: **Hexaammine Cobalt (III)chloride**

ii) $[Pt(NH_3)_2Cl(NH_2\ CH_3)]Cl$: **Diammine chloro Methyl amine platinum(II)chloride**

iii) $[Ti(H_2O)_6]^{3+}$: **Hexaaquotitanium (III) ion**

iv) $[NiCl_4]^{2-}$: **Tetrachloronickelate (II) ion**

[☞ Remaining SAQs are in Page 93]

6 HALOALKANES & HALOARENES

4 Marks

46. Explain the mechanism of S_N1 with example. **[BMP]**

A: SN^1 reaction:

1) Order of SN^1 reaction is one.

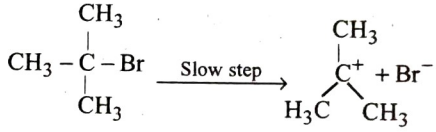
2) SN^1 is a 'unimolecular substitution reaction'.

3) SN^1 mechanism takes place in two steps.

4) **Ex:** Hydrolysis of 2-Bromo 2- methyl propane with aq KOH forms tert-butyl alcohol.

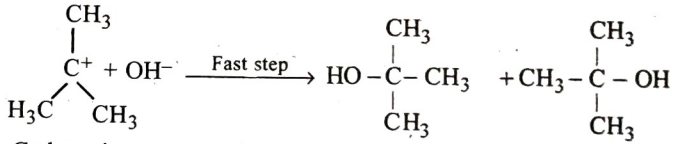
5) Mechanism of S_N1 :

Step 1: Formation of Carbocation .



Carbocation

6) Step 2: Formation of product by the attack of Nucleophile.



Carbocation Inversion Retention

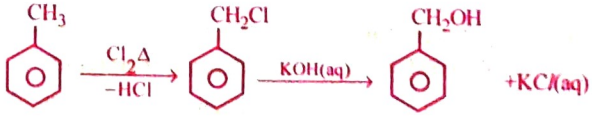
---

47. How will you carry out the following conversions?

(i) Ethane to bromoethane **(ii) Toluene to benzyl alcohol**

A: i) Ethane to bromoethane: $CH_3-CH_3+Br_{2(g)} \xrightarrow{\text{Sunlight}} Br-CH_2-CH_3+HBr$

ii) Toluene to benzyl alcohol: When chlorine is passed into boiling toluene it gives benzyl chloride, which on boiling with KOH (aq) gives benzyl alcohol.



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48. Describe the stereochemical effect on the hydrolysis of 2-bromobutane.

A: The given reaction is S_N2:

1) Order of S_N2 reaction is two.

2) S_N2 is a 'bimolecular substitution reaction'.

3) S_N2 mechanism takes place in single step.

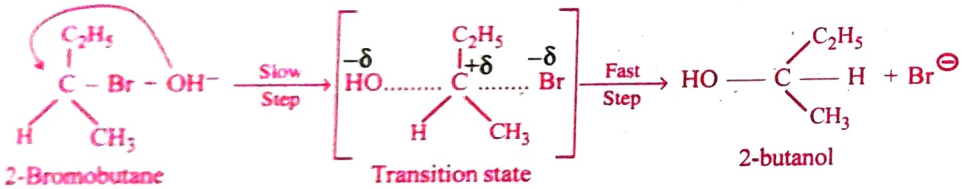
4) Given example Hydrolysis of 2-bromobutane in aqueous KOH forms **2-butanol.**

5) **Reaction:** C_2H_5–CH(Br)–CH_3 + KOH(aq) ⟶ C_2H_5–CH(OH)–CH_3 + KBr

2-Bromobutane → 2-butanol

6) **Mechanism (S_N2 reaction):** The nucleophile (OH^-) approaches the carbon from the opposite side of the halogen atom (Br) to form transition state. It is the slowest step.

7) The product has inverted configuration (Walder inversion).



2-Bromobutane — Transition state — 2-butanol

---

49. Write the differences between S_N1 and S_N2 reactions.

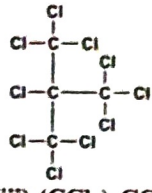
A:

S_N1 reaction	S_N2 reaction
1) **Order** of S_N1 reaction is **one.**	1) **Order** of S_N2 reaction is **two.**
2) S_N1 mechanism takes place in **two steps.**	2) S_N2 mechanism takes place in **one step.**
3) S_N1 is a 'unimolecular substitution reaction'.	3) S_N2 is a 'bimolecular substitution reaction'.
4) Order of reactivity of alkyl halides :3°>2°>1°	4) Order of reactivity of alkyl halides : 1°>2°>3°
5) Rate of reaction of S_N1 depends on concentration of reactants only.	5) Rate of reaction of S_N2 depends on concentration of reactants and nucleophile.
6) **Ex:** Hydrolysis of **2-Bromo 2-methyl propane** with aqueous KOH forms **tert-butyl alcohol.**	6) **Ex:** Hydrolysis of **2-bromobutane** in aqueous KOH forms **2-butanol.**
7) When optically active alkyl halide is subjected to S_N1 reaction, the product formed is **racemic mixture** (d & *l* isomers)	7) When optically active alkyl halide is subjected to S_N2 reaction, the product formed is an **inverted product.** (Walden Inversion)

50. Give the IUPAC names of the following compounds.

(i) $CH_3–CH(Cl)–CH(I)CH_3$ (ii) $ClCH_2.CH=CH–CH_2Br$

(iii) $(CCl_3)_3CCl$ (iv) $CH_3C(p–Cl–C_6H_4)_2–CH(Br)CH_3$



(iii) $(CCl_3)_3CCl$

A: (i) $CH_3–CH(Cl)–CH(I)CH_3$: **2-chloro, 3- iodobutane**

(ii) $ClCH_2.CH=CH–CH_2Br$: **1-bromo-4-chlorobut-2ene**

(iii) $(CCl_3)_3CCl$: **1,1,1,2,3,3,3 hepta chloro 2–(trichloromethyl)propane**

(iv) $CH_3C(p–Cl–C_6H_4)_2–CH(Br)CH_3$: **3–bromo-2,2-bis(4-chlorophenyl)butane**

---

51. Write the structures of the following organic halides.

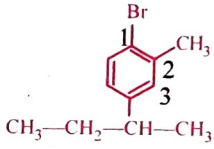
(i) 1-bromo-4-sec-butyl-2 methylbenzene

(ii) 2-chloro-1-phenylbutane

(iii) p-bromochlorobenzene **[AP 15]**

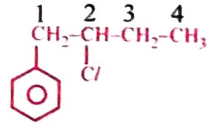
(iv) 4-t-butyl-3-iodoheptane

A: (i)



1-bromo-4-sec-butyl-2 methylbenzene

(ii)

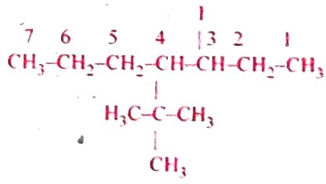


2-chloro-1-phenylbutane

(iii)

p-bromochlorobenzene

(iv)



4-t-butyl-3-iodoheptane

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[☞ Remaining SAQs are in Page 94]

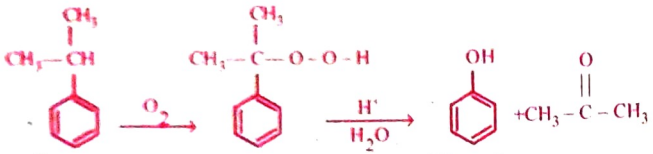
7 ALCOHOLS, PHENOLS & ETHERS

4+4 Marks

52. Give the equations for the preparation of phenol from Cumene. **[BMP]**

A: **1) Cumene** is the common name for Isopropylbenzene .

2) Preparation: When Cumene is oxidised in the presence of air, it is converted into cumene hydroperoxide. When treated with dilute acid, it is then converted into phenol and acetone.



Cumene **Phenol**

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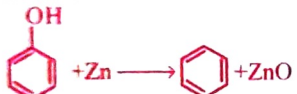
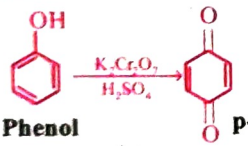
53. Write the products formed by the reduction and oxidation of phenol.

A: **a) Reduction:** Phenols on distillation with zinc dust undergo reduction to form **benzene**



Phenol **Benzene**

b) Oxidation: Phenol undergo oxidation with chromic acid to produce conjugated diketone which is known as p-benzoquinone.



Phenol **p-benzoquinone**

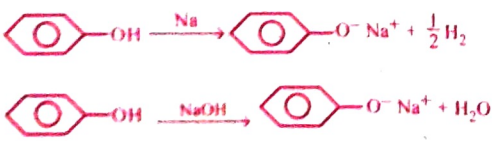
---

54. Explain the acidic nature of phenols and compare with that of alcohols. [AP 17] **[BMP]**

A: **I) Acidic nature of phenols:** Phenols are weakly acidic

a) Phenols turn blue litmus to red.

b) Phenols react with sodium metal and NaOH to form H_2 and H_2O.



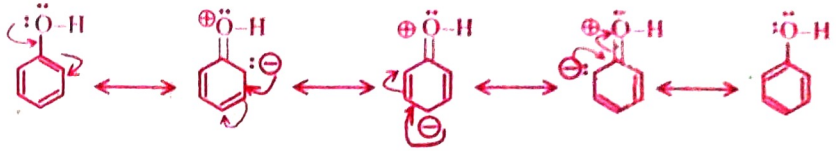
II) Acidic nature of phenols Vs alcohols:

Phenols are relatively more acidic than alcohols and also water.

a) The hydroxyl group in phenol is directly attached to SP^2 carbon on benzene ring. The carbon acts as an electron withdrawing group and Phenol exhibits resonance.

In these structures oxygen carries positive charge. This oxygen attracts electron pair of O–H bond strongly towards itself and facilitates the release of H^+ ion and $C_6H_5O^-$.

The formed phenoxide ion is more stable than phenol.



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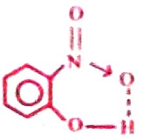
55. While separating a mixture of ortho and para nitrophenols by steam distillation, name the isomer which will be steam volatile. Give reason.

A: 1) The ortho and para isomers can be separated by steam distillation.
The ortho nitro phenol is steam volatile due to intra molecular hydrogen bonding.

2) The para-nitrophenol is less volatile due to intermolecular hydrogen bonding which results in the association of molecules.

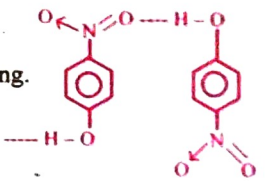


3) o- nitrophenol shows intra molecular hydrogen bonding.

So, it is relatively more volatile than p-isomer.

4) p-nitrophenol shows inter molecular hydrogen bonding.

So, this is less volatile than o-isomer.



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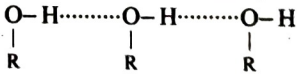
56. Account for the statement: Alcohols boil at highest temperature than hydrocarbons and ethers of comparable molecular masses.

A: Alcohols have intermolecular attractions due to intermolecular hydrogen bonds. This type of intermolecular hydrogen bonding do not exist in hydrocarbons and ethers. Hence alcohols boil at higher temperature than hydrocarbons and ethers of comparable molecular masses.



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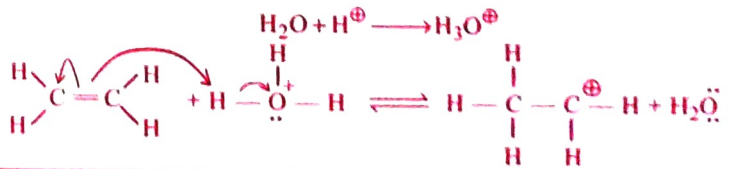
57. Write the mechanism of hydration of ethene to yield ethanol.

A: 1) Ethene is converted into alcohol by direct addition of water in the presence of phosphoric Acid (H_3PO_4).

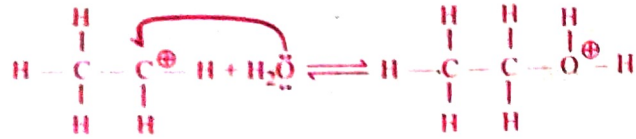
$$CH_2 = CH_2 + H_2O \xrightarrow{H^+} CH_3CH_2OH$$

2) **Mechanism(S_N1):** The mechanism of the reaction involves the following three steps:

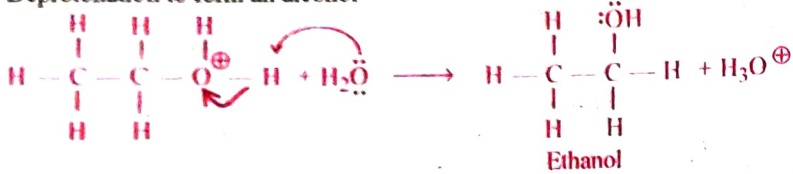
3) **Step 1:** Protonation of alkene to form carbocation by electrophilic attack of H_3O^+



Step 2: Nucleophilic attack of water on carbocation.



Step 3: Deprotonation to form an alcohol



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58. Draw the structures of all isomeric alcohols of molecular formula $C_5H_{12}O$ and give their IUPAC names and classify them as primary, secondary, and tertiary alcohols.

A: The structures, IUPAC names and the classification of all isomeric alcohols of molecular formula $C_5H_{12}O$ are eight and follows as

a) $CH_3 - CH_2 - CH_2 - CH_2 - CH_2OH$	Pentan-1-ol	Primary
b) $CH_3 - CH_2 - CH_2 - CH(OH) - CH_3$	Pentan-2-ol	Secondary
c) $CH_3 - CH_2 - CH(OH) - CH_2 - CH_3$	Pentan-3-ol	Secondary
d) $CH_3 - C(CH_3)_2 - CH_2OH$	2,2, dimethyl Propan-1-ol	Primary
e) $CH_3 - CH_2 - C(CH_3)_2 - OH$	2-methylbutan-2-ol	Tertiary
f) $CH_3 - CH(CH_3) - CH_2 - CH_2 - OH$	3-Methylbutan-1-ol	Primary
g) $CH_3 - CH(CH_3) - CH(OH) - CH_3$	3-Methylbutan-2-ol	Secondary
h) $CH_3 - CH_2 - CH(CH_3) - CH_2 - OH$	2-Methylbutan-1-ol	Primary

8 ALDEHYDES, KETONES & CARBOXYLIC ACIDS

4 Marks

59. What is Tollen's reagent? Explain its reaction with Aldehydes. **[BMP]**

A: **1) Tollen's reagent:** Ammonical silver nitrate solution is called Tollen's reagent.

Its formula is $[Ag(NH_3)_2]^+OH^-$

2) Reaction of toulene with aldehydes (Silver mirror test):

When Tollen's reagent is warmed with an aldehyde, it gets reduced to metallic silver.

This metallic silver gets deposited on the inner wall of the test tube to form a silver mirror.

$$RCHO + 2[Ag(NH_3)_2]^+ + 3OH^- \longrightarrow RCOO^- + 2Ag\downarrow + 4NH_3 + 2H_2O$$

Silver mirror

---

60. Write the oxidation products of acetaldehyde, acetone and acetophenone.

A: **1) Oxidation of acetaldehyde:** $\underset{\text{Acetaldehyde}}{CH_3CHO} \xrightarrow{(O)} \underset{\text{Acetic acid}}{CH_3COOH}$

2) Oxidation of acetone: $\underset{\text{Acetone}}{CH_3-\overset{\overset{O}{\|}}{C}-CH_3} \xrightarrow[K_2Cr_2O_7/H^+]{3(O)} \underset{\text{Acetic acid}}{CH_3COOH} + \underset{\text{Formic acid}}{HCOOH}$

3) Oxidation of acetophenone: $\underset{\text{Acetophenone}}{C_6H_5COCH_3} \xrightarrow{I_2/NaOH} \underset{\text{Idoform}}{CHI_3} + \underset{\text{Benzoicacid}}{C_6H_5COOH}$

---

61. Write the equations of any aldehyde with Fehling's reagent.

A: **1) Fehling's solution** is an alkaline solution of copper sulphate containing Rochelle salt(sodium potassium tartrate + sodium hydroxide).

2) Fehling's test: When Fehling's solution is heated with an aldehyde, a reddish brown precipitate of (Cu_2O) is formed.

3) Reaction: $RCHO + 2Cu^{+2} + 5OH^- \longrightarrow RCOO^- + Cu_2O + 3H_2O$

62. Write the IUPAC names of the following

(i) $CH_3CH_2CH(Br)CH_2COOH$

(ii) $PhCH_2COCH_2COOH$

(iii) $CH_3CH(CH_3)CH_2COOC_2H_5$

A: i) $CH_3CH_2CH(Br)CH_2COOH$: **3-Bromo1-pentanoic acid**

ii) $PhCH_2COCH_2COOH$: **4-Phenyl-3-oxo butanoic acid**

iii) $CH_3CH(CH_3)CH_2COOC_2H_5$: **Ethyl-3-Methyl butanote**

---

63. Describe the following:

i) Crossed aldol condensation ii) Decarboxylation

A: i) Crossed aldol condensation: It is the condensation of two different carbonyl compounds (one of which must have one α-hydrogen) in the presence of alkali. It is the condensation of an aldehyde with a ketone in the presence of an alkali.

CH_3CHO + CH_3CH_2CHO → $CH_3-CH=CH-CHO$ + $CH_3-CH_2-CH=C(CH_3)-CHO$

Acetaldehyde (Ethanal) **(A)** + Propionaldehyde (Propanal) **(B)** →

But-2-enal **(AA)**

2-MethylPent-2-enal **(BB)**

(Self-Aldol Products)

$CH_3-CH=C(CH_3)-CHO$ + $CH_3-CH_2-CH=CH-CHO$

2-Methylbut-2-enal

Pent-2-enal **(BA)**

(AB) (Cross Aldol Products)

ii) Decarboxylation: Sodium Ethanoate is heated with **Sodalime** to form **Methane.**

It removes a carboxyl group and releases CO_2.

Ex: $RCOONa + NaOH + CaO \xrightarrow{\Delta} R-H + Na_2CO_3$

$CH_3COONa + NaOH \xrightarrow[\Delta]{CaO} CH_4 + Na_2CO_3$

Sodium Ethanoate; Soda lime; Methane

[☞Remaining SAQs are in Page 96]

MAGICAL MEMORY MNEMONICS

- **Crossed Aldol:** Different Carbonyls + α-H
- **Decarboxylation:** DSESM [Dear SESMa ನಮ್ಮ] (Remove CO_2 (One Carbon Less)

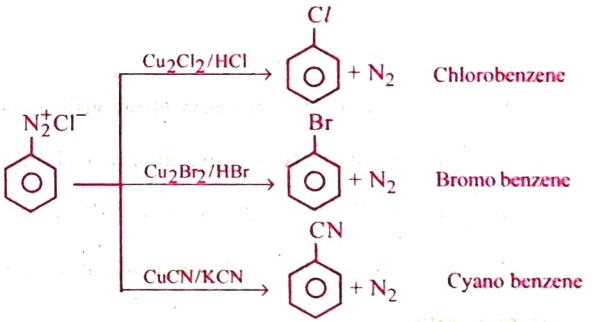
9 AMINES

4 Marks

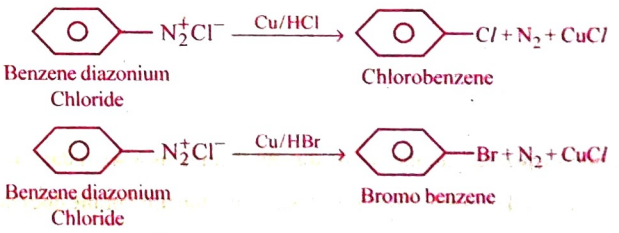
64. Explain the following name reactions: **[BMP]** [AP 15,16,17,18]

i) Sandmeyer reaction **ii) Gattermann reaction**

A: i) Sandmeyer reaction: Benzene diazonium chloride is treated with (Cu_2Cl_2 / HCl) (or) Cu_2Br_2 / HBr (or) CuCN/KCNto form Chloro Benzene (or) Bromo Benzene (or) Cyano Benzene.



ii) Gattermann's reaction: Benzene diazonium chloride is treated with Cu/HCl (or) Cu/HBr to form Chloro Benzene (or) Bromo Benzene



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65. Write the reactions of

(i) aromatic and (ii) aliphatic primary amines with nitrous acid, highlighting the differences.

A: i) **Aromatic** amines react with nitrous acid at low temperatures (273-278K) to form diazonium salts.

$$C_6H_5-NH_2 \xrightarrow[273-278K]{NaNO_2+2HCl} C_6H_5-N_2^+Cl^- + NaCl + 2H_2O$$

ii) Aliphatic primary amines react with nitrous acid to form aliphatic diazonium salts which being unstable, liberate nitrogen gas and alcohols. Quantitative evolution of nitrogenis used for the estimation of amino acids and proteins.

$$R-NH_2 + HNO_2 \xrightarrow{[NaNO_2+HCl]} [R-N_2^+Cl^-] \xrightarrow{H_2O} R-OH + N_2 + HCl$$

---

66. Write the equations involved in the reaction of Nitrous acid with ethylamine and aniline.

A: 1) Primary aliphatic amines like $C_2H_5NH_2$ react with nitrous acid and form aliphatic diazonium salts. But it being unstable, dissociates to give ethyl alcohol and N_2 gas.

$$C_2H_5NH_2 + HNO_2 \xrightarrow{[NaNO_2+HCl]} [C_2H_5-N_2^+Cl^-] \xrightarrow{H_2O} C_2H_5OH + N_2 + HCl$$

2) Aromatic amines like $C_6H_5NH_2$ react with nitrous acid at low temperatures (273-278K) and form **diazonium salts.**

$$\underset{\text{Aniline}}{C_6H_5NH_2} \xrightarrow[273-278K]{NaNO_2+2HCl} \underset{\text{Diazonium chloride}}{C_6H_5N_2^+Cl^-} + NaCl + 2H_2O$$

---

67. Give one chemical test to distinguish between the following pairs of compounds:

i) Methylamine and dimethylamine ii) Aniline and N-methylaniline

iii) Ethylamine and aniline

A: i) **Methylamine** is a primary amine, while dimethyl amine is a secondary amine.

So, methyl amine undergoes carbylamine test, while dimethylamine does not.

When methylamine is heated with chloroform and alcohalic KOH, an offensive smelling compound methyl carbylamine is formed.

$$\underset{\text{Methylamine}}{CH_3NH_2} + \underset{\text{Chloroform}}{CHCl_3} + 3KOH \xrightarrow[\text{heat}]{\text{alc}} \underset{\text{Methyl carbylamine}}{CH_3NC} + 3KCl + 3H_2O$$

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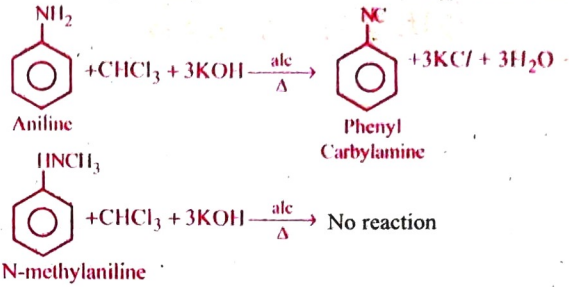
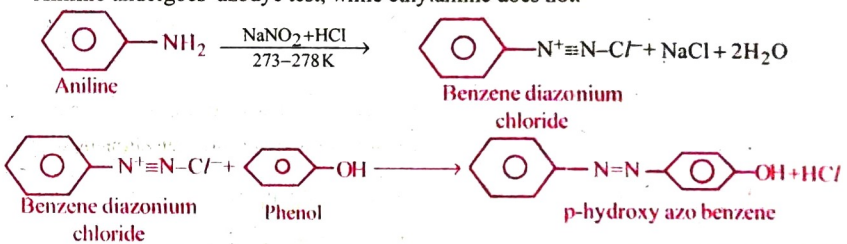
ii) **Aniline** is a primary amine, while **N-methylaniline** is a secondary amine.
So aniline undergoes carbylamine test, while N-methylaniline does not.



iii) **Ethylamine** and aniline are primary aromatic amines.
Aniline undergoes azodye test, while ethylamine does not.



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68. Compare the basicity of the following in gaseous state and in aqueous state and arrange them in increasing order of basicity.

CH_3NH_2, $(CH_3)_2NH$, $(CH_3)_3N$ and NH_3

A: 1) P^{kb} values of NH_3, CH_3NH_2, $(CH_3)_2NH$, $(CH_3)_3N$ are 4.74, 3.35, 3.27, 4.22 respectively.

2) "Smaller is the value of P^{kb}, greater is the basic nature of amine. .

Hence, **increasing order of basicity:** $NH_3 < (CH_3)_3N < CH_3NH_2 < (CH_3)_2NH$

3) In gaseous phase, basic character of amines increases with the increase in number of electron releasing groups(+I groups). Due to +I effect, the order of basicity of amines is as follows

$[(CH_3)_3N > (CH_3)_2NH > CH_3NH_2 > NH_3]$

In aqueous state, basicity depends on stability of protonated amines.

4) Greater the strength of H-bonding greater will be stability and more is the basic strength.

Hence order of basicity : $NH_3 > CH_3NH_2 > (CH_3)_2NH > (CH_3)_3N$

5) But overall basic strength of Amines depends on combined effect of steric, solvation and Inductive effect.

6) Hence overall order is **$(CH_3)_2NH > CH_3NH_2 > (CH_3)_3N > NH_3$**

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10 BIOMOLECULES

4 Marks

69. Give the dietary sources of the following vitamins and name the diseases caused by their deficiency (a) A (b) D (c) E and (d) K **[BMP][AP 15,16,17,19,24]**

A:

Sl.No.	Vitamin	Sources	Deficiency diseases
1.	Vitamin A	Carrots, Fish liver oil, butter and milk	Night blindness, Xerophthalmia
2.	Vitamin D	Sunlight, fish and egg yolk	Rickets (bone deformities in children) and osteo-malacia (soft bones and joint pains in adults)
3.	Vitamin E	Vegetable oils like Sunflower oil, wheat germ oil.	Increased fragility of RBC's and muscular weakness.
4.	Vitamin K	Green leafy vegetables	Increased blood clotting time.

AC DS EV KG

---

70. What are hormones? Give one example for each. **[AP 16,18,19,23,25]**

(i) steroid hormones (ii) Polypeptide hormones and (iii) amino acid derivatives.

A: **1) Hormones :** These are the molecules which act as intracellular messengers.

They transfer biological information from one group of cells to target organs.

These are produced by Endocrine glands. They are directly released into blood.

They are basically two types (i) Animal hormones (ii) Plant hormones.

2) On the basis of their chemical nature, hormones are classified into three types.

i) Steroid hormones. Ex: Estrogens, androgens.

ii) Protein hormones. Ex: Insulin, endorphins

iii) Amino acid derivatives. Ex: Thyroxine, epinephrine, norepinephrine

☺Three Hormones- 4 Marks☺

71. Write notes on vitamins.

A: 1) Vitamins are organic compounds required in small amounts in the diet.

They are useful in the 'normal maintenance of optimum growth' and 'health of the organism'.

2) Vitamins are designated by A,B,C,D etc.,

B vitamin is further divided into subgroups. **Ex:** B_1, B_2, B_6, B_{12} etc.,

3) Vitamins are classified into two groups depending on their solubility in fat or water .

4) **Fat soluble vitamins:** These are the vitamins soluble in fat and oils but insoluble in water .

Ex: Vitamins A,D,E and K. They are stored in liver and adipose tissues.

5) **Water soluble vitamins:** These are the vitamins soluble in water.

Ex: Vitamin B, C.

They must be supplied regularly in diet because they are readily excreted in urine and cannot be stored (except vitamin B_{12}) in our body.

Additional Information (Not for Exam)

Vitamin - Sources - Deficiency diseases

Sl. No.	Name of Vitamins	Sources	Deficiency diseases
1)	Vitamin A (Retinol)	Fish liver oil, carrots, butter and milk	Night blindness, Xerophthalmia (hardening or cornea of eye)
2)	Vitamin D (Calciferol)	Exposure to Sunlight, fish and egg yolk	Rickets (bone deformities in children) and osteo-malacia (soft bones and joint pain in adults)
3)	Vitamin E (Tocopherol)	Vegetable oils like wheat germ oil, Sun-flower oil, etc.,	Increased fragility of RBC's and muscular weakness.
4)	Vitamin K	Green leafy vegetables	Increased blood clotting time.
5)	Vitamin C (Ascorbic acid)	Citrus fruits, amla and green leafy vegetables	Scurvy (bleeding gums)
6)	Vitamin B_1 (Thiamine)	Yeast, milk, green vegetables and cereals	Beri beri (loss of appetite, retarded growth)
7)	Vitamin B_2 (Riboflavin)	Milk, eggwhite, liver, kidney	Cheilosis (cracking at corners of mouth and lips), digestive disorder and burning sensation of the skin
8)	Vitamin B_6 (Pyridoxine)	Yeast, Milk, egg yolk, cereals and grams	Convulsions/Fits (rapid, involuntary muscle contraction)
9)	Vitamin B_{12}	Meat, fish, egg, curd	Pernicious anaemia (RBC deficient in haemoglobin)

72. Write the importance of carbohydrates.

A: 1) Carbohydrates are essential for the life of both plants and animals.

2) **Ex:** Honey, Starch, Cellulose.

3) Honey is an instant source of energy.

4) Starch in seen in plants and glycogen is in animals.

5) Cell walls of bacteria and plant are made up of 'cellulose'.

6) Wood and cotton fibre contain cellulose.

7) "Streptomycin" is a carbohydrate antibiotic.

8) Anticarcinogenic carbohydrates are synthesised to fight against cancer.

---

73. Explain the denaturation of proteins.

A: 1) **Denaturation:** When a protein in its native form, is subjected to physical change such as change in temperature, or chemical change like change in pH, the hydrogen bonds are disturbed.

2) The protein globules unfold, helix get uncoiled and protein loses its biological activity.

3) This is called **denaturation of proteins**.

4) Secondary and tertiary structures are destroyed but the primary structure remains as such.

Ex: Curdling of milk, Coagulation of egg by boiling etc.,

---

74. What are enzymes? Give examples.

A: 1) **Enzymes** are biological catalysts. Each biological system requires a different enzyme. Enzymes are specific and efficient in their action.

2) They are required in only small quantity and work at optimum temperature (310K) and pH (7.4) under one atmospheric pressure. Chemically, they are globular proteins.

3) However some enzymes are also associated with some non-protein components called the co-factors for their activity.

4) **Ex**: Invertase, zymase, Diastase, Maltase, Urease, Pepsin

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20 Good Things a BBQ Student Can Do

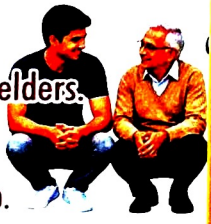
Practice Integrity

1. Return something that does not belong to you.
2. Never cheat in an examination.
3. Give credit when you use someone else's idea.
4. Admit your mistake instead of hiding it.

Spread Kindness

5. Thank someone who helps you.
6. Appreciate a classmate's achievement.
7. Help a new student feel welcome.
8. Perform one secret good deed every week.

Care for People



9. Respect & Greet Your Teachers, Parents
10. Visit an old-age home and spend time with elders.
11. Donate books you no longer need.
12. Teach something useful to a child who needs help.

Serve Society

13. Organise a small food or clothing donation drive.
14. Help keep a public place clean.
15. Volunteer for a community service activity.
16. Donate blood when legally eligible and medically suitable.

Care for Nature

17. Plant a sapling on your birthday.
18. Adopt and care for one tree.
19. Keep your surroundings litter-free.
20. Avoid unnecessary plastic.

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4. A few answers are 'reduced or simplified' to make them ' easy to read ' and 'easy to wirte '.
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7. Various features such as Hit List, Selection of Q's & orderly arrangement of sections, questions. Model Guess Papers etc are prepared just keeping in view of the needs of Rising Stars.
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BLUE PRINT of CHEMISTRY Q' Paper

CHAPTER (Total Marks)	OMQ(1M) 9×1= 9	VSAQ(2M) 14×2 = 28	SAQ(4M) 8×4 = 32	LAQ(8M) 2×8 = 16	Total 85 M
1. Solutions **(9 M)**	1	2+2	4	–	**9**
2. Electrochemistry **(13 M)**	1	2+2	4+4	–	**13**
3*. Chemical Kinetics **(15 M)**	1	2	4	8	**15**
4. The d&f -Block Elements **(8 M)**	--	2+2	4	–	**8**
5*. Coordination Compounds **(15 M)**	1	2	4	8	**15**
6. Haloalkanes & Haloarenes **(7 M)**	1	2	4	–	**7**
7. Alcohols, Phenols & Ethers **(11 M)**	1	2	4+4	–	**11**
8*. Aldehydes, Ketones & Carboxylic acids **(15 M)**	1	2	4	8	**15**
9. Amines **(7 M)**	1	2	4	–	**7**
10. Biomolecules **(9 M)**	1	2+2	4	–	**9**
Section wise Marks/ Total No.of Q's	**9M/9Q**	**28 M/14Q**	**32M/12Q**	**16M/3Q**	**85 /38Q**

☛ * indicates LAQ Chapters.

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4. THE d&f BLOCK ELEMENTS		62-63	37-38		80
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☛ This chapter index helps faculty to revise lessons chapterwise.

☛ To get **'Synopsis & Chem Beats'** of respective chapters please scan the above **'QR Code'** through **Google Lens**

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30 DAYS ACTION PLAN

DAY	CHAPTER	REMARKS
Day-1	1) SOLUTIONS (1M + 2M +2M + 4M = 9 M)	OMQ + VSAQ
Day-2	SOLUTIONS	VSAQ + SAQ
Day-3	SOLUTIONS	SAQ + PROBLEMS
Day-4	2) ELECTRO CHEMISTRY (1M + 2M +2M + 4M + 4M = 13 M)	OMQ + VSAQ
Day-5	ELECTRO CHEMISTRY	SAQ
Day-6	ELECTRO CHEMISTRY	SAQ + PROBLEMS
Day-7	3) CHEMICAL KINETICS(1M + 2M + 4M + 8M = 15 M)	LAQ
Day-8	CHEMICAL KINETICS	SAQ
Day-9	CHEMICAL KINETICS	VSAQ + OMQ
Day-10	CHEMICAL KINETICS **Total Marks in this unit : 37 Marks**	**REVISION & SLIP TEST -1**
Day-11	4) d&f BLOCK ELEMENTS (1M + 2M + 2M + 4M = 9 M)	OMQ + VSAQ
Day-12	d&f BLOCK ELEMENTS	VSAQ + SAQ
Day-13	d&f BLOCK ELEMENTS	REVISION
Day-14	5) CO-ORDINATION COMPOUNDS(1M + 2M + 4M + 8M = 15 M)	LAQ
Day-15	CO-ORDINATION COMPOUNDS	SAQ
Day-16	CO-ORDINATION COMPOUNDS	VSAQ + OMQ
Day-17	CO-ORDINATION COMPOUNDS	REVISION
Day-18	10) BIOMOLECULES (1M + 2M + 2M + 4M = 9 M)	OMQ + VSAQ + SAQ
Day-19	BIOMOLECULES **Total Marks in this unit : 33 Marks**	**REVISION & SLIP TEST -2**
Day-20	6) HALOALKANES & HALOARENES (1M + 2M + 4M = 7 M)	OMQ + VSAQ
Day-21	HALOALKANES & HALOARENES	SAQ Total: 36 Marks
Day-22	7) ALCOHOLS, PHENOLS & ETHERS(1M + 2M + 4M + 4M = 11 M)	OMQ + VSAQ
Day-23	ALCOHOLS, PHENOLS & ETHERS	SAQ
Day-24	ALCOHOLS, PHENOLS & ETHERS	REVISION
Day-25	8) ALDEHYDES, KETONES & CA (1M + 2M + 4M + 8M = 15 M)	LAQ
Day-26	ALDEHYDES, KETONES & CARBOXYLIC ACIDS	SAQ
Day-27	ALDEHYDES, KETONES & CARBOXYLIC ACIDS	VSAQ + OMQ
Day-28	ALDEHYDES, KETONES & CARBOXYLIC ACIDS	REVISION
Day-29	9) AMINES (1M + 2M + 4M = 7 M)	OMQ + VSAQ
Day-30	AMINES **Total Marks in this unit ; 40 Marks**	**SAQ REVISION & SLIP TEST -3**

☞ **30 Days** x 1 Hour Reading Time (or) **3 Days** x 10 Hours Reading Time

★ 8 MARKS LAQ HIT LIST

TOP 9

Ans-Page

★★★ CHEMICAL KINETICS [8 MARKS] ★★★

LAQ 1. Give a detailed account of the collision theory of reaction rates of bimolecular gaseous reactions. [☛P 14(1)]

LAQ 2. a)Write the differences between order & molecularity. [☛P 15(2)]
b) Describe the salient features of collision theory.

LAQ 3. Explain the following terms with suitable examples. [☛P 16(3)]
a)Average rate of a reaction b) Slow and fast reactions
c)Order of a reaction d)Molecularity of a reaction
e)Activation energy of a reaction

★★★ COORDINATION COMPOUNDS [8 MARKS] ★★★

LAQ 4. a)Write four postulates of Werner's theory of coordination compounds. [☛P 17(4)]
b)Explain cis-trans isomerism in $[Pt(NH_3)_2Cl_2]$ complex.

LAQ 5. Explain the applications of coordination compounds in different fields. [☛P 18(5)]

LAQ 6. Explain different types of isomerism exhibited by coordination compounds, giving suitable examples. [☛P 19(6)]

★★★ ALDEHYDES, KETONES & CARBOXYLIC ACIDS [8 MARKS] ★★★

LAQ 7. Explain the following reactions [☛P 22(8)]
a) Aldol reaction b) Esterification
c) H.V.Z reaction d) Bromination of Benzoic acid

LAQ 8. Describe the following: [☛P 23(9)]
i)Acetylation ii) Cannizzaro reaction
iii)Cross aldol condensation iv) Decarboxylation

LAQ 9. How do you distinguish the following pairs of compounds? [☛P 24(10)]
i)Propanal and propanone ii) Acetophenone and benzophenone
iii)Phenol and benzoic acid iv) Pentan-2-one and Pentan-3-one

★ 4 MARKS SAQ HIT LIST

TOP 40

Ans-Page

★★★ SOLUTIONS [4 MARKS] ★★★

SAQ 1. What is meant by positive deviations from Raoult's law and how is the sign of Δ_{mix} H related to positive deviation from Raoult's law ? [☞P 27(12)]

SAQ 2. What is relative lowering of vapour pressure? How is it useful to determine the molar mass of a solute? [☞P 28(14)]

SAQ 3. Define mass percentage, volume percentage and mass to volume percentage solutions. [☞P 29(16)]

SAQ 4. A solution of glucose in water is labeled as 10% w/w. What would be the molality of the solution ? [☞P 29(18)]

SAQ 5. Calculate the mass of a non-volatile solute (molar mass 40g mol^{-1}) which should be dissolved in 114g Octane to reduce its vapour pressure to 80%. [☞P 28(15)]

★★★ ELECTRO CHEMISTRY [4+4 MARKS] ★★★

SAQ 6. What are Galvanic cells? Explain the working of Galvanic cell with a neat sketch taking Daniel cell as example. [☞P 30(19)]

SAQ 7. State and explain Kohlrausch's law of independent migration of ions. [☞P 30(20)]

SAQ 8. Give the construction and working of a standard hydrogen electrode with a neat diagram. [☞P 31(21)]

SAQ 9. What are the products obtained at the cathode and the anode during the electrolysis of the following when platinum electrodes are used in the electrolysis (a) Molten KCl (b) Aq. $CuSO_4$ solution (c) Aq.K_2SO_4 solution [☞P 31(22)]

SAQ 10. State and explain Nernst equation with the help of a metallic electrode and a non-metallic electrode. [☞P 32(23)]

SAQ 11. A solution of $CuSO_4$ is electrolysed for 10 minutes with a current of 1.5 amperes. What is the mass of copper deposited at the cathode? [☞P 33(25)]

★★★ CHEMICAL KINETICS [4 MARKS] ★★★

SAQ 12. What is 'molecularity' of a reaction? How is it different from the 'order' of a reaction? Name one bimolecular and one trimolecular gaseous reactions. [☞P 34(28)]

SAQ 13. Derive an integrated rate equation for a first order reaction. [☞P 35(30)]

SAQ 14. What is half-life ($t_{1/2}$) of a reaction? Derive the equations for the 'half-life' value of zero and first order reactions. [☞P 35(31)]

SAQ 15. What is Arrhenius equation? Derive an equation which describes the effect of rise of temperature (T) on the rate constant (k) of a reaction. [☞P 36(33)]

★★★ d&f BLOCK ELEMENTS [4 MARKS] ★★★

SAQ 16. Write the characteristics properties of transition elements. [☞P 37(34)]

SAQ 17. What is lanthanoid contraction? What are the consequences of lanthanoid contraction? [☞P 37(35)]

SAQ 18. Write the characteristics of interstitial compounds. [☞P 38(37)]

SAQ 19. Why do the transition metal ions exhibit characteristic colours in aqueous solution. Explain giving examples. [☞P 38(38)]

★★COORDINATION COMPOUNDS[4 MARKS] ★★

SAQ 20. Explain Werner's theory of coordination compounds with suitable examples. [☞P 39(40)]

SAQ 21. Explain geometrical isomerism in coordination compounds giving suitable examples. [☞P 39(41)]

SAQ 22. What are homoleptic and heteroleptic complexes? Give one example for each. [☞P 40(42)]

SAQ 23. Using IUPAC norms write the systematic names of the following: [☞P 40(45)]

(i) $[Co(NH_3)_6]Cl_3$ (ii) $[Pt(NH_3)_2Cl(NH_2\ CH_3)]Cl$

(iii) $[Ti(H_2O)_6]^{3+}$ (iv) $[NiCl_4]^{2-}$

★★HALOALKANES & HALOARENES[4 MARKS] ★★

SAQ 24. Explain the mechanism of S_N1 with example. [☞P 41(46)]

SAQ 25. Write the differences between S_N1 and S_N2 reactions. [☞P 42(49)]

SAQ 26. Give the IUPAC names of the following compounds. [☞P 43(50)]

(i) $CH_3–CH(Cl)–CH(I)CH_3$ (ii) $ClCH_2.CH=CH–CH_2Br$

(iii) $(CCl_3)_3CCl$ (iv) $CH_3C(p–Cl–C_6H_4)_2–CH(Br)CH_3$

SAQ 27. Write the structures of the following organic halides. [☞P 43(51)]

(i)1-bromo-4-sec-butyl-2 methylbenzene (ii)2-chloro-1-phenylbutane

(iii)p-bromochlorobenzene (iv)4-t-butyl-3-iodoheptane

★★★ALCOHOLS, PHENOLS & ETHERS[4+4 MARKS] ★★★

SAQ 28. Give the equations for the preparation of phenol from Cumene. [☞P 44(52)]

SAQ 29. Write the products formed by the reduction and oxidation of phenol. [☞P 44(53)]

SAQ 30. Explain the acidic nature of phenols and compare with that of alcohols. [☞P 44(54)]

SAQ 31. While separating a mixture of ortho and para nitrophenols by steam distillation, name the isomer which will be steam volatile. Give reason. [☞P 45(55)]

★★ALDEHYDES, KETONES & CARBOXYLIC ACIDS[4 MARKS] ★★

SAQ 32. What is Tollen's reagent? Explain its reaction with Aldehydes. [☞P 47(59)]

SAQ 33. Write the oxidation products of acetaldehyde, acetone and acetophenone. [☞P 47(60)]

SAQ 34. Describe the following: [☞P 48(63)]

i) Crossed aldol condensation ii) Decarboxylation

★★★ AMINES[4 MARKS] ★★★

SAQ 35. Explain i)Sandmeyer reaction ii) Gattermann reaction [☞P 49(64)]

SAQ 36. Write the reactions of (i) aromatic and (ii) aliphatic primary amines with nitrous acid, highlighting the differences. [☞P 50(65)]

SAQ 37. Write the equations involved in the reaction of Nitrous acid with ethylamine and aniline. [☞P 50(66)]

★★★ BIOMOLECULES[4 MARKS] ★★★

SAQ 38. Give the dietary sources of the following vitamins and name the diseases caused by their deficiency (a) A (b) D (c) E and (d) K [☞P 52(69)]

SAQ 39. What are hormones? Give one example for each. [☞P 52(70)]

(i) steroid hormones (ii) Polypeptide hormones and

(iii) amino acid derivatives.

SAQ 40. Write notes on vitamins. [☞P 53(71)]

★ 2 MARKS VSAQ HIT LIST

TOP 50+

★★★ SOLUTIONS [2+2 MARKS] ★★★

VSAQ 1. Define molarity [☛P 56(75)]
VSAQ 2. Define molality. [☛P 56(76)]
VSAQ 3. Define mole fraction. [☛P 56(77)]
VSAQ 4. State Henry's law. [☛P 56(78)]
VSAQ 5. Define osmotic pressure. [☛P 56(79)]
VSAQ 6. What are isotonic solutions? [☛P 56(80)]
VSAQ 7. What is Cryoscopic constant? [☛P 57(82)]
VSAQ 8. Calculate the molarity of a solution containing 5 g of NaOH in 450 ml solution. [☛P 57(84)]
VSAQ 9. Calculate molality of 2.5 g of ethanoic acid (CH_3COOH) in 75g of benzene. [☛P 57(85)]

★★★ ELECTRO CHEMISTRY [2+2 MARKS] ★★★

VSAQ 10. State Faraday's first law of electrolysis. [☛P 58(86)]
VSAQ 11. State Faraday's second law of electrolysis. [☛P 58(87)]
VSAQ 12. What is a fuel cell? How is it different from a conventional galvanic cell? [☛P 58(88)]
VSAQ 13. What is metallic corrosion? Give one example. [☛P 58(89)]
VSAQ 14. Give the electro-chemical reaction that represents the corrosion or rusting of iron. [☛P 58(90)]
VSAQ 15. Define molar conductivity and how is it related to conductivity (k)? [☛P 59(91)]
VSAQ 16. What is standard hydrogen electrode? [☛P 59(94)]

★★ CHEMICAL KINETICS [2 MARKS] ★★

VSAQ 17. What is a pseudo first order reaction? Give one example. [☛P 60(96)]
VSAQ 18. Give two examples for gaseous first order reactions. [☛P 60(97)]
VSAQ 19. Give two examples for zero Order reactions. [☛P 60(98)]
VSAQ 20. Define molecularity of a reaction with an example. [☛P 60(99)]
VSAQ 21. Calculate the overall order of a reaction which has the rate expression [☛P 61(105)]
(a) Rate = $k[A]^{1/2}[B]^{3/2}$ (b) Rate = $k[A]^{3/2}[B]^{-1}$

★★★ The d&f BLOCK ELEMENTS [2+2 MARKS] ★★★

VSAQ 22. What are transition elements? Give examples. [☛P 62(107)]
VSAQ 23. Why Zn^{2+} is diamagnetic whereas Mn^{2+} is paramagnetic? [☛P 62(108)]
VSAQ 24. Calculate the 'spin only' magnetic moment of $Fe^{2+}_{(aq)}$ ion. [☛P 62(109)]
VSAQ 25. Aqueous Cu^{2+} ions are blue in colour, where as Aqueous Zn^{2+} ions are colourless. Why? [☛P 62(110)]
VSAQ 26. What is an alloy? Give example. [☛P 62(111)]
VSAQ 27. What is mischmetal? Give its composition and uses. [☛P 63(112)]
VSAQ 28. What is actinoid contraction? [☛P 63(113)]

Ans-Page

★★ COORDINATION COMPOUNDS [2 MARKS] ★★

VSAQ 29. What are coordination compounds? Give two examples. [☞P 64(117)]
VSAQ 30. What is a double salt? Give example. [☞P 64(118)]
VSAQ 31. What is a chelate ligand ? Give example. [☞P 64(120)]
VSAQ 32. What is an ambidentate ligand? Give example. [☞P 64(121)]

★★ HALOALKANES & HALOARENES [2 MARKS] ★★

VSAQ 33. What are Enantiomers? [☞P 66(127)]
VSAQ 34. What are ambident nucleophiles? [☞P 66(129)]
VSAQ 35. What is the stereochemical outcome of S_N1 and S_N2 reactions? [☞P 66(130)]
VSAQ 36. Write the structure of the following compounds [☞P 66(131)]
(i) 2-chloro-3-methylpentane (ii)1-bromo-4-sec-butyl-2-methylbenzene
VSAQ 37. Write the isomers of the compound having molecule formula C_4H_9Br? [☞P 66(132)]

★★★ ALCOHOLS, PHENOLS & ETHERS [2 MARKS] ★★★

VSAQ 38. Give the reagents used for the preparation of phenol from chlorobenzene. [☞P 68(137)]
VSAQ 39. Explain why propanol has higher boiling point than that of the hydrocarbon butane. [☞P 68(138)]
VSAQ 40. Alcohols are comparatively more soluble in water than hydrocarbons of comparable molecular masses. Explain this fact. [☞P 68(139)]
VSAQ 41. Write the structures for the following compounds [☞P 69(143)]
1)Ethoxy Ethane 2) Ethoxybutane 3) Phenoxy ethane

★★ ALDEHYDES, KETONES & CARBOXYLIC ACIDS [2 MARKS] ★★

VSAQ 42. List the reagents suitable to reduce carboxylic acid to alcohol. [☞P 70(147)]
VSAQ 43. Write the reaction showing α-halogenation of carboxylic acid and give its name. [☞P 70(148)]
VSAQ 44. How do you distinguish acetophenone and benzophenone? [☞P 70(150)]
VSAQ 45. Compare the acidic strength of acetic acid, Chloroacetic acid, benzoic acid and Phenol. [☞P 71(1530)]

★★ AMINES [2 MARKS] ★★

VSAQ 46. Arrange the following bases in increasing order of their basic strength. [☞P 72(155)]
Aniline, p-nitroaniline and p-toluidine.
VSAQ 47. Write equations for carbylamine reaction of an aliphatic amine. [☞P 72(156)]
VSAQ 48. Arrange the following bases in decreasing order of pK_b values [☞P 72(158)]
$C_2H_5NH_2$, $C_6H_5NHCH_3$, $(C_6H_5)_2NH$ and $C_6H_5NH_2$.

★★★ BIOMOLECULES [2+2 MARKS] ★★★

VSAQ 49. What are essential and non-essential amino acids? Give one example for each. [☞P 74(165)]
VSAQ 50. Differentiate between globular and fibrous proteins. [☞P 74(167)]
VSAQ 51. What is glycogen? How is it different from starch? [☞P 74(168)]
VSAQ 52. What is Zwitter ion? Give an example. [☞P 74(170)]
VSAQ 53. Write the names of three types of RNA. [☞P 75(171)]
VSAQ 54. Name the different types of carbohydrates on the basis of their hydrolysis. Give one example for each. [☞P 75(173)]

BBQ - TIPS

LAQ SCORING CHAPTERS

Easy Chapter: CHEMICAL KINETICS

Imp Chapter: COORDINATION COMPOUNDS

Super Scoring Pages (SSP): 14,15,17,22,23

SAQ SCORING CHAPTERS

Easy Chapters: SOLUTIONS, BIOMOLECULES, THE d&f -BLOCK ELEMENTS

Imp Chapters: ELECTRO CHEMISTRY, CHEMICAL KINETICS, COORDINATION COMPOUNDS, ALCOHOLS PHENOLS AND ETHERS

Super Scoring Pages (SSP): 28, 29, 30, 32, 34, 35, 37, 39, 41, 42, 44, 47, 49, 50, 52, 53

VSAQ SCORING CHAPTERS

Easy Chapters: SOLUTIONS, ELECTRO CHEMISTRY, d&f, BIOMOLECULES

Imp Chapters: CHEMICAL KINETICS, COORDINATION COMPOUNDS, ALCOHOLS, PHENOLS

Super Scoring Pages (SSP): 56, 57, 58, 59, 60, 62, 63, 64, 66, 68, 70, 72, 74, 75

DEAR STUDENT

- **Quick Memory Bytes → (i) Answer Clarity (ii) Easy Memory .**
- Collision theory LAQ ను Cricket shots and Class room Analogy లతో పోల్చుకుంటూ చదివితే అది మీకు Lifelong గుర్తుంటుంది. మీరు ఏ పనిలోనైనా Success సాధించాలంటే 'Hardwork తోపాటు Smartwork కూడా చేయాలి' అనే Life Lessons మీకు గుర్తుంటాయి.
- క్లిష్టమైన Werner's theory ను కూడా 5 నిమిషాలలోనే మర్చిపోకుండా నేర్చడానికే **Quick Memory Bytes.**
- Organic Chemistry లోని Named Reactions ను Easy గా గుర్తుపెట్టుకోవడానికి **Mnemonics.**
- Chapter concept పై Motivation కావాలంటే **Chem beats** మీ కోసం రెడీ! (on QR Code).
- **Rising Stars** కోసం **Level I** Questions, **Shining Stars** కోసం **Level II** Questions (Star Q Plus).

☞ **SINCERE SUGGESTION:** After completing this BulletQ, Shining stars are advised to go through the Notes given by your Lecturers; follow the next Higher Level Material.

RISING STARS
MOST FAVOURITE SECTION

LAQ

SECTION-D

BBQ turns, "I can't to I can".

Small book, Big Marks.

LEARN WITH SAI SIR
@myself_saikumar_dasari

LAQ CHAPTERS

16 MARKS - 12 PAGES

★★★ 1.	CHEMICAL KINETICS [3 Qs- 3 Pages]	14 - 16
★★★ 2.	COORDINATION COMPOUNDS [4 Qs- 5 Pages]	17 - 21
★★★ 3.	ALDEHYDES, KETONES [4 Qs- 4 Pages]	22 - 25

TWO BIRDS AT ONE SHOT!

WOW TIP 1: Answer of LAQ 1 repeats in LAQ 2.
Order & Molecularity in LAQ 2 repeats in LAQ 3.

WOW TIP 2: Answer of LAQ 4 (Werner's theory) comes again in SAQ 40.
Answer of LAQ 6 comes again in SAQ 41.

WOW TIP 3: In the **organic chapter** few named reactions are already learned in Jr. Inter.

YouTube @myself_saikumar_dasari | Instagram @learn_with_sai_sir | Telegram t.me/lernwithsaisir

Super Scoring Pages

★★★SSP★★★

Page 14,15,17,22,23

Total SS Pages : 5

1 CHEMICAL KINETICS

8 Marks

1. Give a detailed account of the collision theory of reaction rates of bimolecular gaseous reactions. **[BMP]**[AP 18,24]

A: The **Collision Theory** proposed by **Arrhenius** explains how fast a chemical reaction occurs. It states that a reaction takes place only when the gas molecules collide **effectively.**

Main Postulates of Collision Theory:

1) Gas molecules always move in continuous random motion:

They move in all directions. For a collision to occur, firstly any two molecules must collide.

2) But, every collision does not produce a product:

Many collisions are ineffective. Only a few collisions lead to product formation.

3) Effective collision is necessary for a reaction. **Rate of reaction** depends on the 'number of effective collisions'.

4) For a collision to be effective, two conditions must be satisfied:

(a) Activation Energy (E_a): The colliding molecules must possess E_a.

(b) Proper Orientation (P) : The molecules must collide in the correct direction.

5) Collision frequency (Z) is the number of collisions occurring 'per unit volume per second'.

6) From Arrhenius equation, specific rate constant $k = PZe^{-Ea/RT}$

where **P** = Probability (orientation) factor, **Z** = Collision frequency, E_a = Activation energy

R = Universal gas constant, **T** = Absolute temperature

7) Effect of Temperature: Increasing temperature increases the **rate of reaction.**

8) Ex: Formation of methanol from bromomethane depends upon the **orientation** of reactants.

$$CH_3Br + \bar{O}H \xrightarrow[\text{Improper orientation}]{\text{Low energy}} \text{No product}$$

$$CH_3Br + \bar{O}H \xrightarrow[\text{Proper orientation}]{\text{Activation Energy}} CH_3OH + \bar{B}r$$

DOES EVERY SHOT MAKE A SIXER? **MEMORY BYTES**

- Firstly, Bat & Ball should connect. (Collide)
- Batsman should hit the Ball hard enough to cross the boundary line. (Activation Energy E_a)
- Batsman should also Hit in the Right direction (very high or low means **catch out!**). (Proper orientation P)

2. a) **Write the differences between order & molecularity.**

   b) **Describe the salient features of collision theory.**

A: **a) Differences between order & molecularity:**

Order	Molecularity
i) Order is the sum of powers of concentrations of reactant terms.	i) Molecularity is the number of reactant molecules in a single elementary step.
ii) It is determined **experimentally**.	ii) It is determined **theoretically**.
iii) It can have **fractional values** also.	iii) It can have **integral values** only.
iv) It can be zero order, first order....	iv) It can be unimolecular,bimolecular....

b) Salient features of collision theory:

Molecular collision theory was proposed by **Arrhenius.**

It explains the **rate of gaseous bimolecular reactions.**

Main postulates :

1) All collisions do not form products, only a few collisions form the products.
2) A reaction takes place only when reactant molecules collide with proper orientation (P).
3) **Threshold energy (E_T)** is the minimum energy that the reactant molecules must possess to form products. Molecules having the threshold energy are called **activated molecules.**
4) Very few molecules possess E_T at ordinary temperatures.
5) **Activation Energy (E_a)** is the minimum excess energy that reactant molecules must acquire to participate in a chemical reaction. **Formula:** $E_a = E_T - E_R$.
6) Only collisions between the activated molecules with proper orientation form products. Hence such collisions are called activated collisions or fruitful collisions or effective collisions.
7) Specific rate constant $k = PZ.e^{-Ea/RT}$

☞ If you want, you can write this **Part B answer in LAQ1 and vice-versa**

CLASS ROOM- COLLISION - CAP **MEMORY BYTES**

☞ Every class is a collision, and only **effective collisions** give **Brilliant Results.**

- **Collision (Interaction):** Faculty comes to class and students listen. No class → No learning.
- **Activation Energy (Energetic teaching & active learning):** Faculty should teach with **enthusiasm,** and students should learn actively with curiosity and focus.
  Low energy & Passive learning → Poor results.
- **Proper Orientation (Right teaching & Keen listening):** Faculty should teach according to students' standards, difficulty level of topics, and students should listen carefully to the lessons without any distractions.

Good Faculty + Good Students → Good Results

☞ **Lecturer + Student:** Interaction + Enthusiasm + Right direction = Brilliant Results.

3. **Explain the following terms with suitable examples.**
   a) Average rate of a reaction **b) Slow and fast reactions**
   c) Order of a reaction **d)Molecularity of a reaction**
   e) Activation energy of a reaction

A: **a) Average rate of a reaction:**

It is defined as the 'rate of change of concentration' of reactants or products per unit time.

Formula: Average rate $\frac{\Delta x}{\Delta t} = -\frac{\Delta[R]}{\Delta t} = +\frac{\Delta[P]}{\Delta t}$

Ex: $2HI \rightarrow H_2 + I_2$

b) Slow reaction: It takes a long time to complete. It requires high activation energy.

Ex: Rusting of Iron takes many days or months.

Fast reaction: It occurs within a very short time. It requires low activation energy.

Ex: Reactions between an acid and base . $HCl + NaOH \rightarrow NaCl + H_2O$

c) Order of a reaction:

It is the 'sum of powers of the concentration terms' of the reactants in the rate law equation.

Ex: For a reaction of the form $aA + bB \rightarrow$ products. , Rate $= k[A]^m[B]^n$.

Here the order of the above reaction is $(m+n)$.

d) Molecularity of a reaction:

It is the number of reacting molecules taking part in a single elementary step.

- **Uni molecular:** $A \rightarrow$ products.

  Ex: $C_2H_5Cl \rightarrow C_2H_4 + HCl$. Here the molecularity is 1.

- **Bi molecular :** $A + B \rightarrow$ products.

  Ex: $H_2 + I_2 \rightarrow 2HI$. Here the molecularity is 2.

e) Activation Energy (E_a):

It is the minimum energy that reactant molecules must possess for the reaction to occur.

Formula: $E_a = E_T - E_R$. E_T = Threshold energy , E_R = Energy of reactants.

Ex: When a matchstick is struck against the match box, **friction** supplies the activation energy for the match stick to ignite.

[☞Few More LAQ are in Page 91]

QUICK MEMORY BYTES

- **Average rate** - How fast the reaction proceeds.
- **Slow reaction** (Days/ months) - Rusting of iron.
- **Fast reaction** (Seconds) - Acid + Base reaction.
- **Order** - Sum of powers in the rate reaction.
- **Molecularity** - No.of molecules colliding.
- **Activation energy** - Minimum energy required to start the reaction.

2 COORDINATION COMPOUNDS

8 Marks

4. a) Write four postulates of Werner's theory of coordination compounds.

b) Explain cis-trans isomerism in $[Pt(NH_3)_2Cl_2]$ complex. **[BMP]** [AP 20,22,23,24,25]

A: **Werner's theory** explains the structure, bonding, and isomerism of complex compounds.

a) Postulates of Werner's theory:

1) The 'central metal ion' in the coordination compounds shows two types of valencies.

a) Primary valency b) Secondary valency.

2) Primary Valency:

i) It represents the oxidation state (number) of the central metal ion .

ii) Oxidation state is satisfied (balanced) only by the negative ion $(Cl^-, NO_3^-, ...)$

iii) These negative ions are ionisable in water.

iv) These ions lie just outside the coordination sphere.

3) Secondary Valency:

i) It is equal to the co-ordination number of the central metal ion.

ii) It is satisfied by negative ions (or) neutral molecules-ligands. $[Cl^-, CN^-, NH_3, H_2O, ...]$

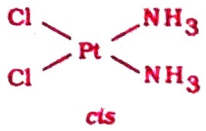
iii) These ligands are not ionisable in water.

iv) These ions lie inside the coordination sphere in fixed positions and exhibit isomerism.

b) Cis-trans isomerism in $[Pt(NH_3)_2Cl_2]$:

The given complex $[Pt(NH_3)_2Cl_2]$ of the type $[MX_2L_2]$.

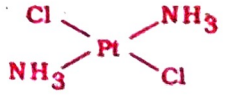
The four ligands occupy four corners of a square.



cis

i) **Cis isomer:** Here, two ligands of same type occupy adjacent positions in the coordination sphere.

ii) **Trans isomer:** Here, two ligands of same type occupy opposite positions in the coordination sphere.



trans

QUICK MEMORY BYTE — **WERNER - Very Powerful SI**

V- Valency is two types (Primary and Secondary).

P - Primary valency is the oxidation number of the central ion. (Outside - Ionisable)

S - Secondary valency is the coordination number of the central ion.(Inside - Non -ionisable)

I - Isomerism occurs due to identical X ligands .

5. Explain the applications of coordination compounds in different fields.

A: **1) Analysis of Chemicals:** Coordination compounds like EDTA, DMG are used in the estimation of qualitative and quantitative analysis of chemical compounds.

2) Hardness of water : Hardwater contains excessive Ca^{+2}, Mg^{+2} ions.

They are estimated by EDTA.

3) Silver and Gold extraction:

'Complex formation' plays a key role in the extraction of Silver and Gold.

Ex: $AgCl + 2NaCN \rightarrow Na[Ag(CN)_2] + NaCl$

$2Na[Ag(CN)_2] + Zn \rightarrow Na_2[Zn(CN)_4] + 2Ag\downarrow$

4) Metal Purification : 'Metal complexes' play a key role in the purification of metals like Ni, Fe.

Ex: Purification of Ni by Mond's process $[Ni(CO)_4] \xrightarrow{\Delta} Ni + 4CO$

5) Biological systems: Coordination complexes are very important in biological systems.

i) Photosynthesis in plants is due to chlorophyll, a complex of Mg^{+2} ion

ii) Vitamin B_{12} is a complex of Co^{+3} ion.

iii) In Hemoglobin, Heme is the complex of Fe^{+2} ion

6) Catalyst compounds : Some complexes act as catalysts by stabilising the activation energy.

Ex: Wilkinson catalyst is a Rhodium complex $[(Ph_3P)_3RhCl]$ used in hydrogenation of alkenes.

7) Photography: In the process of photo developing when AgBr film is dissolved in hypo solution $[Ag(S_2O_3)_2]^{3-}$ is formed

8) Medical Field: Coordination compounds are used in the medicinal chemistry.

Ex: EDTA is used in the treatment of lead poisoning.

QUICK MEMORY BYTES

A Hard Smart Man Became College Popular Master

A →Analysis (Qualitative and quantitative - EDTA, DM)
H → Hardness water - ions Ca^{+2}, Mg^{+2}
S → Silver & Gold extraction
M → Metal purification
B→ Biological
C→ Catalyst
P→ Photography
M → Medicine

Coordination compounds are simply called **Complexes.**

6. Explain different types of isomerism exhibited by coordination compounds, giving suitable examples.

A: Coordination compounds have two types of isomerism: **I) Stereo II) Structural**

I) Stereoisomerism occurs when two complexes have the same molecular formula, but their ligands have different positions in 3D space. This is two types. 1) Geometrical 2) Optical.

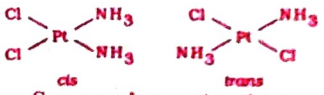
1) Geometrical isomers occur due to different geometric arrangements of the ligands.

i) Geometrical Isomerism in Square planar complexes of the type $[MX_2L_2]$:

Here **4 ligands occupy 4 corner places** of a square around the central ion.

a) cis isomer: Two identical X ligands are arranged adjacent to each other.

b) trans isomer: Two identical X ligands are arranged opposite to each other.

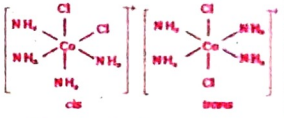


Square planar structure

Ex: cis $[Pt(NH_3)_2Cl_2]$, trans $[Pt(NH_3)_2Cl_2]$

ii) Geometrical Isomerism in Octahedral complexes:

Here **6 ligands occupy 6 corner places** of an octahedral.



Octahedral structure

a) cis & trans isomer of the type $[MX_2L_4]$:

- **cis isomer:** Two X ligands are adjacent to each other.
- **trans isomer:** Two X ligands are opposite to each other.

Ex: cis $[CoCl_2(NH_3)_4]^+$, trans$[CoCl_2(NH_3)_4]^+$

b) fac & mer isomer of the type $[MX_3L_3]$:

- **fac isomer:** Three X ligands occupy one triangular face of the octahedral.
- **mer isomer:** Three X ligands lie along a meridian of the octahedral.

Ex: fac $[Co(NH_3)_3(NO_3)_3]$, mer $[Co(NH_3)_3(NO_3)_3]$.

2) Optical isomers are mirror images that cannot be superimposed on one another(**Enantiomers**.)

i) **d-isomer** rotates the plane polarised light in clock-wise direction (right).

ii) ***l*-isomer** rotates plane polarised light in anticlock-wise direction(left) .

Ex: d-$[Co(en)_3]^{3+}$ and *l*-$[Co(en)_3]^{3+}$

II) Structural isomerism occurs when two complexes have the same molecular formula, but their ligands have different kinds of connections (w.r.to coordination sphere)

1) Ionisation isomers: Here a **ligand inside** the coordination sphere is exchanged with its **counter ion outside** the coordination sphere.

Ex: [$Co(NH_3)_5SO_4$] Br & [$Co(NH_3)_5Br$] SO_4

2) Hydrate isomers: Here we have different number of water molecules H_2O inside and outside the coordination sphere. **Ex:** $[Cr(H_2O)_6]Cl_3$ & $[Cr(H_2O)_5Cl]Cl_2.H_2O$.

3) Linkage isomers: Here the ambidentate ligands present in coordination sphere differ in the point of attachment. **Ex:** [$Co(NH_3)_5(NO_2)$] Cl_2 & [$Co(NH_3)_5(ONO)$] Cl_2

4) Coordination isomers: Here interchange of ligands between cationic and anionic entities occurs. **Ex:** $[Co(NH_3)_6][Cr(CN)_6]$, $[Co(CN)_6][Cr(NH_3)_6]$.

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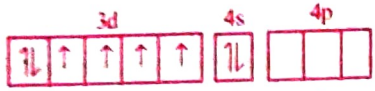
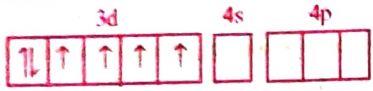
7.1 Discuss the nature of bonding and magnetic behaviour in the following coordination entities on the basis of valence bond theory
(i) $[Fe(CN)_6]^{4-}$ (ii) $[FeF_6]^{3-}$

A: (i) $[Fe(CN)_6]^{4-}$: The oxidation state of Fe in this complex is +2.

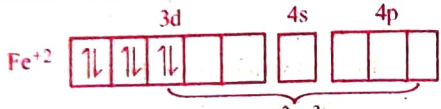
Fe (Z=26)=[Ar]$3d^6 4s^2$



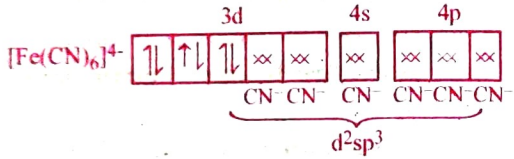
Fe^{+2}(Z=24)=[Ar]$3d^6 4s^0$



CN^- is a strong ligand.
So forced pairing happens and
hence d^2sp^3 hybridization takes place.



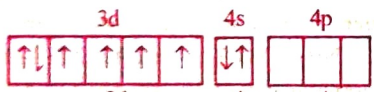
Now six empty d^2sp^3 hybrid orbitals overlap with 6 ligand orbitals (CN^-) to form the complex ion.



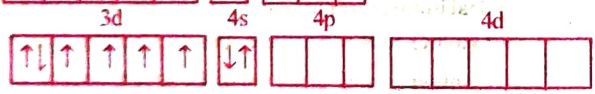
This complex ion is **octahedral**. It is **diamagnetic** as it does not contain any unpaired electron. Also (n–1)d orbitals are involved in hybridization. So it is a inner orbital or low spin complex ion.

ii) $[FeF_6]^{-3}$: The oxidation state of Fe in this complex is +3.

Fe (Z=26)=[Ar]$3d^6 4s^2$



Fe^{+3} (Z=23)= [Ar] $3d^5 4s^0$

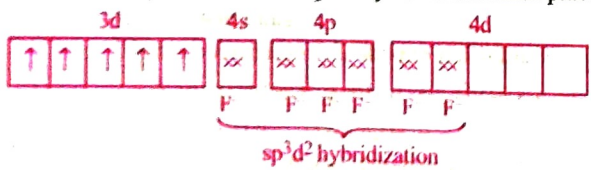


F^- is a weak ligand. So pairing does not happens .

Here 3d orbitals are not available to take part in bonding.

Hence nd orbitals are involved in hybridisation and sp^3d^2 hybridisation takes place.

$[FeF_6]^{-3}$:



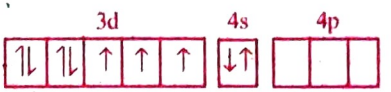
Because of the presence of five unpaired electrons, the complex is **paramagnetic.**

Also outer d orbitals are involved in hybridization, so it is an **outer orbital or high spin complex ion.**

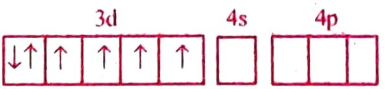
7.2 Discuss the nature of bonding and magnetic behaviour in the following coordination entities on the basis of valence bond theory
(i) $[Co(C_2O_4)_3]^{3-}$ (ii) $[CoF_6]^{3-}$

A: **i) $[Co(C_2O_4)_3]^{3-}$:** The oxidation state of Co in this complex is +3.

$Co(Z=27) = [Ar]3d^74s^2$



$Co^{+3} = [Ar]\ 3d^64s^0$

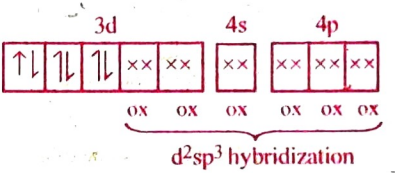


$C_2O_4^{-2}$ is a **strong ligand** .

So pairing happens and hence d^2sp^3 hybridization takes place.

Now 6 empty d^2sp^3 hybrid orbitals overlap with 6 ligand orbitals ($C_2O_4^{-2}$) to form complex ion.

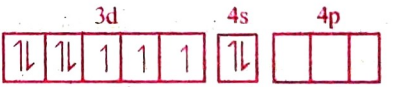
$[Co(C_2O_4)_3]^{3-}$:



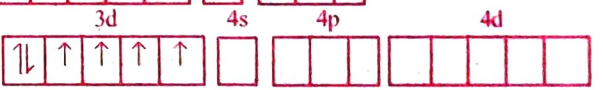
Since, all the electrons are paired, this complex is diamagnetic. It is an inner orbital complex because of the involvement of (n–1)d orbital for bonding.

(ii) $[CoF_6]^{3-}$: The oxidation state of Co in this complex is +3.

$Co(Z=27) : [Ar]\ 3d^74s^2,$



$Co^{+3}\ (Z=24)\ [Ar]\ 3d^64s^0$

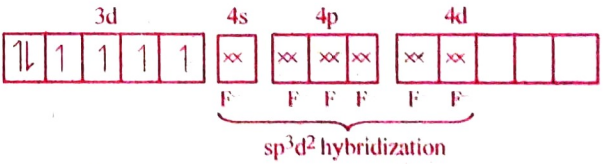


F^- is a weak field ligand. So pairing does not happen

Thus 3d orbitals are not available to take part in bonding.

Hence nd orbitals are involved in hybridisation and sp^3d^2 hybridisation takes place.

$[CoF_6]^{3-}$:



Because of the presence of four unpaired electrons, the complex is **paramagnetic.**
Since outer d orbitals take part in bonding it is an outer orbital complex ion or high spin complex ion.

LEARN WITH SAI SIR
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3 ALDEHYDES, KETONES & CARBOXYLIC ACIDS

8 Marks

8. Explain the following reactions **[BMP][AP 18,23,24,25]**

a) Aldol reaction **b) Esterification**

c) HVZ reaction **d) Bromination of Benzoic acid**

A: a) Aldol reaction:

Two aldehyde or ketone molecules having 'atleast one α-hydrogen' react together in the presence of a 'dilute alkali catalyst' to form **β-hydroxy aldehyde or β-hydroxy ketone (aldol).**

Ex: $\underset{\text{Acetaldehyde}}{2CH_3CHO} \xrightleftharpoons{\text{dil NaOH}} \underset{\text{3-hydroxy butanal(aldol)}}{CH_3-\underset{\displaystyle OH}{\underset{|}{CH}}-CH_2-CHO}$

b) Esterification:

Carboxylic acid reacts with alcohol in the presence of a 'strong acid catalyst' to give **Ester.**

Ex: $\underset{\text{Ethanoicacid}}{CH_3COOH} + \underset{\text{Ethylalcohol}}{C_2H_5OH} \xrightleftharpoons{Conc.H_2SO_4} \underset{\text{(Ester)}}{CH_3COOC_2H_5} + \underset{\text{water}}{H_2O}$

c) HVZ Reaction:

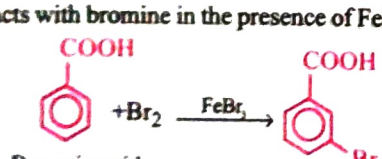
Carboxylic Acid having atleast one α- hydrogen is treated with Halogens (Br (or) Cl) in the presence of red phosphorus undergo Halogenation to form α-Halo Carboxylic Acid.

Ex: $\underset{\text{Carboxylic acid}}{R-CH_2-COOH} \xrightarrow[(ii)\ H_2O]{(i)\ x_2/\text{Redphosphorus}} \underset{\alpha\text{–halo carboxylic acid}}{R-\underset{\displaystyle X}{\underset{|}{CH}}-COOH} + HX$

d) Bromination of Benzoic acid:

Benzoic acid reacts with bromine in the presence of $FeBr_3$ to form **m-bromo benzoic acid.**

Ex:



Benzoic acid m-bromobenzoic acid

QUICK MEMORY BYTES & MNEMONICS

1. **Aldol:** 2 α Aldehydes → β-hydroxy aldehyde (Already read in Jr. Inter)
2. **Esterification** : Acid + Alcohol = Ester + Water
3. **HVZCAH HCA** [Hello Vizag wala! Come And Have Hyderabad Chicken Biryani at Ameerpet]
4. **Bromination:** Ben - Bro with $FeBr_3$ forms Bromo Benzoic acid.

9. Describe the following: [AP 19]

i) Acetylation ii) Cannizzaro reaction

iii) Cross aldol condensation [AP 15,19] iv) Decarboxylation

A: i) Acetylation:

Alcohols when treated with an acyl chloride (acetyl) in the presence of **pyridine** give esters.

Ex: $C_2H_5OH + CH_3-\overset{\overset{O}{\|}}{C}-Cl \xrightarrow{Pyridine} CH_3-\overset{\overset{O}{\|}}{C}-OC_2H_5 + HCl$

(Ethyl alcohol) (Acetyl chloride) (Ethyl acetate)

ii) Cannizzaro Reaction: Formaldehyde in the presence of conc. NaOH undergo self oxidation and reduction to form Methanol and Sodium Formate.

Ex: $\underset{\text{Formaldehyde}}{2HCHO} \xrightarrow[\text{self.Oxdn\&Redn}]{\text{conc.NaOH}} \underset{\text{Methanol}}{CH_3OH} + \underset{\text{Sodium Formate}}{HCOONa}$

iii) Cross aldol condensation: It is the condensation of two different carbonyl compounds (one of which must have one α-hydrogen) in the presence of a strong base.

CH_3CHO + CH_3CH_2CHO →

Acetaldehyde (Ethanal) (A) + Propionaldehyde (Propanal) (B)

→ $CH_3-CH=CH-CHO$ But-2-enal (AA)

+ $CH_3-CH_2-CH=\underset{\underset{CH_3}{|}}{C}-CHO$ 2-MethylPent-2-enal (BB)

(Self-Aldol Products)

→ $CH_3-CH=\underset{\underset{CH_3}{|}}{C}-CHO$ 2-Methylbut-2-enal (AB)

+ $CH_3-CH_2-CH=CH-CHO$ Pent-2-enal (BA)

(Cross Aldol Products)

iv) Decarboxylation : Sodium Ethanoate is heated with **Sodalime** to form **Methane.**

Ex: $\underset{\text{Sodium Ethanoate}}{CH_3COONa} + \underset{\text{Soda lime}}{NaOH} \xrightarrow[\Delta]{CaO} \underset{\text{Methane}}{CH_4} + \underset{\text{Sodium carbonate}}{Na_2CO_3}$

QUICK MEMORY BYTES & MNEMONICS

- **Acetylation** → Mixing of acetyl in alcohol gives ester.
- **Cannizzaro Reaction :** [FM Songs Free]

  Formaldehyde formal గా NaOH వద్దకు వస్తే చాలు చల్లటి Methanol & Sodium Formate form అవుతాయి.
- **Cross Aldol** → CROSS - Carbonyl compounds React with each other,

  One must has α-H Strong base (NaOH) → Successful product.
- **Decarboxylation (Removing CO_2) : DSESM** [Dear SESMa సర్]

10. How do you distinguish the following pairs of compounds?

i) Propanal and propanone **ii) Acetophenone and benzophenone**

iii) Phenol and benzoic acid **iv) Pentan-2-one and Pentan-3-one**

A: i) Propanone and Propanal :

Propanone undergoes **idoform reaction**, but Propanal does not undergo idoform reaction.

$$\underset{\text{Propanone}}{CH_3COCH_3} + 3I_2 + 4NaOH \xrightarrow{\Delta} \underset{\text{Idoform}}{CHI_3} + CH_3COONa + 3H_2O + 3NaI$$

ii) Acetophenone and benzophenone:

Acetophenone undergoes **idoform reaction,** but Bezophenone does not undergo it.

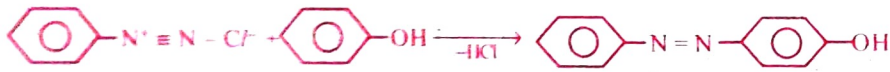
Reason: Acetophenone is a methyl ketone ($C_6H_5COCH_3$)

while benzophenone ($C_6H_5COC_6H_5$) is a diphenyl ketone.

$$\underset{\text{Acetophenone}}{C_6H_5-CO-CH_3} + 3I_2 + 4NaOH \xrightarrow{\Delta} \underset{\text{Idoform}}{CHI_3} + C_6H_5COONa + 3H_2O + 3NaI$$

iii) Phenol and benzoic acid:

- Phenol gives coupling reactions with diazonium compounds, while benzoic acid does not.
- Phenol reacts with benzenediazonium chloride ($C_6H_5N_2Cl$) to give azodyes, but benzoic acid does not react with diazonium salts.
- Phenol gives reddish colour with neutral $FeCl_3$ while benzoic acid gives buff-colour.



Benzene diazonium chloride Phenol P-hydroxy azobenzene (orange dye)

iv) Pentan-2-one and Pentan-3-one:

Pentan-2-one undergoes **idoform reaction**, but pentan-3-one does not give this reaction.

Reason: Pentan-2-one is a methyl ketone.

$$\underset{\text{Pentan-2-one}}{CH_3-\overset{\overset{O}{\|}}{C}-CH_2-CH_2-CH_3} + 3I_2 + 4NaOH \longrightarrow \underset{\text{Idoform}}{CHI_3} + C_3H_7COONa + 3H_2O + 3NaI$$

QUICK MEMORY BYTES

- In (i), (ii), (iv) the first term compounds under go **idoform test** and the second ones do not.

Note: In (i), the terms Propanal and propanone are interchanged in the side heading.

11. Explain the following terms. Give an example of the reaction in each case.
i) Cyanohydrin ii) Acetal
iii) Semicarbazone iv)Aldol

A: i) Cyanohydrin is carbonyl compound + HCN

Hydrogen cyanide (HCN) adds to aldehydes and ketones to form cyanohydrin.

Ex: $HCN + CH_3-\underset{\underset{H}{|}}{C}=O+ \longrightarrow CH_3-\underset{\underset{CN}{|}}{\overset{\overset{H}{|}}{C}}-OH$

Acetaldehyde — acetaldehyde cyanohydrin

ii) Acetal is Aldehyde + 2 alcohol molecules

Aldehydes react with alcohols in the presence of dry hydrogen chloride to form acetals.

Ex: $R-\underset{\underset{\text{Aldehyde}}{}}{\overset{\overset{H}{|}}{C}}=O+ \underset{\text{Alcohol}}{R-OH} \xrightarrow[\text{HCl gas}]{\text{dry}} R-\underset{\underset{OR}{|}}{\overset{\overset{H}{|}}{C}}-OH \xrightarrow[\text{dry HCl}]{+ROH} R-\underset{\underset{OR}{|}}{\overset{\overset{H}{|}}{C}}-OR + H_2O$

Semi Acetal — Acetal

iii) Semicarbozone is a carbonyl compound + semicarbazide

Aldehydes and ketones react with semicarbazide ($NH_2NHCONH_2$) to form semicarbazone.

Ex: $CH_3-\underset{\underset{\text{Acetaldehyde}}{}}{\overset{\overset{H}{|}}{C}}=O+ \underset{\text{Semicarbazide}}{H_2N-NH-CONH_2} \longrightarrow \underset{\text{Ethanal semi carbazone}}{CH_3-\overset{\overset{H}{|}}{C}=NNHCONH_2} + H_2O$

iv)Aldol is a β-hydroxy aldehyde (or) β-hydroxy ketone formed by condensation.

Aldol reaction: Aldehydes and ketones having at least one α-hydrogen undergo a reaction in the presence of dilute alkali as catalyst to form β-hydroxy aldehyde (aldol) or β-hydroxy ketones (ketol) respectively.

Ex: $\underset{\text{Acetaldehyde}}{2CH_3CHO} \xrightleftharpoons{\text{dil NaOH}} \underset{\underset{\text{3-hydroxy butanal(aldol)}}{}}{CH_3-\underset{\underset{OH}{|}}{CH}-CH_2-CHO}$

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Less Reading, More Scoring!

ONE MARK QUESTIONS

OMQ

SECTION-A

Read in Minutes, Remember for Months!

TARGET MARKS:≥5

OMQ CHAPTERS

9 MARKS

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- Being Bullet Q Book, to make it short only 5 selective OMQ [5 MCQ, 5 FIB, 5 OWQ] are taken.

OMQ – Hints & Solutions

1. SOLUTIONS

I. MULTIPLE CHOICE Q'S (MCQ)

1. **Which of the following concentration term is independent of temperature?** [2]

   1. Molarity ✓**2. Molality**
   3. Mass-Volume percentage 4. Volume –Volume percentage

2. **Which of the following is an example of an ideal solution?** [3]

   1. $CS_2+CH_3COCH_3$ 2. $C_6H_5OH+C_6H_5NH_2$
   ✓**3.$C_6H_6+C_6H_5CH_3$** 4. $C_2H_5OH+CH_3COCH_3$

3. **The mole fraction of benzene in solution containing 30% by mass in CCl_4 is** [3]

   1. 0.5611 2. 0.9102 ✓**3. 0.4586** 4. 0.0214

4. **Which of the following solutions has the highest boiling point?** [2]

   1. 0.1 M KNO_3 ✓**2. 0.1 M Na_3PO_4** 3. 0.1 M $BaCl_2$ 4. 0.1 M K_2SO_4

5. **Which law explains the solubility of gases in liquids ?** [3]

   1. Boyle's law 2. Charles' law ✓**3.Henry's law** 4. Raoult's law

II. FILL IN THE BLANKS (FIB)

6. The units of boiling point elevation constant K_b are **K kg mol^{-1}** .
7. Solubility of a gas in liquid **increases** with increase in pressure.
8. For an ideal solution, $\Delta_{mix}H$ is equal to **Zero**.
9. Desalination plants work on the basis of the phenomenon called **reverse osmosis**.
10. Human blood is isotonic with **0.9% (w/v) NaCl**. solution (or) Saline. **[BMP]**

III. ONE WORD ANSWER Q'S (OWQ)

11. **What is the term used for a solution that does not follow Raoult's law?**

A: **Non-ideal solution** does not follow the Raoult's law.

---

12. **In which type of a solution, dynamic equilibrium exists between the dissolved solute and undissolved solute?**

A: In **saturated solution** dynamic equilibrium exists.

---

13. **What is the name given to binary liquid mixtures which have same composition in liquid and vapour phase and boil at a constant temperature?**

A: **Azeotropic mixtures** have same composition in liquid and vapour phase .

---

14. **In the expression $\Delta T_f = K_f m$, what is the name given to the constant K_f?**

A: K_f is called **Cryoscopic constant** (or) Molal depression constant.

---

15. **What is the name given to the solutions, when they have same osmotic pressure at a given temperature?**

A: **Isotonic solutions** have same osmotic pressure at a given temperature.

2. ELECTRO CHEMISTRY

I. MCQ

1. **During electrolytic refining of metals the refined metal is obtained at (on) the** [2]
   1) anode ✓**2)cathode** 3) surface of the electrolyte 4) Both 1 & 2

2. **The standard cell potential of a Daniel cell is** [4]
   1) 1.6 V 2) 1.9 V 3) 0.6 V ✓**4) 1.1V**

3. **In a dry cell which of the following is the electrolyte?** [1]
   ✓**1) $NH_4Cl + ZnCl_2$** 2) H_2SO_4+NaOH 3) NaOH+KOH 4) NaOH+HCl

4. **Which of the following is correct about cathode in an electro chemical cell?** [3]
   1) Oxidation occurs 2) Electrons loss occur ✓**3) Reduction occurs** 4) No reaction occurs

5. **A lead storage battery is mainly used in** [4]
   1) watches 2) automobiles 3) inverters ✓**4) Both 2 & 3**

II. FIB

6. 1 Faraday value is equal to **96487 (or) 96500** C mol^{-1}
7. Corrosion of iron is mainly due to **water** and **air**
8. The units of molar conductivity are **$S cm^2 mol^{-1}$ (or) $S\ m^2 mol^{-1}$**
9. Lead storage battery is an example of **secondary** cell. **[BMP]**
10. In Daniell cell the oxidation half reaction occurs at **anode**

III. OWQ

11. **Write an expression showing the relation between E°_{cell} and Gibbs energy change.**

A: $\Delta G^\circ = -nFE^\circ_{(cell)}$ is the relation between E°_{cell} and Gibbs energy change.

Here n is the no. of electrons involved in oxidation (or) reduction, E°_{cell} = standard EMF

---

12. **Give an example of an inert electrode.**

A: **Gold / Platinum / Graphite** are the examples of an inert electrode.

---

13. **What is the charge of an electron in coulombs?**

A: Charge of an electron in coulombs is **-1.602×10^{-19} coulombs**

---

14. **In a galvanic cell chemical energy is converted to electrical energy by which reaction?**

A: In a galvanic cell chemical energy is converted to electrical energy by **Spontaneous redox reaction.**

---

15. **What happens to the colour of a solution when a zinc wire is dipped in $CuSO_4$ solution?**

A: **Blue colour** of $CuSO_4$ **disappears** when a zinc wire is dipped in $CuSO_4$ solution.

3. CHEMICAL KINETICS

I. MCQ

1. **The unit of rate of a chemical reaction is generally expressed as:** [1]
   ✓**1.mol L^{-1} s^{-1}** 2. mol L^{-1} 3. mol 4. s^{-1}

2. **The rate law for a given reaction is determined from** [3]
   1. balanced chemical equation 2. thermodynamic data
   ✓**3.experimental data** 4. catalyst used

3. **The unit of rate constant of a first order reaction is** [1]
   ✓**1. s^{-1}** 2. L $mol^{-1}s^{-1}$ 3. mol L^{-1} s^{-1} 4. mol s^{-1}

4. **The half-life of a first-order reaction is given by** [3]
   1. $t_{1/2} = 1/k$ 2. $t_{1/2} = [A]_o/2k$ ✓**3. $t_{1/2} = 0.693/k$** 4. $t_{1/2} = k/0.693$

5. **Catalyst increases the rate of a reaction by** [1]
   ✓**1. decreasing activation energy** 2. increasing temperature
   3. increasing activation energy 4. increasing concentration

II. FIB

6. The sum of powers of the concentration terms in the rate law expression is called **order** of that chemical reaction.
7. The overall rate of the reaction is controlled by the slowest step of the reaction and is called the **rate determining step**
8. The number of collisions per second per unit volume of the reaction mixture is known as **collision frequency** **[BMP]**
9. $k = A\, e^{-Ea/RT}$ where A is called **Arrhenius factor or frequency factor**
10. For a first order reaction the following graph is obtained.
    From this graph slope is equal to **k/2.303**

III. OWQ

11. **What is the molecularity of the elementary reaction: A + B → Products?**

A: Molecularity of the given elementary reaction is **2 (two)**

---

12. **Which reactions takes place in one step?**

A: **Elementary reactions** takes place in one step.

---

13. **In a zero - order reaction, if initial concentration of reactant is 0.4 mol/L and rate constant is 0.1 mol $L^{-1}s^{-1}$, what is its $t_{1/2}$?**

A: Given that $[A_0]$=0.4 m/L, K=0.1 m.L^{-1} s^{-1} · For Zero order reaction $t_{1/2} = \frac{[A_0]}{2K} = \frac{0.4}{2\times0.1} = 2$ s

---

14. **What is the unit of rate constant for a zero-order reaction?**

A: Unit of rate constant for a zero-order reaction is **mol L^{-1} s^{-1}**

---

15. **Collision theory is based on which theory of gases?**

A: Collision theory is based on **Kinetic molecular theory of gases.**

4. THE d- AND f- BLOCK ELEMENTS

I. MCQ

1. The correct electronic configuration of Cr is [2]
   1. $[Ar]3d^4\ 4s^2$ **2. $[Ar]3d^5\ 4s^1$** 3. $[Ar]3d^6\ 4s^o$ 4. $[Ar]3d^3\ 4s^2$
2. Which of the following is not regarded as a transition element? [4]
   1. Cr 2. Mn 3. Fe **4. Zn**
3. The common oxidation state of d-block elements is [2]
   1. +3 **2. +2** 3. +1 4. +4
4. Which of the following set of elements represent coinage metals? [1]
   1. Cu, Ag, Au 2. Cu, Fe, Zn 3. Ag, Fe, Cr 4. Au, Fe, Zn
5. Transition metals form large number of complex compounds. This is due to [4]
   1. small size 2. high nuclear charge
   3. presence of vacant d-orbitals **4. All the above**

II. FIB

6. The atomic radii of 5d series of elements is same as that of **4d series** series of elements.
7. In aqueous solution the colour of the Cu^{2+} ion is **blue**
8. Ce^{4+} is a strong **oxidizing** agent.
9. Most of the radioactive elements are present in **5f series / f-block** series/ block of periodic table.
10. The common oxidation state of lanthanides is **+3**

III. OWQ

11. Name the transition element which does not exhibit variable oxidation states?

A: **Sc (Scandium)**does not exhibit variable oxidation states.

---

12. Which substances are attracted very strongly by applied magnetic field?

A: **Ferro Magnetic substances** are attracted **very strongly**.

---

13. What is the other name of f-block elements?

A: **Inner transition elements** is the other name of **f-block elements.**

---

14. What is the catalyst used in the manufacture of high-density polythene?

A: **Zeigler - Natta Catalyst** is used in the manufacture of **high-density polythene.**

---

15. Which metals are involved in the formation of an alloy called Brass?

A: **Cu and Zn** are used in the formation of **Brass.**

5. CO-ORDINATION COMPOUNDS

I. MCQ

1. **Which of the following coordination compound contains cobalt metal?[BMP]** [3]

   1. Chlorophyll 2. Hemoglobin ✓**3. Vitamin B_{12}** 4. Myoglobin

2. **Which of the following is not a double salt?** [3]

   1. $FeSO_4.(NH_4)_2SO_4.6H_2O$ 2. $KCl.MgCl_2 6H_2O$

   ✓**3. $PtCl_4.2HCl$** 4. $KAl(SO_4)_2.12H_2O$

3. **Lewis acid in the complex ion $[CoCl(NH_3)_5]^{2+}$ is** [4]

   1. Cl^- 2. NH_3 3. Co^{2+} ✓**4. Co^{3+}**

4. **Which ligand gives a stable complex?** [4]

   1. SCN^- 2. Cl^- 3. H_2O ✓ **4. Ethylene diamine(en)**

5. **Which complex shows geometrical, optical and also structural isomerism?** [2]

   1. $[Co(en)_3]Cl_3$ ✓**2. $[CoCl_2(en)_2]Br$** 3. $[Co(en)(NH_3)_2Cl_2]Cl$ 4. $[PdCl_2(NH_3)_2$

II. FIB

6. According to Werner's theory of coordination compounds primary valences of metal ion are satisfied by **negative ions**

7. Mohr's salt formula is **$FeSO_4(NH_4)_2SO_4.6H_2O$**

8. The coordination sphere in potassium ferrocyanide complex is **$[Fe(CN)_6]^{4-}$**

9. The IUPAC name of $K_2[PdCl_4]$ is **Potassium tetrachloridopalladate (II)**

10. M-CO bond in metal carbonyls is very strong due to **Synergic** effect. (or) π-back bonding.

III. OWQ

11. **Which platinum complex can effectively inhibit the growth of tumors?**

A: **Cis-platin** can effectively inhibit the growth of tumors. **Ex:** Cis $[Pt(NH_3)_2Cl_2]$

---

12. **Facial and Meridional isomers come under which category of isomerism?**

A: Facial and Meridional isomers show **Geometrical isomerism.**

---

13. **What is the geometry of $Fe(CO)_5$ complex?**

A: Geometry of $Fe(CO)_5$ is **Trigonal bipyramidal.**

---

14. **Which complex ion is formed when undecomposed AgBr of the film dissolves in hypo solution.**

A: **$[Ag(S_2O_3)_2]^{3-}$** is formed when AgBr film dissolves in hypo solution.

---

15. **How many unpaired electrons are there in $[NiCl_4]^{2-}$ ion?**

A: **Two unpaired electrons** are present in $[NiCl_4]^{2-}$ ion.

Ni^{2+} : $[Ar]4s^0 3d^8$

6. HALOALKANES & HALOARENES

I. MCQ

1. What is the general formula for haloalkanes? (X=halogen atom, n=1, 2, 3...) [2]
   1) $C_nH_{2n}X$ ✓2) $C_nH_{2n+1}X$ 3) $C_nH_{2n-1}X$ 4) $C_nH_{2n-3}X$
2. A primary alkyl halide would prefer to undergo [2]
   1) S_N1 reaction ✓2) S_N2 reaction 3) α–elimination 4) racemization
3. Which of the following is the suitable medium for preparing Grignard reagents? **[BMP]** [2]
   1) Dry acetone ✓2) Dry ether 3) Concentrated HCl 4) Alcoholic KOH
4. Which of the following alkyl halide will undergo S_N1 reaction most readily? [4]
   1) $(CH_3)_3C—F$ 2) $(CH_3)_3C—Cl$ 3) $(CH_3)_3C—Br$ ✓4) $(CH_3)_3C—I$
5. Chlorobenzene is [1]
   ✓1) less reactive than benzyl chloride 2) more reactive than ethyl bromide
   3) nearly as reactive as methyl chloride 4) more reactive than isopropyl chloride

II. FIB

6. Our body produces an iodine containing hormone, thyroxine, the deficiency of which causes a disease called **Goiter**
7. Alkyl iodides are often prepared by the reaction of alkyl chlorides/ bromides with NaI in dry acetone. This reaction is known as **Finkelstein** reaction.
8. Groups like cyanides and nitrites possess two nucleophilic centers and are called **ambident nucleophiles**
9. The compounds which rotate the plane polarized light are called **Optically active or Chiral**
10. The compounds which are obtained by the reaction of haloalkanes with magnesium metal in dry ether are called **Grignard Reagents**

III. OWQ

11. Preservation of the spatial arrangement of bonds to an asymmetric center during a chemical reaction or transformation is called

A: Preservation of the spatial arrangement of bonds is called **Retention of Configuration.**

12. Diazonium salt with cuprous chloride or cuprous bromide results in the replacement of the diazonium group by –Cl or –Br. This reaction is called

A: **Sandmeyer reaction** is the replacement of the diazonium group by –Cl or –Br.

13. Synthesis of alkyl fluorides by heating an alkyl chloride/bromide in the presence of a metallic fluoride such as AgF is named as

A: **Swarts reaction** is the synthesis of alkyl fluorides by heating an alkyl chloride/bromide.

14. If the compound rotates the plane of plane polarized light to the right, i.e., clockwise direction, it is called

A: **Dextro rotatory** compound rotates the plane polarized light to the right.

15. The stereoisomers which are related to each other as non- superimposable mirror images are called

A: **Enantiomers** are non- superimposable mirror images

7.ALCOHOLS, PHENOLS & ETHERS

I. MCQ

1. **Glycerol is a: [BMP]** [4]
   1) dihydric alcohol 2) monohydric alcohol 3) hexahydric alcohol ✓**4) trihydric alcohol**
2. **Which of the following can work as a dehydrating agent for alcohols?** [4]
   1) Conc. H_2SO_4 2) Anhydrous Al_2O_3 3) H_3PO_4 ✓**4) All of these**
3. **Which of the following shows strong hydrogen bonding?** [3]
   1) Ethyl amine 2) Ammonia ✓**3)Ethyl alcohol** 4) Diethyl ether
4. **When phenol is treated with excess bromine water it gives** [4]
   1) m-bromophenol 2) o- and p-bromophenol
   3) 2,4-dibromophenol ✓**4) 2,4,6-tribromophenol**
5. **Which of the following is formed when phenol is exposed to air?** [1]
   ✓**1) Benzoquinone** 2) Benzyl quinone 3) Acetophenone 4) Benzene

II. FIB

6. Isopropyl benzene is commonly called **cumene.**
7. Boiling points of alcohols and phenols are higher in comparison to other classes of compounds due to **intermolecular hydrogen bonding**
8. Alcohols and phenols react with carboxylic acids to form **esters**
9. Lucas reagent is **Conc. HCl/$ZnCl_2$** (anhydrous)
10. On treating phenol with chloroform in the presence of sodium hydroxide, salicylaldehyde is formed. This reaction is known as **Reimer-Tiemann reaction**

III. OWQ

11. **Phenol on heating with zinc dust gives a compound X. What is X?**

A: Phenol on heating with zinc dust gives **Benzene (C_6H_6)**

---

12. **When an alkyl halide is allowed to react with sodium alkoxide, ether is formed. What is the name given to this reaction?**

A: **Williamson synthesis:** Alkyl halide + sodium alkoxide→ Ether
(R'–X + R–O–Na → R–O–R' + NaX)

---

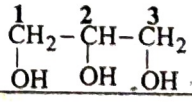
13. **In the preparation of phenol from cumene, what is the byproduct obtained?**

A: The byproduct obtained in the preparation of phenol from cumene is **Acetone (CH_3COCH_3)**

---

14. **What is the IUPAC name of glycerol?**

A: IUPAC name of glycerol is **Propane-1,2,3-triol.**



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15. **Which class of compounds are obtained when alkenes react with water in the presence of acid as catalyst?**

A: Alkenes when react with water in the presence of acid as catalyst gives **Alcohols.**

$$CH_2=CH_2+H_2O \xrightarrow{H^+} CH_3CH_2OH$$

8. ALDEHYDES, KETONES & CABOXYLIC ACIDS

I. MCQ

1. With which of the following reagents both ethanal and propanone react? [4]
   1) Tollens' 2) Schiff's 3) Fehling ✓**4) Grignard**

2. Cannizzaro reaction is not given by [1]
   ✓**1) CH_3CHO** 2) PhCHO 3) HCHO 4) $(CH_3)_3C–CHO$

3. Which of the following is a dicarboxylic acid? [1]
   ✓**1) oxalic acid** 2) benzoic acid 3) salicylic acid 4) picric acid

4. Common reagent used to detect aldehydes and ketones is [2]
   1) Lucas reagent ✓**2) 2, 4-DNP** 3) Baeyer's reagent 4) Tollens' reagent

5. Choose the weakest acid among the following [4]
   1) FCH_2COOH 2) Cl_2CCOOH 3) CH_3COOH ✓**4) CH_3CH_2COOH**

II. FIB

6. IUPAC name of formaldehyde is **methanal (HCHO)**
7. Chromium trioxide (CrO_3) in acidic (H_2SO_4) media is called **Jones** reagent
8. Tollens' reagent is **freshly prepared ammonical silver nitrate solution**
9. Carboxylic acids having an α-hydrogen are halogenated at the α-position on treatment with chlorine or bromine in presence of small amount of phosphorus to give α- halocarboxylic acids. This reaction is known as **Hell-Volhard-Zelinsky reaction (HVZ reaction)**
10. IUPAC name of Mesityl oxide is **4-methylpent-3-en-2-one**

III. OWQ

11. What is name of the product formed when benzene is reacted with CO and HCl in the presence of anhydrous $AlCl_3$?
    A: **Benzaldehyde** is obtained when benzene is reacted with CO and HCl.

---

12. What is Rochelle salt? **[BMP]**
    A: **Rochelle salt:** Sodium potassium tartarate +Sodiumhydroxide
    (A double salt of Tartaric acid)

---

13. Which acid is used as solvent in food industry as vinegar?
    A: Vinegar: **Ethanoic acid (CH_3COOH or acetic acid).**

---

14. Aldehydes and ketones undergo what type of addition reactions?
    A: Aldehydes and ketones undergo **Nucleophilic addition reactions.**

---

15. Phthalic acid on reaction with ammonia followed by strong heating gives a compound 'X' along with elimination of NH_3. What is the name of that compound 'X'?
    A: **phthalimide**

9. AMINES

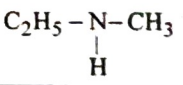
1. **Which of the following amines cannot be prepared by Gabriel phthalimide reaction?**
   1) Benzylamine ✓**2) Aniline** 3) Ethylamine 4) Methylamine [2]
2. **Classification of amines into 1°, 2°, 3° is based on the** [3]
   1) number of amino groups 2) nature of carbon atom
   ✓**3) degree of substitution at nitrogen** 4) degree of unsaturation
3. **The number of primary amines possible for a compound with molecular formula $C_4H_{11}N$ are**
   1) 1 2) 2 3) 3 ✓**4) 4** [4]
4. **Which among the following has the lowest pK_b value in aqueous medium?** [3]
   1) NH_3 2) CH_3NH_2 ✓**3) $(CH_3)_2NH$** 4) $(CH_3)_3N$
5. **Aniline reacts with the following to give benzanilide** [3]
   1) Benzyl chloride 2) Benzal chloride ✓**3) Benzoyl chloride** 4) Benzo tri chloride

II. FIB

6. The shape of trimethyl amine is **pyramidal**
7. Benzenesulphonyl chloride ($C_6H_5SO_2Cl$), is also known as **Hinsberg's reagent**
8. Carbylamine reaction or isocyanide test is used for the identification of **primary amines or 1° amines**
9. The conversion of primary aromatic amines into diazonium salts is known as **diazotisation**
10. Coupling reaction of aryl diazonium salts with phenols or aryl amines gives rise to the formation of **azodyes**

III. OWQ

11. **What is the IUPAC name of ethyl methyl amine?** **[BMP]**



A: IUPAC name of ethyl methyl amine is **N–methyl ethanamine.**

---

12. **The process of cleavage of the C–X bond of alkyl halides by ammonia molecule is known as**

A: **Ammonolysis** is the cleavage of the C–X bond of alkyl halides by ammonia molecule.

---

13. **Aniline and other arylamines are usually colourless but get coloured on storage why?**

A: Aniline and other arylamines get coloured on storage due to **Atmospheric oxidation.**

---

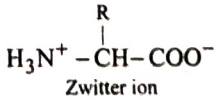
14. **Tertiary amines like trimethylamine are used as**

A: Tertiary amines are used to **attract flies or insects.**

---

15. **A molecule that has both positive and negative regions of charge is called**

A: **Zwitter ion** is the molecule that has both positive and negative regions of charge.



Zwitter ion

10. BIOMOLECULES

I. MCQ

1. **Which of the following is not an example of a hormone?** [3]
   1) Insulin 2) Estrogen ✓**3) Pyridoxine** 4) Androgen
2. **Which of the following produces only glucose units on complete hydrolysis?** [2]
   1) Lactose ✓**2) Maltose** 3) Raffinose 4) Sucrose
3. **Which compound inverts its optical rotation on hydrolysis?** [2]
   1) Glucose ✓**2) Sucrose** 3) Maltose 4) Fructose
4. **Which amino acid is basic in nature?** [2]
   1) Glutamic acid ✓**2) Lysine** 3) Tyrosine 4) Leucine
5. **Which vitamin deficiency causes convulsions?** [4]
   1) Vitamin C 2) Vitamin D 3) Vitamin E ✓**4) Vitamin B_6**

II. FIB

6. In sucrose molecule, glucose and fructose are connected between **C_1 and C_2** atoms respectively.
7. Curdling of milk is an example of **denaturation of protein.**
8. Complementary base for the adenine present in DNA is **thymine.**
9. Rickets disease is caused by the deficiency of vitamin **D** .
10. Hormones are molecules that act as **intercellular messengers.**

III. OWQ

11. **Which linkage connects two monosaccharide units?** **[BMP]**

A: **Glycosidic linkage** connects two monosaccharide units.

---

12. **Which non-essential amino acid is optically inactive?**

A: **Glycine** is optically inactive non-essential amino acid .

---

13. **What type of bonding helps in stabilizing the α-helix structure of proteins?**

A: **Hydrogen bonding** helps in stabilizing the α-helix structure of proteins

---

14. **Name the disease that occurs due to deficiency of ascorbic acid.**

A: **Scurvy (bleeding gums)** occurs due to deficiency of ascorbic acid.

---

15. **Increased level of which hormone causes hyperthyroidism ?**

A: Increased level of **Thyroxine** causes hyperthyroidism.

SHINING STARS

STAR QUESTIONS PLUS

SOLUTIONS

VSAQ

175. What is van't Hoff's factor?

How is it related to 'α' in the case of a binary electrolyte (1:1)?

A: **Van't Hoff's factor(i)**: It is the ratio of the experimental value of the colligative property to the calculated value of the colligative property.

Incase of a binary electrolyte (1:1), the relation between α and i is as follows:

$i = \frac{1+(n-1)\alpha}{1}$. Degree of dissociation, $\alpha = \frac{i-1}{n-1}$

The Van't Hoff factor i accounts for dissociation or association of solute.

SAQ

176. If the osmotic pressure of glucose solution is 1.52 bar at 300K. What would be its concentration if R=0.083 L bar mol^{-1} K^{-1} ?

A: Given osmotic pressure Π=1.52, R = 0.083 L bar mol^{-1} K^{-1}, T= 300K, Concentration C = ?

Formula: $\Pi = CRT \Rightarrow C = \frac{\Pi}{RT} = \frac{1.52}{0.083 \times 300} = 0.061\,M$

---

177. How is molar mass related to the depression in freezing point of a solution?

A: Depression in Freezing point $\Delta T_f = K_f \times m$, where m= molality of the solution

$$\Rightarrow \Delta T_f = K_f \times \frac{w_s}{M} \times \frac{1000}{w_o(g)} \Rightarrow M_s = \frac{K_f \times w_s \times 1000}{\Delta T_f \times w_o}$$

Here, M_s = molar mass of solute, K_f = molal depression constant

w_s = Mass of solute in grams, ΔT_f = depression in freezing point, w_o = Mass of solvent in grams

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ELECTRO CHEMISTRY (SAQ)

178. On what factors the electrical conductance of an aqueous solution of electrolyte depends?

A: The conductance of the solution of an electrolyte depends on the following factors

i) **Nature of the electrolyte:** Strong electrolytes ionize almost completely in the solution and hence conduct electricity to a large extent. Whereas weak electrolytes ionise to a small extent and hence conduct electricity to a small extent.

ii) **Size of the ions :** Greater the size of the ions or greater the solvation of the ions, lesser is the conductance.

iii) **Nature of the solvent and its viscosity:** Electrolytes ionise more in a polar solvent. Hence, greater the polarity of the solvent, greater is the ionization and hence greater is the electrical conductance. Also greater the viscosity of a solvent, lesser is the conductance.

iv) **Concentration of the solution:** Higher the concentration of the solution, lesser is the conductance. This is because in a weak electrolyte, the ionization is less whereas in a strong electrolyte, the interionic attractions are higher at higher concentration.
With dilution, incase of weak electrolytes ionization increases and incase of strong electrolytes, interionic attraction decreases and hence conductance increases.

v) **Temperature:** On increasing the temperature, the dissociation of electrolyte increases and hence the conductance increases.

---

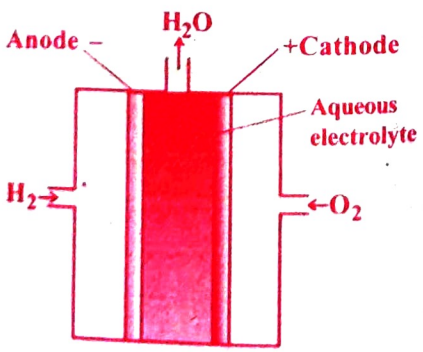
179. What are fuel cells? How are they different from galvanic cells?
Give the construction of H_2-O_2 fuel cell.

A: **A fuel cell** is a device that converts the energy from combustion of fuel into electrical energy. The fuel cell like any other electrochemical cell has two electrodes and an electrolyte.
Oxidation of fuel occurs at the anode: Fuel → oxidation product + ne^-
The oxidant gets reduced at the cathode: Oxidation + ne^- → Reduction product.
Fuel cells are different from conventional galvanic cells in the following aspects:

a) In a fuel cell, reactants are fed into the cell and products are removed from the cell. These do not form integral part of cell as in a conventional galvanic cell.

b) Chemical energy is not stored in a fuel cell, whereas in a galvanic cell chemical energy is converted into electrical energy.

H_2–O_2 fuel cell: This fuel cell consists of porous carbon electrodes suspended in concentrated NaOH solution.



Catalysts like finely divided platinum or palladium is incorporated into the electrodes for increasing the rate of electrode reactions.

H_2 and O_2 gases are bubbled at the surface of the electrodes.

The electrode reactions:

At cathode: $O_{2(g)} + 2H_2O + 4e^- \rightarrow 4OH^-_{aq}$

At anode: $2H_{2(g)} + 4OH^-_{aq} \rightarrow 4H_2O_{(\ell)} + 4e^-$

Overall reaction: $2H_{2(g)} + O_{2(g)} \rightarrow 2H_2O_{(\ell)}$

180. Explain the variation of molar conductivity with the change in the concentration of the electrolyte. Give reasons.

A: Molar conductivity of electrolytes generally, increases with dilution.

(i) For Strong Electrolytes (NaCl, KCl):

Λ^o_m increases **slightly** with dilution.

The increase is **gradual and linear**

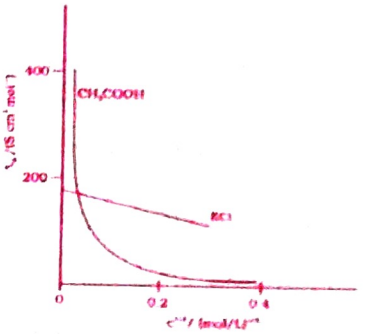
Reason:

Strong electrolytes are **almost completely dissociated** even at higher concentrations

On dilution:

Inter-ionic attractions decrease, Ion mobility increases

Hence, conductivity increases slightly



(ii) For Weak Electrolytes (CH_3COOH, NH_4OH):

Λ^o_m increases **sharply** with dilution

Reason: Weak electrolytes are **partially ionised**

On dilution: Degree of ionisation increases (Le Chatelier's principle)

More ions are produced. Hence, conductivity increases rapidly.

Graphical representation of the variation of Λ_m vs $\sqrt{C}$

For a strong electrolyte (KCl) and a weak electrolyte (CH_3COOH) is shown in the graph.

Strong electrolyte - straight line (small increase)

Weak electrolyte - curved line (steep increase)

---

181. Explain with a suitable example the relation between the Gibbs energy of chemical reaction (G) and the functioning of the electrochemical cell.

A: Electrical work done in one second in an electrochemical cell is equal to decrease in its Gibbs energy. Electrical work done = decrease in free energy.

In the case of an electrochemical cell, the electrical work done is equal to the electrical energy produced. This in turn is equal to the product of the quantity of electricity flowing in the circuit and the EMF of the cell.

Now, for every one mole of electrons transferred in the cell reaction, the quantity of electricity that flows through the cell is one faraday (1F = 96,500 Coulomb).

Hence, if n moles of electrons are transferred, in any cell reaction, the quantity of electricity flowing = nF Faraday

If E_{cell} represents the EMF of the cell, then electrical work done = nF E_{cell}

Hence $-\Delta_r G = nF\ E_{cell}$

For comparing different cells, standard cell potentials are used which are represented by E^0_{cell}

The corresponding free energy change is called the standard change of the reaction, represented by $\Delta_r G^0$.

Hence we can write $-\Delta G^0 = nF\ E^0_{cell}$

Thus for a cell reaction $Zn_{(s)} + Cu^{+2}_{(aq)} \rightarrow Zn^{+2}_{(aq)} + Cu_{(s)}$

$\Delta_r G^0 = -2F\ E^0_{cell}$ [$\because$ n=2 = no.of electrons involved]

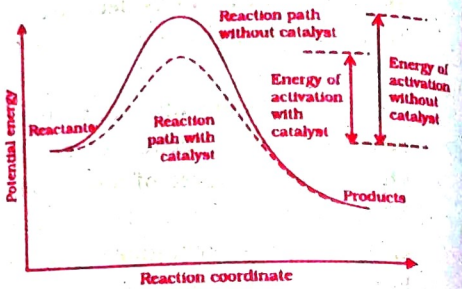
CHEMICAL KINETICS (SAQ)

182. Discuss the effect of catalyst on the kinetics of a chemical reaction with a suitable diagram.

A: A catalyst is a substance which increases the rate of a reaction without itself undergoing any permanent chemical change.

Ex: MnO_2 acts as catalyst in the following reaction to increase its rate .

$$2KClO_3 \xrightarrow{MnO_2} 2KCl + 3O_2$$



The action of the catalyst can be explained by intermediate complex theory. According to this theory, a catalyst participates in a chemical reaction by forming temporary bonds with the reactants . It results in the formation of an intermediate complex. This has a transitory existence and decomposes to yield products and the catalyst.

Catalyst takes the reaction in a new path having low activation energy. When the activation energy decreases, the rate of reaction increases.

---

183. Define average rate of a reaction. How is the rate of reaction expressed in terms of change in the concentration of reactants and products for the following reactions.

1) $2HI_{(g)} \rightarrow H_{2(g)} + I_{2(g)}$ 2) $Hg_{(l)} + Cl_{2(g)} \rightarrow HgCl_{2(s)}$

3) $5Br^-_{(aq)} + BrO^-_{3(aq)} + 6H^+_{(aq)} \rightarrow 3Br_{2(aq)} + 3H_2O_{(l)}$

A: **Average rate of a reaction** is defined as the rate of change of concentration per unit time.

$$\text{Average time} = \frac{\text{Change in concentration in give time}}{\text{Time taken}} = \frac{\Delta x}{\Delta t} \text{ (or) } -\frac{\Delta[R]}{\Delta t} = +\frac{\Delta[P]}{\Delta t}$$

1) $2HI_{(g)} \rightarrow H_{2(g)} + I_{2(g)}$ Average rate $= -\frac{1}{2}\frac{\Delta[HI]}{\Delta t} = +\frac{\Delta[H_2]}{\Delta t} = +\frac{\Delta[I_2]}{\Delta t}$

2) $Hg_{(l)} + Cl_{2(g)} \rightarrow HgCl_{2(s)}$ Average rate $= -\frac{\Delta[Hg]}{\Delta t} = -\frac{\Delta[Cl_2]}{\Delta t} = +\frac{\Delta[HgCl_2]}{\Delta t}$

3) $5Br^-_{(aq)} + BrO^-_{3(aq)} + 6H^+_{(aq)} \rightarrow 3Br_{2(aq)} + 3H_2O_{(l)}$

$$\text{Average rate} = -\frac{1}{5}\frac{\Delta[Br^-]}{\Delta t} = -\frac{\Delta[BrO_3^-]}{\Delta t} = -\frac{1}{6}\frac{\Delta[H^+]}{\Delta t} = +\frac{1}{3}\frac{\Delta[Br_2]}{\Delta t} = +\frac{1}{3}\frac{\Delta[H_2O]}{\Delta t}$$

184. What is rate equation? How is it obtained? Write the rate equations for

1) $2NO_{(g)} + O_{2(g)} \rightarrow 2NO_{2(g)}$ 2) $CHCl_3 + Cl_2 \rightarrow CCl_4 + HCl$

3) $CH_3COOC_2H_{5(l)} + H_2O_{(l)} \rightarrow CH_3COOH_{(aq)} + C_2H_5OH_{(aq)}$

A: **Rate equation (rate law)** is the expression in which reaction rate is given in terms of molar concentration of reactants with each term raised to some power, which may or may not be same as the stoichiometric coefficient of the reacting species, in a balanced chemical equation.

Rate equation is always obtained from the experimental data.

1) $2NO_{(g)} + O_{2(g)} \rightarrow 2NO_{2(g)}$; Rate = $k[NO]^2[O_2]$

2) $CHCl_3 + Cl_2 \rightarrow CCl_4 + HCl$; Rate = $[CHCl_3][Cl_2]^{1/2}$.

3) $CH_3COOC_2H_5 + H_2O \rightarrow CH_3COOH + C_2H_5OH$; Rate = $k[CH_3COOC_2H_5]$

LAQ

185. Give two examples for each of zero order and first order reactions. Write the equations for the rate of a reaction in terms of concentration changes of reactants and products for the following reactions.

a) $A_{(g)} + B_{(g)} \rightarrow C_{(g)} + D_{(g)}$ **b)** $A_{(g)} \rightarrow B_{(g)} + C_{(g)}$ **c)** $A_{(g)} + B_{(g)} \rightarrow C_{(g)}$

A: **Two examples for zero order reaction:**

i) $2NH_{3(g)} \xrightarrow[1130K]{Pt} N_{2(g)} + 3H_{2(g)}$ ii) $2HI_{(g)} \xrightarrow[\Delta]{Au} H_{2(g)} + I_{2(g)}$

Two examples for first order reaction:

i) $NH_4NO_{2(g)} \rightarrow N_{2(g)} + 2H_2O$ ii) $SO_2Cl_{2(g)} \rightarrow SO_{2(g)} + Cl_{2(g)}$

For the reaction (a), $A_{(g)} + B_{(g)} \rightarrow C_{(g)} + D_{(g)}$; Rate = $-\frac{\Delta[A]}{\Delta t} = -\frac{\Delta[B]}{\Delta t} = +\frac{\Delta[C]}{\Delta t} = +\frac{\Delta[D]}{\Delta t}$

For the reaction (b), $A_{(g)} \rightarrow B_{(g)} + C_{(g)}$; Rate = $-\frac{\Delta[A]}{\Delta t} = +\frac{\Delta[B]}{\Delta t} = +\frac{\Delta[C]}{\Delta t}$

For the reaction (c), $A_{(g)} + B_{(g)} \rightarrow C_{(g)}$; Rate = $-\frac{\Delta[A]}{\Delta t} = -\frac{\Delta[B]}{\Delta t} = +\frac{\Delta[C]}{\Delta t}$

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186. Discuss the effect of temperature on the rate of a reaction. Derive necessary equations in this context.

A: Effect of temperature on rate of reaction: In general for every 10°C rise of temperature the rate of a reaction is nearly doubled and in some cases it is tripled.

The ratio of specific rates that differ by 10°C is known as temperature coefficient.

$$\frac{K_{t+10^\circ C}}{K_{t^\circ C}} = 2 \text{ (or) } 3$$

Effect of temperature on rate of reaction is explained by Arrhenius in the form of an equation known as Arrhenius Equation.

1) Arrhenius equation: The arrhenius equation gives accurately the temperature dependence of the rate of a chemical reaction.

2) Rate constant $k = A\, e^{-Ea/RT}$.....(i)

Where A is frequency factor, E_a is activation energy

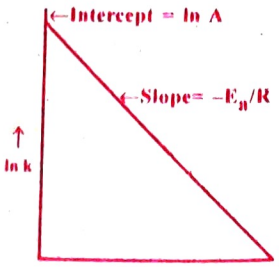
3) Taking natural logarithm of both sides of equation (i), we get $\ln k = \frac{-E_a}{RT} + \ln A$ .....(ii)

4) The plot of ln k vs 1/T is a straight line.

The slope of the line $= -\frac{E_a}{R}$

y-intercept = lnA.

Using these values, E_a and A can be calculated.



A plot between ln k and 1/T

5) At temperature T_1,

equation (ii) is $\ln k_1 = \frac{-E_a}{RT_1} + \ln A$ .....(iii)

6) At temperature T_2, equation (ii) is $\ln k_2 = \frac{-E_a}{RT_2} + \ln A$ .....(iv)

K_1 and K_2 are the values of rate constants at temperature T_1 and T_2 respectively.

7) Subtracting equation (iii) from (iv) we get, $\ln k_2 - \ln k_1 = \frac{E_a}{RT_1} - \frac{E_a}{RT_2}$ $[\because T_2 > T_1]$

8) $\ln \frac{k_2}{k_1} = \frac{E_a}{R}\left[\frac{1}{T_1} - \frac{1}{T_2}\right] \Rightarrow 2.303 \log \frac{k_2}{k_1} = \frac{E_a}{R}\left[\frac{1}{T_1} - \frac{1}{T_2}\right] \Rightarrow \log \frac{k_2}{k_1} = \frac{E_a}{2.303R}\left[\frac{1}{T_1} - \frac{1}{T_2}\right]$

d&f - BLOCK ELEMENTS (SAQ)

187. Explain why Ni and Fe exhibit zero oxidation state in $[Ni(CO)_4]$ and $[Fe(CO)_5]$ respectively.

A: In $[Ni(CO)_4]$ and $[Fe(CO)_5]$ the oxidation state of Ni and Fe is zero.
CO is a neutral ligand, hence the oxidation states of Ni and Fe are zero.
Low oxidation states are found when the complex compound has ligands capable of π-acceptor character in addition to the σ bonding.

---

188. Write down the electronic configuration of (i) Cr^{3+} (ii) Cu^{+} (iii) Co^{2+} (iv) Mn^{2+}

A:
i) E.C of Cr^{3+} : $[Ar]3d^3$
ii) E.C of Cu^{+} : $[Ar]3d^{10}$
iii) E.C of Co^{2+} : $[Ar]3d^7$
iv) E.C of Mn^{2+} : $[Ar]3d^5$

$Cr(24) : Cr^{3+} (21 = 18 + 3)$
$Cu(29) : Cu^{+} (28 = 18 + 10)$
$Co(27) : Co^{2+} (25 = 18 + 7)$
$Mn(25) : Mn^{2+} (23 = 18 + 5)$

---

189. Compare the stability of +2 oxidation state of the elements of the first transition series.

A: The sum $I.E_1 + I.E_2$ of first transition series increases due to increased nuclear charge.
Thus, the standard potentials become less and less negative.
So the tendency to form M^{+2} ion decreases.
The greater stability of +2 state of Mn is due to half-filled d-subshell (d^5).
For zinc, it is due to completely filled d-subshell (d^{10}).
For nickel, it is due to highest negative enthalpy of hydration.

---

190. Compared to the changes in atomic and ionic sizes of elements of 3d and 4d series, the change in radii of elements of 4d and 5d series is virtually the same. Comment.

A: Compared to the change in atomic and ionic sizes of elements 3d and 4d series, the change in radii of elements 4d and 5d series is virtually the same.

This is due to the intervention of the 4f orbitals which must be filled before the 5d series of elements. This filling of 4f before 5d orbitals results in a regular decrease in atomic radii called Lanthanoid contraction. Due to Lanthanoid contraction the 4d and 5d series elements exhibit same size.

COORDINATION COMPOUNDS (SAQ)

191. What is meant by chelate effect ? Give example.

A: **Chelate effect:** It is the stabilization of complex compounds due to ring formation (chelation) by bidentate or polydentate ligands. Complexes containing chelating ligands are more stable than complexes containing unidentate ligands.

Ex: $[Co(en)_3]^{3+}$ is more stable than $[Co(NH_3)_6]^{3+}$

Note: en= Ethylenediamine $\Rightarrow$bidentate ligand (chelating ligand)

192. Give the oxidation numbers of the central metal atoms in the following complex entities
(i) $[Ni(CO)_4]$ (ii) $[Co(NH_3)_6]^{3+}$ (iii) $[Fe(CN)_6]^{4-}$ (iv) $[Fe(C_2O_4)_3]^{3-}$ [AP 19]

A: **i)** $[Ni(CO)_4]$

$x+4(0)=0$

$\Rightarrow x=0$

ii) $[Co(NH_3)_6]^{+3}$

$x+6(0) = +3$

$\Rightarrow x = +3$

In metal carbonyls metal oxidation state is equal to zero. (synergic effect)

iii) $[Fe(CN)_6]^{4-}$

$x+6(-1)= -4$

$\Rightarrow x=6-4=+2$

iv) $[Fe(C_2O_4)_3]^{3-}$

$x+3(-2) = -3$

$\Rightarrow x = +3$

---

193. Explain the terms (i) Ligand (ii) Coordination number
(iii) Coordination entity (iv) Central metal atom/ ion

A: **i) Ligand:** The ions or molecules bound to the central atom / ion in the complex by donating lone pair of electrons are called ligands. a) simple ions such as Cl^-

b) small molecules such as H_2O

c) large molecules such as $H_2N–CH_2–CH_2–NH_2$

d) macro molecules such as proteins.

ii) Coordination Number (CN) : It is the number of ligands to which the metal is directly bonded. **Ex:** In $[Pt\ Cl_6]^{-2}$, C.N of Pt = 6

iii) Coordination entity: It is the central metal atom/ ion together with the ligands directly bonded to it. It is enclosed between square brackets. **Ex:** $[COCl_3(NH_3)_3]$, $[Ni(CO)_4]$

iv) Central metal atom /ion: The atom / ion to which a fixed number of ligands/ ions/ molecules are bound in a complex is called central metal atom or ion.

Ex: In $K_4[Fe(CN)_6]$ the central metal is Fe

HALOALKANES & HALOARENES (SAQ)

194. Write the mechanism of dehydrohalogenation of 2-bromobutane.

A: **Dehydrohalogenation of 2-bromobutane** is an **Elimination reaction.**

It involves the removal of one molecule of HX, from the molecule of a haloalkane.

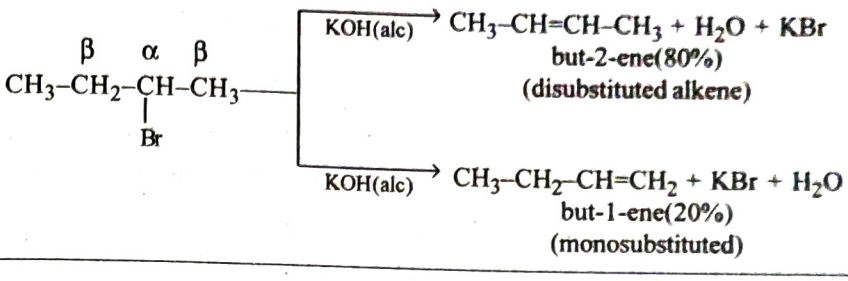
In elimination reactions, the halogen atom of the haloalkane (2-bromobutane) and a hydrogen atom from the adjacent carbon atom (β-carbon) are involved. Hence these reactions are also called β-elimination reactions. These reactions follow Saytzeff's rule.

Saytzeff rule: In dehydrohalogenation reactions, the preferred product is that alkene which has greater number of alkyl groups attached to the doubly bonded carbon atoms. i.e., more substituted alkane is the **major product** (but-2-ene(80%)).

Mechanism: In this reaction, but-2-ene is more substituted alkene than but-1-ene.

Hence, **but-2-ene is the major product.**

2-bromobutane can eliminate one molecule of HBr in the following two ways.



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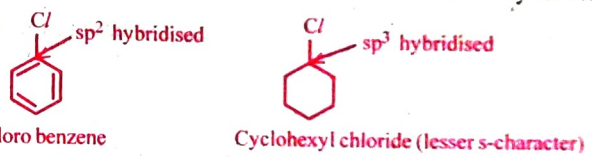
195. Explain why the dipole moment of chlorobenzene is lower than that of cyclohexyl chloride.

A: Carbon atom in chlorobenzene is sp^2 hybridised which is more electronegative than sp^3 carbon atom of cyclohexylchloride. Less electronegative leads to lesser s-character.
So, polarity of C–Cl bond in chlorobenzene is less than the C-Cl bond in cyclohexyl chloride.
Hence the dipole moment of chlorobenzene is lower than that of cyclohexyl chloride.



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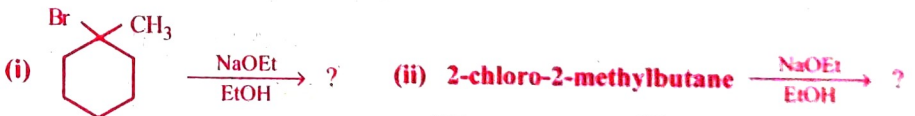
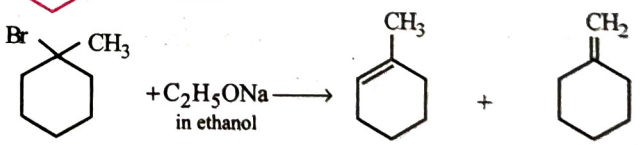
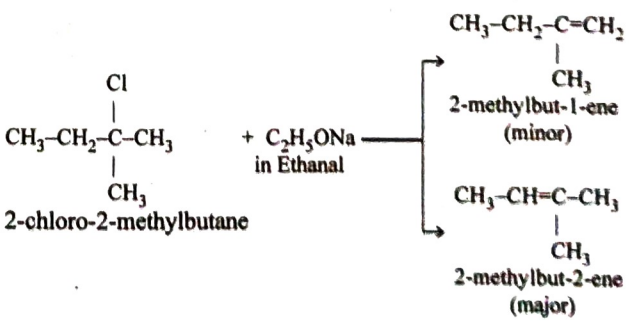
196. Predict the alkenes that would be formed in the following reactions and identify the major alkene.



A: (i)



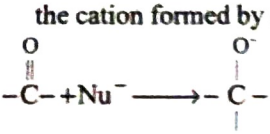
1-bromo 1-methyl cyclo hexane | 1-methyl cyclohex 1-ene (major) | Methylene cyclohexane (minor)



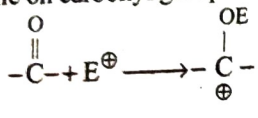
ALDEHYDES & KETONES (SAQ)

197. Explain why Aldehydes and ketones undergo nucleophilic addition while alkenes undergoes electrophilic addition though both are unsaturated compounds.

A: **1) Nucleophilic addition:** Aldehydes and ketones undergo nucleophilic addition because the intermediate anion formed by the attack of nucleophile on carbonyl group is more stable than the cation formed by attack of electrophile on carbonyl group.

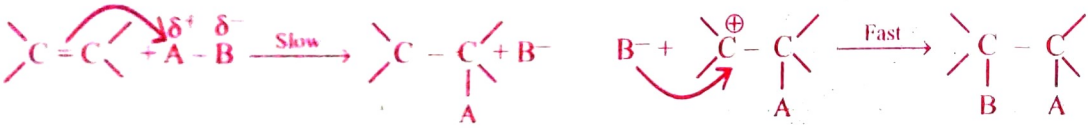


Anion(more stable)



Cation(less stable)

2) Electrophilic addition: Alkenes undergoes electrophilic addition beacuse in alkenes there is double bond between two carbon atoms and is electron rich. This double bond is highly reactive. So it is attacked by electrophilic first. So alkenes undergo electrophilic addition reactions.



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198. Arrange the following in the increasing order of their acidic strength: benzoic acid, 4-methoxybenzoic acid, 4-nitrobenzoic acid and 4-nethylbenzoic acid.

A: Acidic strength of electron with drawing $-NO_2$ group increases

Acidic strength of electron with releasing – $-OCH_3$, $-CH_3$ groups decreases

But releasing power of $-OCH_3$ is more than that of $-CH_3$.

Lower P^{ka} values indicate higher acidity.

Hence the increasing order (low to high) acidic strengths are given below:

4-methoxy benzoic acid < 4-methyl benzoic acid < benzoic acid < 4-nitrobenzoic acid.

P^{ka} values: (4.46) (4.36) (4.19) (3.41)

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CHAPTERS: 1,2,3

Q 1 UNIT TEST-1

Time : 1 Hour

SR. CHEMISTRY

Max.Marks : 25

Ans-Page Index

SECTION-A

I. Answer ALL the following questions. $3 \times 1 = 3$

1. Which law explains the solubility of gases in liquids ? [☞P 77(5)]
   1. Boyle's law 2. Charles' law 3.Henry's law 4.Raoult's law
2. Lead storage battery is an example of _____cell. [☞P 78(9)]
3. What is the term used for a solution that does not follow Raoult's law? [☞P 77(11)]

SECTION-B

II. Answer ALL the following questions $3 \times 2 = 6$

4. What are isotonic solutions? [☞P 56(80)]
5. State Faraday's first law of electrolysis. [☞P 58(86)]
6. What is standard hydrogen electrode? [☞P 59(94)]

SECTION-C

III. Answer ANY TWO of the following questions. $2 \times 4 = 8$

7. What is relative lowering of vapour pressure? How is it useful to determine the molar mass of a solute? [☞P 28(14)]
8. A solution of glucose in water is labeled as 10% w/w. What would be the molality of the solution ? [☞P 29(18)]
9. State and explain Nernst equation with the help of a metallic electrode and a non-metallic electrode. [☞P 32(23)]

(or)

10. What are Galvanic cells? Explain the working of Galvanic cell with a neat sketch taking Daniel cell as example. [☞P 30(19)]
11. State and explain Kohlrausch's law of independent migration of ions. [☞P 30(20)]
12. Define mass percentage, volume percentage and mass to volume percentage. [☞P 29(16)]

SECTION-D

IV. Answer ANY ONE of the following questions. $1 \times 8 = 8$

13. Give a detailed account of the collision theory of reaction rates of bimolecular gaseous reactions. [☞P 14(1)]

(or)

14. Explain the following terms with suitable examples. [☞P 16(3)]
    a) Average rate of a reaction b) Slow and fast reactions
    c) Order of a reaction d)Molecularity of a reaction
    e) Activation energy of a reaction

☞ **'Extra choice' is given in Section-C just to cover all the IMP questions**

CHAPTERS: 4,5,10

Q 2 UNIT TEST-2

SR. CHEMISTRY

Time : 1 Hour Max.Marks : 25

Ans-Page Index

SECTION-A

I. Answer ALL the following questions. $3 \times 1 = 3$

1. Lewis acid in the complex ion $[CoCl(NH_3)_5]^{2+}$ is [☞P 81(3)]

   1. Cl^- 2. NH_3 3. Co^{2+} 4. Co^{3+}

2. Mohr's salt formula is _____ [☞P 81(7)]

3. Name the disease that occurs due to deficiency of ascorbic acid. [☞P 86(14)]

SECTION-B

II. Answer ALL the following questions $3 \times 2 = 6$

4. Calculate the 'spin only' magnetic moment of $Fe^{2+}_{(aq)}$ ion. [☞P 62(109)]
5. $CuSO_4.5H_2O$ is blue in colour where as anhydrous $CuSO_4$ is colourless. Why? [☞P 65(122)]
6. What is Zwitter ion? Give an example. [☞P 74(170)]

SECTION-C

III. Answer ANY TWO of the following questions. $2 \times 4 = 8$

7. Write the characteristics properties of transition elements. [☞P 37(34)]
8. Explain Werner's theory of coordination compounds with suitable examples. [☞P 39(40)]
9. What are hormones? Give one example for each. [☞P 52(70)]
   (i) steroid hormones (ii) Polypeptide hormones and (iii) amino acid derivatives.

   (or)

10. What is lanthanoid contraction? What are the consequences of lanthanoid contraction? [☞P 37(35)]
11. Explain geometrical isomerism in coordination compounds giving suitable examples. [☞P 39(41)]
12. Write notes on vitamins. [☞P 53(71)]

SECTION-D

IV. Answer ANY ONE of the following questions. $1 \times 8 = 8$

13. a) Write four postulates of Werner's theory of coordination compounds. **(or)** [☞P 17(4)]
    b) Explain cis-trans isomerism in $[Pt(NH_3)_2Cl_2]$ complex.
14. Explain different types of isomerism exhibited by coordination compounds, giving suitable examples. [☞P 19(6)]

☞ **'Extra choice' is given in Section-C just to cover all the IMP questions**

CHAPTERS: 6,7,8,9

Q 3 UNIT TEST-3

Time : 1 Hour

SR. CHEMISTRY

Max.Marks : 25

SECTION-A

I. Answer ALL the following questions. $3 \times 1 = 3$

1. A primary alkyl halide would prefer to undergo
   1) S_N1 reaction 2) S_N2 reaction
   3) α–elimination 4) racemization
2. Alcohols and phenols react with carboxylic acids to form _____
3. The conversion of primary aromatic amines into diazonium salts is known as _____

SECTION-B

II. Answer ALL the following questions $3 \times 2 = 6$

4. What are ambident nucleophiles?
5. Identify the reactant needed to form t-butyl alcohol from acetone.
6. Write equations for carbylamine reaction of an aliphatic amine.

SECTION-C

III. Answer ANY TWO of the following questions. $2 \times 4 = 8$

7. Explain the mechanism of S_N1 with example.
8. Explain the following name reactions:
   i)Sandmeyer reaction ii) Gattermann reaction
9. Give the equations for the preparation of phenol from Cumene.

(or)

10. Write the products formed by the reduction and oxidation of phenol.
11. What is Tollen's reagent? Explain its reaction with Aldehydes.
12. Describe the following:
    i) Crossed aldol condensation ii) Decarboxylation

SECTION-D

IV. Answer ANY ONE of the following questions. $1 \times 8 = 8$

13. Explain the folliwng reactions
    a) Aldol reaction b) Esterification
    c) H.V.Z reaction d) Bromination of Benzoic acid
14. Describe the following:
    i) Acetylation ii) Cannizzaro reaction
    iii) Cross aldol condensation iv) Decarboxylation

☛ **'Extra choice' is given in Section-C just to cover all the IMP questions**

Ans-Page Index

[☛P 82(2)]
[☛P 83(8)]
[☛P 85(9)]
[☛P 66(129)]
[☛P 69(146)]
[☛P 72(156)]
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[☛P 49(64)]
[☛P 44(52)]
[☛P 44(53)]
[☛P 47(59)]
[☛P 48(63)]
[☛P 22(8)]
[☛P 23(9)]

Q 4 BOARD MODEL PAPER

Time : 3 Hours **SR. CHEMISTRY** Max.Marks : 85

SECTION-A

I. Answer ALL the following questions. $9 \times 1 = 9$

Ans-Page Index

1. Human blood is isotonic with _____solution (or) Saline. [☞P 77(10)]
2. Lead storage battery is an example of _____cell. [☞P 78(9)]
3. The number of collisions per second per unit volume of the reaction mixture is known as _____ [☞P 79(8)]
4. Which of the following coordination compound contains cobalt metal? [☞P 81(1)]
   1. Chlorophyll 2. Hemoglobin 3. Vitamin B_{12} 4. Myoglobin
5. Which of the following is the suitable medium for preparing Grignard reagents? [☞P 82(3)]
   1)Dry acetone 2) Dry ether 3)Concentrated HCl 4)Alcoholic KOH
6. Glycerol is a: [☞P 83(1)]
   1)dihydric alcohol 2) monohydric alcohol
   3) hexahydric alcohol 4) trihydric alcohol
7. What is Rochelle salt? [☞P 84(12)]
8. What is the IUPAC name of ethyl methyl amine? [☞P 85(11)]
9. Which linkage connects two monosaccharide units? [☞P 86(11)]

SECTION-B

II. Answer ALL the following questions $14 \times 2 = 28$

10. State Henry's law. [☞P 56(78)]
11. Define osmotic pressure. [☞P 56(79)]
12. State Faraday's first law of electrolysis. [☞P 58(86)]
13. What is metallic corrosion? Give one example. [☞P 58(89)]
14. What is a pseudo first order reaction? Give one example. [☞P 60(96)]
15. Why Zn^{2+} is diamagnetic whereas Mn^{2+} is paramagnetic? [☞P 62(108)]
16. Calculate the 'spin only' magnetic moment of $Fe^{2+}_{(aq)}$ ion. [☞P 62(109)]
17. What is an ambidentate ligand? Give example. [☞P 64(121)]
18. What are Enantiomers? [☞P 66(127)]
19. Give the reagents used for the preparation of phenol from chlorobenzene. [☞P 68(137)]
20. List the reagents suitable to reduce carboxylic acid to alcohol. [☞P 70(147)]
21. Arrange the following bases in increasing order of their basic strength. Aniline, p-nitroaniline and p-toluidine. [☞P 72(155)]
22. Differentiate between globular and fibrous proteins. [☞P 74(167)]
23. What are essential and non-essential amino acids? Give one example for each. [☞P 74(165)]

SECTION-C

III. Answer ANY EIGHT of the following questions. $8 \times 4 = 32$

24. What is meant by positive deviations from Raoult's law and how is the sign of Δ_{mix} H related to positive deviation from Raoult's law ? |☞P 27(12)|
25. State and explain Kohlrausch's law of independent migration of ions. |☞P 30(20)|
26. What is 'molecularity' of a reaction? How is it different from the 'order' of a reaction? Name one bimolecular and one trimolecular gaseous reactions. |☞P 34(28)|
27. What are Galvanic cells? Explain the working of Galvanic cell with a neat sketch taking Daniel cell as example. |☞P 30(19)|
28. Write the characteristics properties of transition elements. |☞P 37(34)|
29. What are homoleptic and heteroleptic complexes? Give one example for each. |☞P 40(42)|
30. Explain the mechanism of S_N1 with example. |☞P 41(46)|
31. Explain the acidic nature of phenols and compare with that of alcohols. |☞P 44(54)|
32. Give the equations for the preparation of phenol from Cumene. |☞P 44(52)|
33. What is Tollen's reagent? Explain its reaction with Aldehydes. |☞P 47(59)|
34. Explain the following reactions: |☞P 49(64)|
    i)Sandmeyer reaction ii) Gattermann reaction
35. Give the dietary sources of the following vitamins and name the diseases caused by their deficiency (a) A (b) D (c) E and (d) K |☞P 52(69)|

SECTION-D

IV. Answer ANY TWO of the following questions. $2 \times 8 = 16$

36. Give a detailed account of the collision theory of reaction rates of bimolecular gaseous reactions. |☞P 14(1)|
37. a)Write four postulates of Werner's theory of coordination compounds. |☞P 17(4)|
    b)Explain cis-trans isomerism in $[Pt(NH_3)_2Cl_2]$ complex.
37. Explain the folliwng reactions |☞P 22(8)|
    a) Aldol reaction b) Esterification
    c) H.V.Z reaction d) Bromination of Benzoic acid

GUESS PAPERS

Q 5 GUESS PAPER -1

SR. CHEMISTRY

Time : 3 Hours | Max.Marks : 85

SECTION-A

Ans-Page Index

I. Answer ALL the following questions. $9 \times 1 = 9$

1. Solubility of a gas in liquid _____ with increase in pressure. [☞P 77(7)]
2. 1 Faraday value is equal to _____ C mol^{-1} [☞P 78(6)]
3. Which reactions takes place in one step? [☞P 79(12)]
4. Lewis acid in the complex ion $[CoCl(NH_3)_5]^{2+}$ is [☞P 81(3)]
   1. Cl^- 2. NH_3 3. Co^{2+} 4. Co^{3+}
5. What is the general formula for haloalkanes? (X=halogen atom, n=1, 2, 3…) [☞P 82(1)]
   1) $C_nH_{2n}X$ 2) $C_nH_{2n+1}X$ 3) $C_nH_{2n-1}X$ 4) $C_nH_{2n-3}X$
6. Which of the following shows strong hydrogen bonding? [☞P 83(3)]
   1) Ethyl amine 2)Ammonia 3)Ethyl alcohol 4)Diethyl ether
7. Which acid is used as solvent in food industry as vinegar? [☞P 84(13)]
8. Coupling reaction of aryl diazonium salts with phenols or aryl amines gives rise to the formation of_____ [☞P 85(10)]
9. Which non-essential amino acid is optically inactive? [☞P 86(12)]

SECTION-B

II. Answer ALL the following questions $14 \times 2 = 28$

10. What are isotonic solutions? [☞P 56(80)]
11. Calculate the molarity of a solution containing 5 g of NaOH in 450 ml solution. [☞P 57(84)]
12. State Faraday's second law of electrolysis. [☞P 58(87)]
13. What is cell constant of a conductivity cell? [☞P 59(92)]
14. Give two examples for gaseous first order reactions. [☞P 60(97)]
15. Aqueous Cu^{2+} ions are blue in colour, where as Aqueous Zn^{2+} ions are colourless. Why? [☞P 62(110)]
16. What is mischmetal? Give its composition and uses. [☞P 63(112)]
17. $CuSO_4.5H_2O$ is blue in colour where as anhydrous $CuSO_4$ is colourless. Why? [☞P 65(122)]
18. What are ambident nucleophiles? [☞P 66(129)]
19. Explain why propanol has higher boiling point than that of the hydrocarbon butane. [☞P 68(138)]
20. Write the reaction showing α-halogenation of carboxylic acid and give its name. [☞P 70(148)]
21. Write equations for carbylamine reaction of an aliphatic amine. [☞P 72(156)]
22. What are anomers? [☞P 74(169)]
23. Why are sugars classified as reducing and non-reducing sugars? [☞P 75(172)]

SECTION-C

III. Answer ANY EIGHT of the following questions. **8 × 4 = 32**

24. What is relative lowering of vapour pressure? How is it useful to determine the molar mass of a solute? [☞P 28(14)]
25. Give the construction and working of a standard hydrogen electrode with a neat diagram. [☞P 31(21)]
26. Derive the integrated rate equation for a zero order reaction. [☞P 35(30)]
27. State and explain Kohlrausch's law of independent migration of ions. [☞P 30(20)]
28. What is lanthanoid contraction? What are the consequences of lanthanoid contraction? [☞P 37(35)]
29. Explain Werner's theory of coordination compounds with suitable examples. [☞P 39(40)]
30. How will you carry out the following conversions? [☞P 41(47)]
    (i) Ethane to bromoethane (ii) Toluene to benzyl alcohol
31. Write the products formed by the reduction and oxidation of phenol. [☞P 44(53)]
32. Account for the statement: Alcohols boil at highest temperature than hydrocarbons and ethers of comparable molecular masses. [☞P 45(56)]
33. Write the oxidation products of acetaldehyde, acetone and acetophenone. [☞P 47(60)]
34. Write the reactions of [☞P 50(65)]
    (i) aromatic and (ii) aliphatic primary amines with nitrous acid, highlighting the differences.
35. What are hormones? Give one example for each. [☞P 52(70)]
    (i) steroid hormones (ii) Polypeptide hormones and (iii) amino acid derivatives.

SECTION-D

IV. Answer ANY TWO of the following questions. **2 × 8 = 16**

36. a) Write the differences between order & molecularity. [☞P 15(2)]
    b) Describe the salient features of collision theory.
37 Discuss the nature of bonding and magnetic behaviour in the following coordination entities on the basis of valence bond theory [☞P 20(7.1)]
    (i) $[Fe(CN)_6]^{4-}$ (ii) $[FeF_6]^{3-}$
38. Describe the following: [☞P 23(9)]
    i) Acetylation ii) Cannizzaro reaction
    iii) Cross aldol condensation iv) Decarboxylation

GUESS PAPERS

Q 6 GUESS PAPER -2

Time : 3 Hours

SR. CHEMISTRY

Max.Marks : 85

SECTION-A

I. Answer ALL the following questions. $9 \times 1 = 9$

1. Desalination plants work on the basis of the phenomenon called _____ [☞P 77(9)]
2. In Daniell cell the oxidation half reaction occurs at _____ [☞P 78(10)]
3. What is the unit of rate constant for a zero-order reaction? [☞P 79(14)]
4. Which ligand gives a stable complex? [☞P 81(4)]
   1. SCN^- 2. Cl^- 3. H_2O 4. Ethylene diamine(en)
5. A primary alkyl halide would prefer to undergo [☞P 82(2)]
   1) S_N1 reaction 2) S_N2 reaction 3) α–elimination 4) racemization
6. Which of the following can work as a dehydrating agent for alcohols? [☞P 83(2)]
   1) Conc. H_2SO_4 2) Anhydrous Al_2O_3 3) H_3PO_4 4) All of these
7. Aldehydes and ketones undergo what type of addition reactions? [☞P 84(14)]
8. A molecule that has both positive and negative regions of charge is called [☞P 85(15)]
9. Name the disease that occurs due to deficiency of ascorbic acid. [☞P 86(14)]

SECTION-B

II. Answer ALL the following questions $14 \times 2 = 28$

10. What is Cryoscopic constant? [☞P 57(82)]
11. Define mole fraction. [☞P 56(77)]
12. What is a fuel cell? How is it different from a conventional galvanic cell? [☞P 58(88)]
13. Define molar conductivity Λ_m and how is it related to conductivity (k)? [☞P 59(91)]
14. Give two examples for zero Order reactions. [☞P 60(98)]
15. What is an alloy? Give example. [☞P 62(111)]
16. What is actinoid contraction? [☞P 63(113)]
17. What is a double salt? Give example. [☞P 64(118)]
18. What type of isomerism is exhibited by o-, m- and p-dichlorobenzenes? [☞P 66(128)]
19. Alcohols are comparatively more soluble in water than hydrocarbons of comparable molecular masses. Explain this fact. [☞P 68(139)]
20. Compare the acidic strength of acetic acid, Chloroacetic acid, benzoic acid and Phenol. [☞P 71(153)]
21. Gabriel phthalimide synthesis exclusively forms primary amines . Explain. [☞P 72(157)]
22. Name the vitamin responsible for the coagulation of blood. [☞P 74(166)]
23. What is Zwitter ion? Give an example. [☞P 74(170)]

Ans-Page Index

SECTION-C

III. Answer ANY EIGHT of the following questions. $8 \times 4 = 32$

24. What is meant by negative deviation from Raoult's law and how is the sign of Δ_{mix} H related to negative deviation from Raoult's law ? [☞P 27(13)]
25. What are the products obtained at the cathode and the anode during the electrolysis of the following when platinum electrodes are used in the electrolysis(a) Molten KCl(b) Aq. $CuSO_4$ solution (c) Aq.K_2SO_4 solution [☞P 31(22)]
26. What is half-life ($t_{1/2}$) of a reaction? Derive the equations for the 'half-life' value of zero and first order reactions. [☞P 35(31)]
27. What are Galvanic cells? Explain the working of Galvanic cell with a neat sketch taking Daniel cell as example. [☞P 30(19)]
28. What are interstitial compounds? How are they formed? Give two examples. [☞P 37(36)]
29. Explain geometrical isomerism in coordination compounds giving suitable examples. [☞P 39(41)]
30. Describe the stereochemical effect on the hydrolysis of 2-bromobutane. [☞P 42(48)]
31. Write the mechanism of hydration of ethene to yield ethanol. [☞P 45(57)]
32. While separating a mixture of ortho and para nitrophenols by steam distillation, name the isomer which will be steam volatile. Give reason. [☞P 45(55)]
33. Write the equations of any aldehyde with Fehling's reagent. [☞P 47(61)]
34. Write the equations involved in the reaction of Nitrous acid with ethylamine and aniline. [☞P 50(66)]
35. Explain the denaturation of proteins. [☞P 54(73)]

SECTION-D

IV. Answer ANY TWO of the following questions. $2 \times 8 = 16$

36. Explain the following terms with suitable examples. [☞P 16(3)]
    a) Average rate of a reaction
    b) Slow and fast reactions
    c) Order of a reaction
    d)Molecularity of a reaction
    e) Activation energy of a reaction
37. Explain different types of isomerism exhibited by coordination compounds, giving suitable examples. [☞P 19(6)]
38. Explain the following terms. Give an example of the reaction in each case. [☞P 25(11)]
    i) Cyanohydrin
    ii) Acetal
    iii) Semicarbazone
    iv)Aldol

Q 7

GUESS PAPERS

GUESS PAPER -3

Time : 3 Hours

SR. CHEMISTRY

Max.Marks : 85

Ans-Page Index

SECTION-A

I. Answer ALL the following questions. $9 \times 1 = 9$

1. For an ideal solution, $\Delta_{mix}H$ is equal to _____ [☞P 77(8)]
2. What is the charge of an electron in coulombs? [☞P 78(13)]
3. The sum of powers of the concentration terms in the rate law expression is called _____ of that chemical reaction. [☞P 79(6)]
4. Which of the following is not a double salt? [☞P 81(2)]
   1) $FeSO_4.(NH_4)_2SO_4.6H_2O$ 2) $KCl.MgCl_2 6H_2O$
   3) $PtCl_4.2HCl$ 4) $KAl(SO_4)_2.12H_2O$
5. Which of the following alkyl halide will undergo S_N1 reaction most readily? [☞P 82(4)]
   1) $(CH_3)_3C—F$ 2) $(CH_3)_3C—Cl$ 3) $(CH_3)_3C—Br$ 4) $(CH_3)_3C—I$
6. When phenol is treated with excess bromine water it gives [☞P 83(4)]
   1) m-bromophenol 2) o- and p-bromophenol
   3) 2,4-dibromophenol 4) 2,4,6-tribromophenol
7. What is name of the product formed when benzene is reacted with CO and HCl in the presence of anhydrous $AlCl_3$? [☞P 84(11)]
8. Aniline and other arylamines are usually colourless but get coloured on storage why? [☞P 85(13)]
9. Increased level of which hormone causes hyperthyroidism ? [☞P 86(15)]

SECTION-B

II. Answer ALL the following questions $14 \times 2 = 28$

10. Define molarity. [☞P 56(75)]
11. Calculate the amount of benzoic acid (C_6H_5COOH) required for preparing 250 mL of 0.15 M solution of benzoic acid in methanol. [☞P 57(83)]
12. State Faraday's first law of electrolysis. [☞P 58(86)]
13. What is standard hydrogen electrode? [☞P 59(94)]
14. Write the equation for the rate of the reaction [☞P 60(100)]

$$5Br^-_{(aq)} + BrO^-_{3(aq)} + 6H^+_{(aq)} \rightarrow 3Br_{2(aq)} + 3H_2O(1)$$

15. What are transition elements? Give examples. [☞P 62(107)]
16. Calculate the 'spin only' magnetic moment of $Fe^{2+}_{(aq)}$ ion. [☞P 62(109)]
17. What are coordination compounds? Give two examples. [☞P 64(117)]
18. Write the isomers of the compound having molecule formula C_4H_9Br? [☞P 66(132)]
19. Preparation of ethers by acid dehydration of secondary or tertiary alcohols is not a suitable method. Give reason. [☞P 68(141)]

20. How do you distinguish acetophenone and benzophenone? [☛P 70(150)]
21. Explain why ethylamine is more soluble in water where as aniline is sparingly soluble. [☛P 73(161)]
22. Write the names of three types of RNA. [☛P 75(171)]
23. What is glycogen? How is it different from starch? [☛P 74(168)]

SECTION-C

III. Answer ANY EIGHT of the following questions. $8 \times 4 = 32$

24. Calculate the mass of a non-volatile solute (molar mass 40g mol^{-1}) which should be dissolved in 114g Octane to reduce its vapour pressure to 80%. [☛P 28(15)]
25. State and explain Nernst equation with the help of a metallic electrode and a non-metallic electrode. [☛P 32(23)]
26. Derive an integrated rate equation for a first order reaction. [☛P 35(30)]
27. Give the construction and working of a standard hydrogen electrode with a neat diagram. [☛P 31(21)]
28. Write the characteristics of interstitial compounds. [☛P 38(37)]
29. Using IUPAC norms write the formulas for the following : [☛P 40(44)]
    (i) Tetrahydroxozincate (II) ion (ii) Hexaaminecobalt (III) sulphate
    (iii) Potassium tetrachloropalladate(II)
    (iv) Potassium tri(oxoalato)chromate(III)
30. Write the structures of the following organic halides. [☛P 43(51)]
    (i) 1-bromo-4-sec-butyl-2 methylbenzene (ii)2-chloro-1-phenylbutane
    (iii) p-bromochlorobenzene (iv)4-t-butyl-3-iodoheptane
31. Draw the structures of all isomeric alcohols of molecular formula $C_5H_{12}O$ and give their IUPAC names and classify them as primary, secondary, and tertiary alcohols. [☛P 46(58)]
32. Write the products formed by the reduction and oxidation of phenol. [☛P 44(53)]
33. Write the IUPAC names of the following [☛P 48(62)]
    (i) $CH_3CH_2CH(Br)CH_2COOH$ (ii) $PhCH_2COCH_2COOH$
    (iii) $CH_3CH(CH_3)CH_2COOC_2H_5$
34. Compare the basicity of the following in gaseous state and in aqueous state and arrange them in increasing order of basicity. [☛P 51(68)]
    CH_3NH_2, $(CH_3)_2NH$, $(CH_3)_3N$ and NH_3
35. Write notes on vitamins. [☛P 53(71)]

SECTION-D

IV. Answer ANY TWO of the following questions. $2 \times 8 = 16$

36. Give a detailed account of the collision theory of reaction rates of bimolecular gaseous reactions. [☛P 14(1)]
37 Discuss the nature of bonding and magnetic behaviour in the following coordination entities on the basis of valence bond theory [☛P 21(7.2)]
    (i) $[Co(C_2O_4)_3]^{3-}$ (ii) $[CoF_6]^{3-}$
38. How do you distinguish the following pairs of compounds? [☛P 24(10)]
    i) Propanal and propanone ii) Acetophenone and benzophenone
    iii) Phenol and benzoic acid iv) Pentan-2-one and Pentan-3-one

Q 8

GUESS PAPERS

GUESS PAPER -4

SR. CHEMISTRY

Time : 3 Hours | Max.Marks : 85

	Ans-Page Index

SECTION-A

I. Answer ALL the following questions. $9 \times 1 = 9$

1. The units of boiling point elevation constant K_b are _____ [☞P 77(6)]
2. Corrosion of iron is mainly due to _____ and _____ [☞P 78(7)]
3. The overall rate of the reaction is controlled by the slowest step of the reaction and is called the _____ [☞P 79(7)]
4. Which complex shows geometrical, optical and also structural isomerism? [☞P 81(5)]
   1. $[Co(en)_3]Cl_3$ 2. $[CoCl_2(en)_2]Br$ 3. $[Co(en)(NH_3)_2Cl_2]Cl$ 4. $[PdCl_2(NH_3)_2$
5. Chlorobenzene is [☞P 82(5)]
   1) less reactive than benzyl chloride 2) more reactive than ethyl bromide
   3) nearly as reactive as methyl chloride 4) more reactive than isopropyl chloride
6. Which of the following is formed when phenol is exposed to air? [☞P 83(5)]
   1) Benzoquinone 2) Benzyl quinone 3) Acetophenone 4) Benzene
7. Phthalic acid on reaction with ammonia followed by strong heating gives a compound 'X' along with elimination of NH_3. What is the name of that compound 'X'? [☞P 84(15)]
8. The process of cleavage of the C–X bond of alkyl halides by ammonia molecule is known as [☞P 85(12)]
9. What type of bonding helps in stabilizing the α-helix structure of proteins? [☞P 86(13)]

SECTION-B

II. Answer ALL the following questions $14 \times 2 = 28$

10. Define molality. [☞P 56(76)]
11. What is Ebullioscopic constant? [☞P 57(81)]
12. Give the electro-chemical reaction that represents the corrosion or rusting of iron. [☞P 58(90)]
13. How is a galvanic cell represented on paper as per IUPAC convention? Give one example. [☞P 59(93)]
14. What are complex reactions? Name one complex reaction. [☞P 61(103)]
15. What is mischmetal? Give its composition and uses. [☞P 63(112)]
16. Why Zn^{2+} is diamagnetic whereas Mn^{2+} is paramagnetic? [☞P 62(108)]
17. What is the oxidation state of cobalt in (i) $K[Co(CO)_4]$ and (ii) $[Co(NH_3)_6]^{3+}$ [☞P 65(123)]
18. Write the structure of the following compounds [☞P 66(131)]
    (i) 2-chloro-3-methylpentane (ii) 1-bromo-4-sec-butyl-2-methylbenzene
19. Write the structures for the following compounds [☞P 69(143)]
    1) Ethoxy Ethane 2) Ethoxybutane 3) Phenoxy ethane
20. An organic acid with molecular formula $C_8H_8O_2$ on decarboxylation forms Toluene. Identify the organic acid. [☞P 70(151)]

21. Arrange the following bases in decreasing order of pK_b values $C_2H_5NH_2$, $C_6H_5NHCH_3$, $(C_6H_5)_2NH$ and $C_6H_5NH_2$. [☛P 72(158)]
22. What is Zwitter ion? Give an example. [☛P 74(170)]
23. What is the difference between a nucleoside and a nucleotide? [☛P 75(174)]

SECTION-C

III. Answer ANY EIGHT of the following questions. $8 \times 4 = 32$

24. A solution of glucose in water is labeled as 10% w/w. What would be the molality of the solution ? [☛P 29(18)]
25. What is metallic corrosion? Explain it with respect to iron corrosion. [☛P 32(24)]
26. What is Arrhenius equation? Derive an equation which describes the effect of rise of temperature (T) on the rate constant (k) of a reaction. [☛P 36(33)]
27. What are Galvanic cells? Explain the working of Galvanic cell with a neat sketch taking Daniel cell as example. [☛P 30(19)]
28. Why do the transition metal ions exhibit characteristic colours in aqueous solution. Explain giving examples. [☛P 38(38)]
29. What are homoleptic and heteroleptic complexes? Give one example for each. [☛P 40(42)]
30. Write the differences between S_N1 and S_N2 reactions. [☛P 42(49)]
31. Account for the statement: Alcohols boil at highest temperature than hydrocarbons and ethers of comparable molecular masses. [☛P 45(56)]
32. Write the mechanism of hydration of ethene to yield ethanol. [☛P 45(57)]
33. Describe the following: [☛P 48(63)]
    i) Crossed aldol condensation ii) Decarboxylation
34. Give one chemical test to distinguish between the following pairs of compounds: [☛P 50(67)]
    i) Methylamine and dimethylamine ii) Aniline and N-methylaniline
    iii) Ethylamine and aniline
35. What are enzymes?Give examples. [☛P 54(74)]

SECTION-D

IV. Answer ANY TWO of the following questions. $2 \times 8 = 16$

36. a)Write the differences between order & molecularity. [☛P 15(2)]
    b)Describe the salient features of collision theory.
37. a)Write four postulates of Werner's theory of coordination compounds. [☛P 17(4)]
    b) Explain cis-trans isomerism in $[Pt(NH_3)_2Cl_2]$ complex.
38. Explain the folliwng reactions [☛P 22(8)]
    a) Aldol reaction b) Esterification
    c) H.V.Z reaction d) Bromination of Benzoic acid

Q 9

GUESS PAPERS

GUESS PAPER -5

SR. CHEMISTRY

Time : 3 Hours Max.Marks : 85

Ans-Page Index

SECTION-A

I. Answer ALL the following questions. 9 × 1 = 9

1. What is the name given to the solutions, when they have same osmotic pressure at a given temperature? [☛P 77(15)]
2. The units of molar conductivity are _____ [☛P 78(8)]
3. $k = A\ e^{-Ea/RT}$ where A is called _____ [☛P 79(9)]
4. The IUPAC name of $K_2[PdCl_4]$ is _____ [☛P 81(9)]
5. A primary alkyl halide would prefer to undergo [☛P 82(2)]
   1) S_N1 reaction 2) S_N2 reaction
   3) α–elimination 4) racemization
6. When phenol is treated with excess bromine water it gives [☛P 83(4)]
   1) m-bromophenol 2) o- and p-bromophenol
   3) 2,4-dibromophenol 4) 2,4,6-tribromophenol
7. Which of the following is a dicarboxylic acid? [☛P 84(3)]
   1) oxalic acid 2) benzoic acid 3) salicylic acid 4) picric acid
8. The process of cleavage of the C–X bond of alkyl halides by ammonia molecule is known as [☛P 85(12)]
9. Name the disease that occurs due to deficiency of ascorbic acid. [☛P 86(14)]

SECTION-B

II. Answer ALL the following questions 14 × 2 = 28

10. Define mole fraction. [☛P 56(77)]
11. Calculate molality of 2.5 g of ethanoic acid (CH_3COOH) in 75g of benzene. [☛P 57(85)]
12. State Faraday's first law of electrolysis. [☛P 58(86)]
13. Write the Nernst equation for the EMF of the cell [☛P 59(95)]
    $Ni(s) / Ni^{2+}(aq) // Ag^{+}(aq) / Ag(s)$
14. A first order reaction is found to have a rate constant $k = 5.5 \times 10^{-14}\ s^{-1}$. Find the half-life of the reaction. [☛P 61(106)]
15. How do transition metals exhibit catalytic activity? [☛P 63(115)]
16. Why do the transition metals form a large number of complex compounds? [☛P 63(114)]
17. $[Cr(NH_3)_6]^{3+}$ is paramagnetic while $[Ni(CN)_4]^{2-}$ is diamagnetic. Why? [☛P 65(124)]
18. What is the stereochemical outcome of S_N1 and S_N2 reactions? [☛P 66(130)]
19. Identify the reactant needed to form t-butyl alcohol from acetone. [☛P 69(146)]
20. Write equations showing the conversion of [☛P 71(154)]
    i) acetic acid to acetyl chloride ii) benzoic acid to benzamide
21. Give structures of A,B and C in the following reaction. [☛P 72(160)]

$$C_6H_5N_2Cl \xrightarrow{CuCN} A \xrightarrow{H_2O/H^+} B \xrightarrow{NH_3} C$$

22. Write the names of three types of RNA. [☛P 75(171)]
23. Name the different types of carbohydrates on the basis of their hydrolysis. Give one example for each. [☛P 75(173)]

SECTION-C

III. Answer ANY EIGHT of the following questions. $8 \times 4 = 32$

24. Define mass percentage, volume percentage and mass to volume percentage solutions. [☛P 29(16)]
25. What are fuel cells? How are they different from galvanic cells? Give the construction of H_2-O_2 fuel cell. [☛P 88(179)]
26. Explain the terms [☛P 36(32)]
    a) Activation energy (E_a) b) Collision frequency (Z)
    c) Probability factor (P) with respect to Arrhenius equation.
27. What is lanthanoid contraction? What are the consequences of lanthanoid contraction? [☛P 37(35)]
28. A solution of $CuSO_4$ is electrolysed for 10 minutes with a current of 1.5 amperes. What is the mass of copper deposited at the cathode? [☛P 33(25)]
29. Explain the following terms by giving an example [☛P 40(43)]
    i) Unidentate ligand ii)Bidentate ligand
    iii) Polydentate ligand iv)Ambidentate ligand
30. Give the IUPAC names of the following compounds. [☛P 43(50)]
    (i) CH_3–CH(Cl)–CH(I)CH_3 (ii) $ClCH_2$.CH=CH–CH_2Br
    (iii) $(CCl_3)_3CCl$ (iv) $CH_3C(p\text{–}Cl\text{–}C_6H_4)_2$–CH(Br)$CH_3$
31. Give the equations for the preparation of phenol from Cumene. [☛P 44(52)]
32. Draw the structures of all isomeric alcohols of molecular formula $C_5H_{12}O$ and give their IUPAC names and classify them as primary, secondary, and tertiary alcohols. [☛P 46(58)]
33. Arrange the following in the increasing order of their acidic strength: benzoic acid, 4-methoxybenzoic acid, 4-nitrobenzoic acid and 4-nethylbenzoic acid. [☛P 96(198)]
34. Write the equations involved in the reaction of Nitrous acid with ethylamine and aniline. [☛P 50(66)]
35. Write the importance of carbohydrates. [☛P 54(72)]

SECTION-D

IV. Answer ANY TWO of the following questions. $2 \times 8 = 16$

36. Give a detailed account of the collision theory of reaction rates of bimolecular gaseous reactions. [☛P 14(1)]
37. Explain the applications of coordination compounds in different fields. [☛P 18(5)]
38. Describe the following: [☛P 23(9)]
    i) Acetylation ii) Cannizzaro reaction
    iii) Cross aldol condensation iv) Decarboxylation

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KALPANA CHAWLA

Dreams Take You Far

From Karnal to space, she pursued her dream and became the first woman of Indian origin to fly in space.

LESSON: Dream beyond your surroundings.

YOUR TURN: Write down one big dream of yours.

P. V. SINDHU

Keep Fighting

She faced many defeats and setbacks but never stopped. She became an Olympic medallist and World Champion.

LESSON: Champions don't fear another attempt.

YOUR TURN: Don't quit. Try once more.

D. GUKESH

Youngest World Champion

At just 18, he became the youngest-ever undisputed World Chess Champion.

LESSON: Age is not a limit to achievement.

YOUR TURN: Believe in yourself. Great things are possible!

KONERU HUMPY

Patience Wins

A chess prodigy at a young age, she continued to compete at the highest level and inspired millions.

LESSON: Talent grows through patience and persistence.

YOUR TURN: Be patient. Keep learning every day.

ARUNIMA SINHA

Nothing Is Impossible

After losing a leg in an accident, she set a goal and became the first Indian amputee to climb Mount Everest.

LESSON: Don't let one setback define your life.

YOUR TURN: Decide your goal and never give up.

MITHALI RAJ

Let Performance Speak

She faced the challenges of being a woman in cricket and became one of India's greatest cricketers and a long-serving captain.

LESSON: Stay disciplined. Let your performance speak.

YOUR TURN: Do your best in everything you do.

SUDHA CHANDRAN

Dance of Courage

After losing a leg, she learned to dance again with a prosthetic limb and returned to the stage and screen.

LESSON: What you lose need not take away your dream.

YOUR TURN: Face your challenges with courage.

SACHIN TENDULKAR

Practice. Patience. Passion.

Started young, faced failures and injuries, but built his greatness through years of practice, patience and discipline.

LESSON: Greatness is built one day at a time.

YOUR TURN: Be consistent every single day.

SAINA NEHWAL

Fall. Recover. Return Stronger.

She faced injuries and tough phases but worked hard and came back to reach the top of world badminton.

LESSON: Fall. Recover. Return stronger.

YOUR TURN: Come back stronger from every fall.

NAMBI NARAYANAN

Truth. Patience. Purpose.

A brilliant space scientist, he faced a false case and years of hardship. He never gave up and continued to serve the nation.

LESSON: Truth may be delayed, but never stop standing for it.

YOUR TURN: Always choose honesty and hard work.

GOODNESS ACTIONS (NO MONEY NEEDED!)

- Plant a Sapling
- Make Someone Smile
- Read & Share Good Books
- Save Water
- Help Someone
- Keep It Clean
- Listen Patiently
- Encourage Others
- Give Without Expecting
- Respect Your Nation

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